

The quartic QRH boundary

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Abstract

We prove nonvanishing for finite-order Hecke L -functions of trivial infinite type over $\mathbf{Q}(\sqrt{-3})$, and for all Dirichlet L -functions, in the strict half-plane $\Re s > b_0$, apart from principal poles. Here $b_0 = 11/12 - \ell_0/4 = 0.8748866470399641\dots$, where ℓ_0 is the root of $927X^4 - 3135X^3 + 2433X^2 + 275X - 100$ in $[1/6, 1/5]$. The corresponding value is $c_0 = 1 - b_0 = 0.1251133529600359\dots$. The proof uses a variable-parameter moment estimate, the physical weighted numerator and an exact quartic optimization, with the established $7/8$ theorem as initialization. An ordinary Lean formalization supplies the exact boundary and a rational corollary at $\theta = 683505193/781250000$.

T0. Exact statement and a noncircular initialization

Let $K = \mathbf{Q}(\omega)$, $\omega = e^{2\pi i/3}$, and $\mathcal{O} = \mathbf{Z}[\omega]$. Let ℓ_0 be the unique root in

$$I = (16712007850680/10^{14}, 16712007850682/10^{14})$$

of $p(X) = 927X^4 - 3135X^3 + 2433X^2 + 275X - 100$. Set

$$b_0 = 11/12 - \ell_0/4, \quad c_0 = 1 - b_0.$$

Theorem T0. Every finite-order Hecke character of trivial infinite type over K , in any nonzero integral modulus, has nonzero ordinary L -function for $b_0 < \Re s$, except at its principal pole. Consequently, for every $q \geq 1$, every $\chi \in \text{DirChar}(\mathbf{C}, q)$, and every $s \in \mathbf{C}$,

$$b_0 < \Re s, \quad \neg(\chi = 1 \wedge s = 1) \implies L(s, \chi) \neq 0.$$

The boundary is strict; no assertion of nonvanishing on $\Re s = b_0$ is made.

We use the ordinary Hecke objects of the formal source. In `OpenAI/math` at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, `OAI.SevenEighths.HeckeFamily.LFunction_ne_zero_of_seven_eighths_lt_re` proves nonvanishing for every `HeckeFamily.Character` when $7/8 < \Re s$, with the principal pole excluded. We use this coarse theorem to initialize the family supremum.

For exact agreement with `OAI.SevenEighths.HeckeZeroSupremum.beta`, take the supremum of $\{1/2\}$ together with the real parts of all positive-real-part Hecke zeros outside the principal pole. The set is nonempty and bounded above. Primitive reduction in T2 shows that imprimitive characters add no zeros in $\Re s > 0$. The upstream theorem therefore gives $\beta_* \leq 7/8$. The source's primitive-family supremum and this definition agree whenever $\beta_* > b_0$; the sentinel is then irrelevant.

Suppose for contradiction that $\beta_* > b_0$, and put $\kappa = 2\beta_* - 1$. Then

$$7/10 < \kappa_0 = 2b_0 - 1 < \kappa \leq 3/4.$$

Only this compact range is needed for the modified positive-slot moment estimate. No assumption $\beta_* \leq b_0$ occurs. The hypothesis $\beta_* \leq (1 + \kappa)/2$ used to estimate the small prime slots is an equality by this definition. B3 proves the moment induction with this hypothesis and a positive comparison margin, not with the desired improved conclusion.

The only initial zero-free bound used in the argument is $\beta_* \leq 7/8$. The inverse amplification estimates below remain necessary for the nonprincipal rows.

T1. Dependency order, conventions and exact endpoint

Read B1 for finite arithmetic; B2 for completed reflection, local branches and its energy bound; B3 for inverse and plain moments; B4 for physical Euler factors and coefficient mass. B0 assembles these results using the detector, signed Gram bound, full numerator inversion and a uniform improvement of the zero supremum. The forward references are acyclic.

The normalizations are fixed throughout: ideal columns use one primary generator, element rows keep all six units, $q_a = |a|^2$, the additive character is $e(z) = \exp(2\pi i \operatorname{Tr}(z/\lambda))$, and the measure is $2 dx dy/\sqrt{3}$. Symbols at nonunits are zero even for exponent zero or six. The Hecke ideal-series normalization is ordinary L , not completed L ; upstream Hecke/IdealBridge.lean divides its lattice continuation by six. No extra factor six is introduced into the unique frequency factorization $H = ur^6$.

The following elementary facts complete the exact numerical part of B0.6. The formal counterpart is `Cycle25.Arithmetic.Endpoint`, using exact rational and polynomial identities.

T1.1 (root). Exact substitution gives $p(\inf I) < 0 < p(\sup I)$. On $[1/6, 1/5]$,

$$p'(x) = 3708x^3 - 9405x^2 + 4866x + 275 \geq -9405/25 + 4866/6 + 275 = 3549/5 > 0.$$

The intermediate value theorem and strict monotonicity prove existence and uniqueness of the chosen root. Other roots of the quartic are irrelevant.

T1.2 (geometry without numerical optimization). In B0.45–B0.49 use

$$\begin{aligned} \kappa_0 &= (5 - 3\ell_0)/6, \\ \delta_0 &= \frac{25 - 60\ell_0 + 27\ell_0^2}{6(5 + 4\ell_0 - 3\ell_0^2)}, \\ C_0 &= \frac{6\kappa_0 - 1}{15\kappa_0 - 2}, \quad V_0 = 5/6 - \delta_0, \\ R'_0 &= -1 + \frac{C_0(V_0^2 - \delta_0^2 C_0)}{2(V_0 + \delta_0 C_0)^2}, \\ h_0 &= -\frac{(1 + \ell_0)/2 + \ell_0/(6\kappa_0)}{3R'_0/4 - 1/(24\kappa_0)}, \quad x_0 = 1 + \ell_0 - h_0, \quad y_0 = h_0 - 2\ell_0. \end{aligned}$$

Every denominator is separated from zero by the same rational interval calculation. Reducing polynomials modulo p verifies both binding equations B0.48 and the full two-variable identity B0.49. The latter identity, with its strictly positive coefficient minorants, proves the bound on the entire parameter rectangle; a floating-point grid would not suffice. In particular, $36 - 174\delta + 347\delta^2 > 0$ follows by completing the square, since $36 - 174^2/(4 \cdot 347) > 0$. The nonnegative terms in B0.49 prove $F_{\kappa_0}^{(4)} \leq b_0$, with equality only at its binding point.

T1.3 (uniform perturbation). On $0 \leq p \leq 1/2$, $\kappa \geq 7/10$, the counting coefficient of §B0.5 has derivative

$$\partial_\kappa C_\kappa(p) = \frac{6p(1-p)^2}{[3\kappa(3-p) - 2p]^2} \leq \frac{8/9}{[(15/2)(7/10) - 1]^2} < 1/18.$$

For $V = 5/6 - \delta \geq 0$, $\partial_C R = \delta V^2/[2(V + \delta C)^2] \leq 5/12$. The exact intervals give $h_0 < 4/5$ and $0 < h_0 - 4\ell_0 < 13/100$. Differentiating B0.47 now gives

$$\partial_\kappa F^{(4)} \leq \frac{3}{4} \frac{4}{5} \frac{5}{12} \frac{1}{18} + \frac{5}{12} \frac{13/100}{12(7/10)^2} < 1/10.$$

As $\kappa - \kappa_0 = 2(\beta_* - b_0)$,

$$F_\kappa^{(4)} - \beta_* \leq -\frac{4}{5}(\beta_* - b_0).$$

The other dyad endpoint, floor bin and small rows have respectively margins exceeding .06, .01, .05, all checked with rational intervals. The same exact interval identities check the supply and

geometry inequalities. Thus the strict losses can be chosen uniformly before a target character is fixed.

T1.4 (minimal generalized damping interface). For $\kappa \geq 7/10$, $t, z, \delta, q \geq 0$, $q \leq \delta/2$, let $z_\kappa(t) = \max(0, z - t/(3\kappa))$. Since $0 \leq z - z_\kappa(t) \leq t/(3\kappa)$,

$$-\delta t/4 - q z_\kappa(t)/2 \leq -q z/2 - \delta t(1/4 - 1/(12\kappa)) \leq -q z/2.$$

This is the variable- κ form of the weaker damping used by the weighted-numerator Lean route. B0.5 proves its own stronger $\delta/2$ -weighted coefficient statement. The physical coefficient dictionary and combined-conductor mass bound are proved in `Cycle25.Numerator.Imported.NumeratorLocalCoefficients`, `Cycle25.Numerator.Imported.NumeratorPhysicalPacketMass`, `Cycle25.Numerator.ReflectionMass`, and `Cycle25.Numerator.ActualBin`; these supply the scalar-to-source transfer.

T1.5 (rational target). The root interval implies

$$b_0 < \theta = 683505193/781250000, \quad 0 < \theta - b_0 < 4 \cdot 10^{-14}.$$

Exact cross-multiplication proves

$$\theta < 10499/12000 < 874957019420098946128604623/10^{27} < 7/8.$$

The second intermediate value is the pinned public catalogue threshold. No finite decimal approximation to b_0 is treated as its definition. Proving T0 implies the rational target by transitivity of the strict inequalities.

Analytic argument

B0 gives the compensated-probe argument. Appendices B1–B4 supply finite arithmetic, completed reflection, moments and local coefficient estimates. The classical theta, sieve and functional-equation inputs are identified in the references and in the named source declarations below.

B0. The compensated-probe proof

Theorem B0 (Hecke boundary). Let

$$c_0 = \frac{1}{12} + \frac{\ell_0}{4} = 0.1251133529600359\dots, \quad b_0 = 1 - c_0, \quad 927\ell_0^4 - 3135\ell_0^3 + 2433\ell_0^2 + 275\ell_0 - 100 = 0, \quad (\text{B0.1})$$

where $0.16712007850680 < \ell_0 < 0.16712007850682$. Every primitive finite-order Hecke L -function over $\mathbb{K} = \mathbb{Q}(\omega)$ is zero-free in $\Re s > b_0$, with its principal pole allowed. In particular, $\zeta(s) \neq 0$ for $\Re s > b_0$, with its pole at one excluded.

We prove this theorem using the dependency order in T1 and the weaker initialization in T0. Reflection bounds a compensated probe; Poisson summation exposes its reciprocal Hecke principal term. Moments and the exact optimization prevent the remaining rows from cancelling that term. Appendices B1–B4 supply the finite arithmetic, reflection, moments and local factors. Notation in these appendices is local: in particular, M_u denotes the sextic probe rather than the cubic polynomial in the main text.

B0.1. Notation and the four estimates used in the proof

Let $\mathcal{O} = \mathbf{Z}[\omega]$, $\omega = e^{2\pi i/3}$, $\lambda = 1 + 2\omega$, $q_a = Na$, $\alpha(a) = a/|a|$, and

$$e(z) = \exp(2\pi i \operatorname{Tr}_{\mathbb{K}/\mathbf{Q}}(z/\lambda)), \quad d\mu(z) = 2 \, dx \, dy / \sqrt{3}.$$

Fourier transforms use $e(-zy)$. Ideal sums use the unique primary generator congruent to one modulo $3\mathcal{O}$; element rows retain six units. The fixed excluded set S contains primes above six, fixed conductors and a sufficiently large fixed small-prime cutoff. Columns avoid S . A star denotes squarefree; (6) denotes row valuations $0, \dots, 5$.

The sextic symbol χ_n is multiplicative in its denominator. Every character power, including zero and six, vanishes at nonunits. Put $\psi_u(n) = \chi_n(u)$, and, for squarefree n ,

$$\gamma_j(n) = q_n^{-1/2} \sum_{d \bmod n} \chi_n(d)^j e(d/n), \quad a_\nu(n) = \overline{\alpha(n)} \gamma_2(n) \nu(n).$$

Appendix B1 proves the finite identities

$$\begin{aligned} \chi_b(a) &= R(a, b) \chi_a(b), \quad R(a, b) = R(b, a) \in \{\pm 1\}, \quad (a, b) = 1, \\ \gamma_2(n)^3 &= \mu(n) \alpha(n), \quad \gamma_1(n) \gamma_2(n) = \mu(n) \alpha(n) G(n), \\ \mu(n) \gamma_{-1}(n) &= \chi_n(-1) G(n)^{-1} \overline{\alpha(n)} \gamma_2(n), \\ a_\nu(ab) &= a_\nu(a) a_\nu(b) \chi_b(a)^4 \quad ((a, b) = 1). \end{aligned} \tag{B0.2}$$

For odd $a + b\omega$, one convention is $\Gamma_q(a + b\omega) = (1 + i^{-b} + i^a + i^{b-a})/2$, $G(c) = \overline{\chi_c(4)} \Gamma_q(c)$, and $R(v, w) = \Gamma_q(vw)/(\Gamma_q(v)\Gamma_q(w))$. Here γ_{-1} means the inverse-character Gauss sum. Fixed finite phases need not have order dividing six.

For $\varphi_u = \nu \psi_u^{\pm 1}$, define

$$M_u(D; W) = D^{-1/2} \sum_n \mu(n) \varphi_u(n) W(q_n/D), \quad S_u(N; W) = N^{-1/2} \sum_n \varphi_u(n) W(q_n/N). \tag{B0.3}$$

Profiles are common smooth functions on fixed annuli; bounded real norm powers and independent height twists have finite seminorm and polynomial height costs. Moving arithmetic coefficients are never absorbed into these costs.

For disjoint row-independent prime supports $\mathcal{P}_i$, with $q_p \asymp U^{z_i}$, set

$$Q_{u,i} = U^{-z_i/2} \sum_{p \in \mathcal{P}_i} a_i(p) \varphi_u(p) W_i(q_p/U^{z_i}), \quad Q_{u,I} = \prod_{i \in I} Q_{u,i}, \quad z_I = \sum_{i \in I} z_i.$$

The slot count is fixed before $U \rightarrow \infty$. The following estimates are proved in Appendix B3. Here β_* is the family supremum in T0.

- **UM**: $\sum_{0 < q_u \leq H} |M_u(D)|^2 \ll H(HD)^\epsilon$ for $H \geq D^{1+\vartheta}$, every fixed $\vartheta, \epsilon > 0$, and fixed polynomial scale ranges. All rows, including sixth powers, occur.
- **MI**: for arbitrary bounded row-independent a_i ,

$$r + 2z_I \leq 1 - c_1, \quad 2r + 8z_I \leq 3 - c_2 \implies \sum_{0 < q_u \ll U} |M_u(U^r) Q_{u,I}|^2 \ll U^{1+\epsilon}, \tag{B0.4}$$

for fixed positive margins c_1, c_2 , either common orientation and every subcollection. There is no mesh condition.

- **MP**: the prime sums run over the genuine canonical good-prime pools of a fixed ray modulus, with one common smooth annular profile W . Each slot coefficient is an individual fixed ray character; its bounded real norm power and height may vary by slot. On the original natural eligible rows,

$$2m + 6\kappa z_I \leq 1 \implies \sum_{u \text{ eligible}} |S_u(U^m)|^4 |Q_{u,I}|^2 \ll U^{1+\epsilon}. \tag{B0.5}$$

Assume $7/10 \leq \kappa \leq 3/4$, $\beta_* \geq 51/100$, and $2\beta_* - 1 \leq \kappa$. Eligibility, the row zeros and the common puncture are exactly those described in B3.4. The mesh depends only on the bounded width, puncture and length ranges and the requested loss; it is chosen before the arithmetic family, eventual slot count, profiles and target height. Constants and finite seminorm and height orders may depend on the subsequently fixed slot system and common profile. With no slots, **MP0** holds for all bounded m , excluding only the principal inducing character, without a zero-free hypothesis.

The contradiction uses $\kappa = 2\beta_* - 1$, which satisfies these hypotheses by T0.

The completed theta sum is

$$\mathcal{C}_V(N; m, \nu) = \sum_{c \text{ sf}, n} \frac{\overline{\alpha(cn^3)} \gamma_2(c) \nu(cn^3) \chi_{cn^3}(m)}{\sqrt{q_c} q_n} V(q_c q_n^3 / N). \quad (\text{B0.6})$$

The indices c, n may share primes. A mark means divisibility of the **entire** index cn^3 . The unmarked energy is $(HN)^\epsilon (H + H^2/N)$.

RE is the marked reflection estimate proved in Appendix B2. We use its consequence (B0.26) below. A mark is imposed on the whole completed index; its prime support may meet the row. The proof keeps every cusp source, the fixed ray phases and the natural character zeros.

B0.2. Finite reconstruction and common zero witnesses

The original zero extensions give

$$\varphi_{ua^6}(n) = \varphi_u(n) \mathbf{1}_{(n,a)=1}.$$

Separating the primes of a squarefree column which divide a gives the exact finite identity

$$M_u(D; W) = \sum_{d|\text{rad}(a)} \mu(d) \varphi_u(d) q_d^{-1/2} M_{ua^6}(D/q_d; W). \quad (\text{B0.7})$$

There is no requirement $(u, a) = 1$. The character zeros make (B0.7) valid also at shared primes. The coefficient mass is at most $d(a) \ll q_a^\epsilon$, where $d(a)$ counts the ideal divisors of a .

Differentiating in log scale replaces W by $-W/2 - yW'$. Sobolev on unit log intervals, followed by UM for these common profiles, therefore gives

$$\sum_{0 < q_v \ll H} \sup_{0 < E \leq D} |M_v(E; W)|^2 \ll H(HD)^\epsilon, \quad H \geq D^{1+\vartheta}. \quad (\text{B0.8})$$

The range of subunit scales contains only boundedly many columns and is empty below a fixed positive scale. The $O(\log(2 + D))$ intervals cost only a small power. This argument never assigns a moving arithmetic coefficient to a common-profile moment.

For sixth-power-free $q_u \asymp U$, choose

$$H = \max(2U, D^{1+\vartheta}), \quad P = (H/U)^{1/6}.$$

There are $\gg_S P$ primary ideals a , prime to S , with $q_a \leq P$. The map $(u, a) \mapsto ua^6$ is injective: valuations modulo six recover u , the quotients recover a , and its primary generator fixes the unit of u . Averaging (B0.7), summing the rows, and applying (B0.8) gives

$$\sum_{q_u \asymp U}^{(6)} |M_u(D; W)|^2 \ll U^{1/6} H^{5/6} (UD)^\epsilon. \quad (\text{B0.9})$$

For bounded r and $D = U^r$, choose ϑ sufficiently small in terms of the requested final loss. Thus the exponent in the sixth-power-free inverse moment is $e(r) = \max\{1, (1 + 5r)/6\}$. Sobolev in the real power and height variables gives the same bound for separate rowwise parameters in a fixed real range and a height interval of length T , at a fixed polynomial cost $(1 + T)^A$. Only the smooth tests vary in that argument.

Buffered zero bins

Fix a finite conjugation-stable group Θ of characters supported on S , containing the numerator, denominator, and fixed phase presentations needed for the chosen target. For a sixth-power-free row having a prime factor outside S , define

$$\mathcal{X}(u) = \{\nu\psi_u^{\pm 1} : \nu \in \Theta\}.$$

Every member is nonprincipal: a good prime of valuation $1, \dots, 5$ in u has nontrivial tame inertia. Its primitive conductor norm is $O_S(q_u)$.

Let $a_* = 51/100$, take a small buffer $e > 0$, and put $I = \lceil 1/e \rceil + 2$. For a height T , let $B_j(u)$ be the maximum of a_* and the real parts of zeros of members of $\mathcal{X}(u)$ at height at most $3^j T$. Some $i < I$ satisfies $B_{i+1} - B_i \leq e$. Round B_i down to the e -grid, obtaining a , so that

$$a \leq B_i < a + e, \quad B_{i+1} < a + 2e. \quad (\text{B0.13})$$

There are only finitely many labels, independently of U . A nonfloor bin supplies an actual zero $\rho = \sigma + i\gamma$, with $a \leq \sigma < a + e$, $|\gamma| \leq 3^i T$. Put $\delta = 2a - 1$. The floor bin is counted trivially.

Here is the analytic estimate used in these bins. The functional equation and Euler convergence give a polynomial conductor-height bound for $L_{\mathbb{K}}$ on a fixed strip. In a zero-free disk about $2 + it$, continue its Euler logarithm. Borel–Carathéodory bounds the logarithm by $O_e(\log(Q(3 + |t|)^2))$ on a slightly smaller disk; it is bounded on the concentric radius $49/100$ by Euler convergence. Three circles gives a bound of the form $O_e((\log(Q(3 + |t|)^2))^{\theta_e})$, with $\theta_e < 1$, on a disk smaller by another fixed buffer. Exponentiating either sign yields

$$|L_{\mathbb{K}}(s, \varphi)| + |L_{\mathbb{K}}(s, \varphi)^{-1}| \ll_{e, \varepsilon} \{2Q(3 + |t|)^2\}^\varepsilon \quad (\text{B0.14})$$

on the retained zero-free region. Cauchy’s formula gives the corresponding bounds for finitely many derivatives of $\log L_{\mathbb{K}}$. A principal pole is handled by applying the logarithm argument to $(s - 1)\zeta_{\mathbb{K}}(s)/(s + 1)$.

Deleting primes of radical norm $O_S(U)$ costs U^ε on a fixed positive half-plane. On $\Re s \geq -v$, the direct deleted product costs $U^{v+\varepsilon}$. These follow primewise from $\log |1 - z|^{\pm 1} = O(|z|)$, treating a fixed finite set separately.

Mellin inversion, (B0.14), and the primitive functional equation give, for the required bounded real powers and retained heights,

$$|M_u(U^r)|^2 \ll U^{\delta r + \varepsilon_0}, \quad |S_u(U^m)|^2 \ll U^{\delta \min(m, 1-m) + \varepsilon_0}. \quad (\text{B0.15})$$

For the inverse polynomial, move the retained Mellin segment only to $\Re s = a + 6e$. For the plain polynomial, use either that line or $1 - a - 6e$; reflection supplies $U^{a-1/2+O(e)}$ on the second line. This gives exactly the two exponents in (B0.15). The original character zeros are restored as deleted Euler factors, not removed from the polynomials.

For a centrally normalized smooth prime polynomial of length P , the same logarithm estimate and the Dirichlet series of $-L'/L$ give

$$\left| P^{-1/2} \sum_{\mathfrak{p}} \overline{\psi_u(\mathfrak{p})} w(q_{\mathfrak{p}}/P) (q_{\mathfrak{p}}/P)^{it} \right| \ll P^{\delta/2+O(e)+\varepsilon} U^\varepsilon (1 + T)^A. \quad (\text{B0.16})$$

Insert the factor $1/\log q_{\mathfrak{p}}$ into the smooth test after using $-L'/L$. Prime powers contribute $O(P^{1/2+\varepsilon})$ before central normalization and are harmless. Primes in the fixed exclusion are removed; the row’s nonunit zeros remain. Formula (B0.16) needs only the numerator presentation and its conjugate, both included in $\mathcal{X}(u)$.

For precision about heights, first fix the real losses and smooth orders. On a retained segment allocate a fixed fraction of T to each of the finitely many independent height coordinates. Outside it, leave reciprocal contours on an absolute or globally zero-free line. The discarded tails are $O(U^B T^{-N})$, with B fixed before N . Taking $U^{\tau_-} \leq T \leq U^{\tau_+}$, with fixed sufficiently small positive exponents, makes the retained bounds and discarded tails compatible. The final choices in §B0.7 impose these requirements simultaneously.

A common pair of saturated witnesses

Lemma 1W (shared zero witnesses). Choose $t_0 \in [1, 3/2]$ before choosing witnesses. Each row in a nonfloor bin has one presentation, one real power $\sigma \in [a, a + e)$, one common height, and dyadic exponents r, m such that

$$r \leq t_0 + o(1), \quad r + m \geq t_0 - o(1), \quad 0 \leq m \leq \frac{1}{2} + O(\varepsilon_0), \quad (\text{B0.17})$$

$$|M_u(U^r)|^2 \gg U^{\delta r - O(\varepsilon_0)}, \quad |S_u(U^m)|^2 \gg U^{\delta m - O(\varepsilon_0)}. \quad (\text{B0.18})$$

Once the presentation and dyads are fixed, the base profiles are common to the rows, with uniform required seminorms.

Proof. Choose a smooth cutoff $V_{\leq}$, equal to one on $[0, 1]$ and zero on $[2, \infty)$, and set

$$C_{\varphi}(s) = \sum_l \mu(l) \varphi(l) V_{\leq}(q_l / U^{t_0}) q_l^{-s}.$$

At the buffered zero ρ , consider

$$\frac{1}{2\pi i} \int_{(2)} U^{40z} \Gamma(z) L_{\mathbb{K}}^S(\rho + z, \varphi) C_{\varphi}(\rho + z) dz. \quad (\text{B0.19})$$

Move to $\Re z = -1/4$. The residue at zero vanishes. A weak polynomial Hecke bound, ideal counting, and gamma decay make the remaining integral $o(1)$; the factor U^{-10} dominates these fixed powers when the analysis height exponent is sufficiently small.

On the initial line, expand the two Dirichlet series. The unit coefficient is $1 + o(1)$, and every coefficient at $1 < q_{lk} \leq U^{t_0}$ is zero by $\mu * 1 = \mathbf{1}_1$. Removing the unit therefore leaves a sum of magnitude $1 + o(1)$, supported smoothly at product norm at least a fixed multiple of U^{t_0} . A smooth terminal cutoff at U^{41} costs $o(1)$.

Split the two factors into dyads and separate their product weight by Fourier inversion in log norm. Its common Fourier measure has bounded mass and arbitrary polynomial decay. Some pair and frequency satisfies

$$|M_u(U^r) S_u(U^m)|^2 \gg U^{\delta(r+m) - O(\varepsilon_0)}. \quad (\text{B0.20})$$

The two factors have the same character, real power, and height. Comparing (B0.20) with (B0.15) forces $m \leq 1/2 + O(\varepsilon_0)$, since $\delta \geq 1/50$. Dividing by each of the two pointwise bounds proves (B0.18). The inverse profile is a fixed dyadic profile times $V_{\leq}(U^r y / U^{t_0})$; derivatives of the latter are supported on a fixed annulus. The $O((\log U)^2)$ dyadic choices and the discarded height tails fit the previously prescribed losses. $\square$

This proof permits t_0 to depend on the fixed bin and on a subsequently fixed amplitude class, provided it is fixed before that class's witnesses are chosen. It does not select a different truncation separately for each row inside a moment.

B0.3. The physical probe, signed Gram bound and compensator

Fix primitive η , let b_* be the product of fixed excluded prime generators, and choose a primitive residue character $\xi \bmod b_*$, locally quadratic at odd primes and cubic at norm four. Then ξ^6 is principal on units. Put $\tau = q_{b_*}^{-1/2} g_\xi(b_*, 1)$, $\Xi(A) = \xi(A) \chi_A(b_*)$, and $V(r) = e^{-(\log r)^2/4}/(2\sqrt{\pi})$, so $\widehat{V}(t) = e^{t^2}$. For nonnegative nonzero smooth annular W_0, W_1 , define

$$\begin{aligned} T_{m,a,\eta}(t) &= \sum_{c \text{ sf}, n} \gamma_2(c) \overline{\alpha(A)G(A)} \eta(A) \Xi(A)^{-1} R(A, a) \chi_A(m) q_c^{-1/2-t} q_n^{-1-3t}, \quad A = cn^3, \\ I_\eta(X, Y, Z) &= \frac{1}{Y} \sum_a \frac{W_1(q_a/Y) \chi_a(b_*)}{\tau \xi(a) \sqrt{q_{b_*} q_a X}} \\ &\times \sum_m \xi(m) q_a^{-1/2} g_{\chi_a}(a, -m) W_0\left(\frac{q_m}{q_{b_*} q_a X}\right) \\ &\times \frac{1}{2\pi i} \int_{(4)} Z^t e^{t^2} T_{m,a,\eta}(t) dt. \end{aligned} \quad (\text{B0.21})$$

Expanding fixed ray phases and separating the radial weight gives finitely many expressions $Q^{-1/2} \int \widehat{W}_0(iv) \sum_m \Omega(q_m/Q) \xi(m) (q_m/Q)^{-iv} A_{m,\sigma,v}(Y) B_{m,\sigma}(Z) dv / (2\pi)$, where $Q = q_{b_*} XY$, B is (B0.6), and

$$\begin{aligned} A_{m,\sigma,v} &= Y^{-3/2} \sum_{a \in \sigma} \frac{\chi_a(b_*)}{\tau \xi(a)} W_1(q_a/Y) (q_a/Y)^{-1+iv} \\ &\times g_{\chi_a}(a, -m). \end{aligned}$$

The $Y^{-3/2}$ normalization is essential.

For $P_a = Y^2/Q \geq 1$, the following signed Gram estimate uses the displayed coefficients of A :

$$\sum_{q_m \ll Q} |A_{m,\sigma,v}|^2 \ll (1 + |v|)^J \frac{Q}{Y} \left(1 + P_a^{1/6} + P_a^2/Y\right) Z^\epsilon. \quad (\text{B0.22})$$

Proof. Apply Poisson summation to the physical row after expanding the square. The complete finite transform of the two unnormalized Gauss sums is $q_u q_v F(u, v; j)$, where F is (B1.14). This cancels the reciprocal modulus $Q/(q_u q_v)$ from row Poisson and leaves the exact prefactor Q/Y^3 . The zero frequency is $\mathbf{1}_{u=v} \phi(u)$, by (B1.16), and contributes Q/Y .

For nonzero frequency write $u = Cn_1, v = Cn_2, j = Ck$. Equation (B1.17) expresses the correlation through a bounded local coefficient modulo $\text{rad } C$. Extend that coefficient by the local table in Appendix B1 and insert the full projector (B1.18) before separating the columns. On its d -term, the natural zeros force $(d, Ck) = 1$. After fixed ray splitting, the remaining joint coefficient is periodic in the two columns modulo an ideal of norm $\ll q_C q_k$. Its complete mean is zero unless every good valuation of k outside C is divisible by six: otherwise one complete nonprincipal character sum vanishes.

Dyadically let $q_C \asymp C_0$, $q_d \asymp D_0$, $q_C q_k \asymp R P_a$, and $N = Y/(C_0 D_0)$. For a mean-zero coefficient, Poisson in the two complex column variables has no zero mode. Repeated integration by parts in its common smooth profile gives, for any fixed $A, B > 0$,

$$R^{-B} C_0 N^2 \min\{1, (R P_a/N)^A\}. \quad (\text{B0.23})$$

The factor C_0 is the local bound $|L_C| \leq q_C$; a nonzero Fourier mode has size at least the reciprocal modulus. There are $O(D_0 R P_a)$ choices of C, d, k on these dyads, with divisor factors absorbed by Z^ϵ . The sum reduces to

$$\sum_{i,j \geq 0} 2^{-i-j} \min\{1, (2^{i+j}/\Lambda)^A\} \ll_A \frac{1 + \log(2 + \Lambda)}{\Lambda}, \quad \Lambda = Y/(R P_a), \quad A > 1. \quad (\text{B0.24})$$

Indeed grouping by $i + j = n$ leaves $(n + 1)2^{-n}$; splitting at $2^n = \Lambda$ proves the bound, with the same upper bound when $\Lambda < 1$. After the prefactor Q/Y^3 , this gives $(Q/Y)P_a^2/Y$. Taking B large sums the Fourier shells.

In the exceptional case, write $k = er^6$, with e supported on C and the fixed primes and having sixth-power-free valuations. There are $\ll Z^\epsilon(RP_a/C_0)^{1/6}$ possible roots. The remaining dyadic weights are $C_0^{-1/6+\epsilon}D_0^{-1+\epsilon}$; their sums give $(Q/Y)P_a^{1/6}$. This proves (B0.22), with the height factor coming from finitely many profile derivatives. The exceptional term is retained, and the proof does not give the same bound for arbitrary column coefficients. $\square$

Take disjoint prime slots of lengths $P_i = Z^{\ell_i}$, $\ell = \sum \ell_i$, with nonnegative profiles w_i . If $I_{\eta;D}$ inserts $D \mid cn^3$, the exact finite compensator is

$$I_\eta^{\text{comp}}(Z) = \sum_{(p_i)} \prod_i w_i(q_{p_i}/P_i) \sum_J (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \quad (\text{B0.25})$$

All good primes are allowed; $\bar{\eta}$, rather than η , is necessary.

The reflected-energy corollary in Appendix B2 says that, for completed length Z^N , row range Z^M , and product-form prime marks of total length z ,

$$M + 2z \leq N \implies \sum_{q_m \ll Z^M} |\mathcal{T}(m)|^2 \ll Z^{M+(N+2z-2M)_+/4+\epsilon}. \quad (\text{B0.26})$$

The proof uses the full reflected-block exponent, so it also covers collisions between a row and a mark.

Consequently, for $X = Z^x, Y = Z^y, M = x + y$,

$$0 < \ell < x \leq y, \quad M + \ell \leq 1, \quad y - 2x + \ell < 0 \implies |I_\eta^{\text{comp}}| \ll Z^{x/2+(y-x)/12+(1+3\ell-2M)_+/8+\epsilon}. \quad (\text{B0.27})$$

A rescaled subset of total length d has $M' = M - 2d, z' = \ell - d, N' = 1 + \ell - d$. Combining (B0.22)–(B0.26) with its outside coefficient and tuple count adds $-d + (1 + 3\ell - 2M + d)_+/8$, decreasing in d . Its maximum is at zero. Requiring every rescaled energy separately to have exponent M' would lose this saving.

B0.4. Exact Euler replacement and principal signal

Write $\mathcal{M}(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr$ for the radial Fourier Mellin transform. Poisson in the physical row and $s = t + 1 - z$ give

$$\begin{aligned} I_\eta &= \frac{1}{(2\pi i)^3} \int_{(2)} \int_{(3)} \int_{(3)} \\ &\times X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} \mathcal{M}(z) \widehat{W}_1(w) \\ &\times \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \frac{\zeta_{\mathbb{K}}^S(6z) L_{\mathbb{K}}^S(w, \psi_u)}{L_{\mathbb{K}}^S(s, \eta \overline{\psi_u})} H_{\eta,u} dz dw ds. \end{aligned} \quad (\text{B0.28})$$

The fixed primitive factor excludes zero frequency. Writing a nonzero frequency as ua^6 gives $\zeta_{\mathbb{K}}^S(6z)$; no unit row is discarded.

The complete local table is as follows. For $p \notin S$, set

$$\begin{aligned} Q &= q_p, \quad j = v_p(u), \quad \rho = \chi_p(u/p^j), \quad \mathfrak{a} = \overline{\alpha(p)}^3 \eta(p)^3, \quad \mathfrak{b} = \overline{\alpha(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \quad \mathfrak{b}^3 = \mathfrak{a}^2, \\ V &= Q^{-6z}, \quad R = \mathfrak{a}^2 Q^{4-6s-6z}, \quad W^\circ = \rho Q^{-w}, \quad W = \chi_p(u) Q^{-w}, \quad D = \eta(p) \overline{\chi_p(u)} Q^{-s}. \end{aligned}$$

In particular $W = D = 0$ if $j > 0$. Then

$$P_p^* = \frac{1}{1-R} \left\{ \frac{R(1-Q^{-1}) - \eta(p)(Q-1)Q^{-s-w} V^{\mathbf{1}_{j \leq 1}}}{1-V} + J_j \right\},$$

$$P_p = (1 - V)^{-1} + \mathbf{1}_{j=0} W^\circ / (1 - W^\circ) + P_p^*, \quad H_p = P_p(1 - V)(1 - W)/(1 - D), \quad H_{\eta,u} = \prod_{p \notin S} H_p. \quad (\text{B0.29})$$

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-s} + W^\circ R$
1	$\eta(p)Q^{-s-w}$
2	$\mathfrak{a}\rho^{-3}Q^{3/2-3s}$
3	$-\eta(p)\mathfrak{b}\rho^{-2}Q^{2-3s-w} + \mathfrak{b}^2\rho^{-4}Q^{2-4s}$
4	$-\eta(p)\mathfrak{a}\rho^{-3}Q^{5/2-4s-w}$
5	$-\mathfrak{a}^2Q^{3-6s}$

The exponent $V^{1_{j \leq 1}}$ is V at $j = 0, 1$, and one otherwise. Appendix B4 derives this table by full-modulus Poisson summation and CRT. For reference, its finite local sum is

$$C_p(t, k, j') = \sum_{\substack{d \bmod p^k \\ p^{j'} \equiv p^t d \pmod{p^k}}} \chi_p(d)^k g_{\chi_p^t} \left(p^t, \frac{p^{j'} - p^t d}{p^k} \right), \quad t = e + 3l, \quad (\text{B0.30})$$

with $e = 0, 1$. The appendix evaluates every case, restores the unit phases, and sums the resulting geometric series. The local table therefore includes all shared primes and imprimitive rows.

Set $c_p = \overline{\eta(p)}Q^s$, $h_p = Q^{-w}$, and define, without division by P_p or H_p ,

$$G_p = \frac{(c_p - h_p)(1 - V)(1 - W)P_p^* - h_p(1 - VW)}{1 - D}. \quad (\text{B0.31})$$

Then $\overline{\eta(p)}Q^{s+z-1}P_p^* - Q^{z-w-1}P_p = Q^{z-1}(1 - D)G_p / ((1 - V)(1 - W))$. Thus compensation replaces $H_{\eta,u}$ in (B0.28) by

$$\mathcal{H}_{\eta,u,Z} = \sum_{(p_i)} \prod_i w_i(Q_i/P_i) Q_i^{z-1} G_{p_i} \prod_{p \notin S \cup \{p_i\}} H_p. \quad (\text{B0.32})$$

For $p \nmid u$, put $v = \chi_p(u)$, $E_p = P_p^* + D$. The phase identity $c_p D = v^{-1}$ gives

$$G_p + v^{-1}H_p = \frac{(1 - V)(1 - W)}{1 - D} \left[(v^{-1} - h_p) \left(\frac{V}{1 - V} - D \right) + (c_p + v^{-1} - h_p)E_p \right]. \quad (\text{B0.33})$$

Define $\Delta_p = G_p + \overline{\psi_u(p)}H_p$ also at row primes. Expanding $G_p = -\overline{\psi_u(p)}H_p + \Delta_p$ factors every main slot into $-\sum_p w_i(Q/P_i)Q^{z-1}\overline{\psi_u(p)}$, while error slots replace the corresponding H_p by Δ_p . This exact expansion keeps row collisions and never assumes local nonvanishing.

The local products converge normally and satisfy $H_{\eta,u} \ll q_u^\epsilon$ on

$$\Re s \geq .51, \quad \Re z \geq .34, \quad \Re w \geq -.01, \quad \Re(s + w) \geq 1 + \epsilon_1. \quad (\text{B0.34})$$

For the principal row they also converge on $\Re s \geq .87, \Re w \geq 3/4, \Re z \geq 1/8$. The entire correction has a fixed raw polynomial bound on every contour box. On $(\beta_* + e, 1/2, .34)$ and $(2, 2, v)$, respectively, it is $\ll q_u^\epsilon Z^{.34\ell}$ and $\ll q_u^\epsilon Z^{v\ell}$.

Only $u = 1$ has principal numerator. A nonunit sixth-power-free row coprime to S has nontrivial tame inertia. For a unit $u \neq 1$, choose a primary $n \equiv -2 - 3\omega \pmod{36\mathcal{O}}$, coprime to the remaining fixed primes by CRT. Then $q_n \equiv 7 \pmod{36}$, so $\chi_n(u) = u \neq 1$; this proves that its numerator character is also nonprincipal. At $w = 1, z = 1/6$, write $t = Q^{-1}$, $D = \eta(p)Q^{-s}$, $R = \overline{\alpha(p)}^6 \eta(p)^6 Q^{3-6s}$. Then

$$H_{\eta,p}(s) = \frac{1 - t^2}{1 - R} \left(1 + \frac{t(D - R)}{1 - D} \right). \quad (\text{B0.35})$$

Enlarging the fixed cutoff gives $|H_\eta(s) - 1| \leq 1/2$ for $\Re s \geq .87$. At this residue, $G_p/H_p = -1 + O(Q^{-.87})$. With r_{slot} fixed prime slots, define

$$c_S = \frac{(\text{Res}_{s=1} \zeta_{\mathbb{K}}^S(s))^2}{6} \widehat{W}_1(1) \mathcal{M}(1/6) > 0, \quad A(Z) = (-1)^{r_{\text{slot}}} Z^{-\ell/6} \prod_i \sum_p w_i(Q/P_i) Q^{-5/6}.$$

Writing the radial Riesz kernel as a positive integral of Gaussians and Fourier-transforming gives, with $\kappa_{\text{rad}} = 2\pi/\sqrt{3}$,

$$\mathcal{M}(\sigma) = \kappa_{\text{rad}}^{1-2\sigma} \frac{\Gamma(\sigma)}{\Gamma(1-\sigma)} \int_0^\infty W_0(r) r^{-\sigma} dr > 0, \quad 0 < \sigma < 1.$$

In particular, $\mathcal{M}(1/6) > 0$. The prime ideal theorem gives $|A(Z)| \asymp (\log Z)^{-r_{\text{slot}}}$, and $\mathcal{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A(Z) (1 + O(Z^{-\mu_P}))$ for any $0 < \mu_P < .87 \min \ell_i$. Put

$$h = 1 + \ell - x, \quad k = 5/6 - x/3 - \ell/6, \quad J_\eta = I_\eta^{\text{comp}} / (c_S A(Z)),$$

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-k} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_{\mathbb{K}}^S(s, \eta)} ds. \quad (\text{B0.36})$$

The main term of the double residue is f_η , with the prime-product approximation error estimated in §B0.7; the power $Z^{\ell/6}$, rather than the logarithmic normalizer, is the compensator's quantitative gain.

B0.5. Amplitude-sensitive witness and numerator estimates

For the contradiction argument, suppose $\beta_* > b_0$. For a bin $q_u \asymp U = Z^d$, let $\kappa = 2\beta_* - 1 \in [7/10, 3/4]$, $L = \ell/d$. Partition physical main-slot amplitudes into finite classes with exponents $g_i \in [0, \delta/2]$; error slots and small amplitudes have $g_i = 0$. Put $q = L^{-1} \sum_i (\ell_i/d) g_i = \delta p$, $0 \leq p \leq 1/2$. Greedy filling in decreasing g_i supplies, at capacity $z \leq L$, a genuine subcollection with

$$\prod_{i \in I} |Q_{u,i}|^2 \geq U^{2qz - \delta \max_i (\ell_i/d) - o(1)}. \quad (\text{B0.37})$$

Discarding one fractional slot accounts for the mesh loss. Adjusting its coefficient by the inverse fixed character, or conjugating the whole witness, matches the physical $\overline{\psi_u}$ orientation without introducing row-dependent coefficients.

For $r \leq 1$, use inverse capacity $(1-r)/2$; for $m \leq 1/2$, use plain capacity $(1-2m)/(6\kappa)$. MI, MP and (B0.18) give count lines

$$A_I(r) = 1 - q - (\delta - q)r, \quad A_P(m) = 1 - q/(3\kappa) - 2(\delta - q/(3\kappa))m. \quad (\text{B0.38})$$

Proof of the row count. The short-witness crossing is $r_*(t) = [2(1-p/(3\kappa))t + p/(3\kappa) - p]/[3 - p - 2p/(3\kappa)]$, with $r_* \geq 13/21$, $1/3 \leq t - r_* \leq 1/2$. Thus inverse and plain capacities are at most $4/21$ and $1/12$. If $L > 4/21$, the supply suffices. On the inverse branch, decrease capacity to $z = (1-r)/2 - \nu_0$. Then

$$1 - r - 2z = 2\nu_0, \quad 3 - 2r - 8z = 8\nu_0 + (2r - 1) \geq 8\nu_0 + 5/21. \quad (\text{B0.39})$$

Thus both MI inequalities have positive margins. Smaller capacities use the unmarked estimate with an $O(\nu_0)$ loss; decreasing the plain capacity is handled in the same way. Set

$$C_\kappa(p) = \frac{2(1-p)(1-p/(3\kappa))}{3 - p - 2p/(3\kappa)}, \quad V = 5/6 - \delta.$$

The short envelope is $1 - \delta + \delta C_\kappa(p)(3/2 - t)$; the long inverse envelope from (B0.9) is $1 - \delta + V(t - 1)$. Balancing at $t = 1 + \delta C_\kappa(p)/(2(V + \delta C_\kappa(p)))$ gives

$$R_\kappa(\delta, p) = 1 - \delta + \frac{\delta C_\kappa(p)V}{2(V + \delta C_\kappa(p))}, \quad \#\text{rows} \ll U^{R_\kappa + \epsilon} (1 + T)^A. \quad (\text{B0.40})$$

The endpoint $V = 0$ uses $t = 3/2$. Without marks, the same balance gives $R_0(\delta) = 1 - 10\delta/[3(5 - 2\delta)]$, $t = 5/(5 - 2\delta)$.

The physical numerator needs its **own** moment. For one main/error term of (B0.32), define

$$\mathcal{N}_u(Y; s, z) = \frac{1}{2\pi i} \int_{(3)} Y^{w-1} \widehat{W}_1(w) L_{\mathbb{K}}^S(w, \psi_u) \mathcal{H}_{\eta, u, Z}^{\text{term}}(s, w, z) dw.$$

At $\Re s = a + 16e$, $\Re z = z_0 = .34$, with $m = \log Y / \log U > 1/2$, set $z_A = [2 \min(m, 1) - 1]/(6\kappa) \leq L$. Then

$$\sum_{u \text{ in class}} |\mathcal{N}_u| \ll Y^{-1/2} Z^{\ell(z_0-1/2)+\ell q} U^{1/4+3R/4-qz_A/2+\epsilon} (1+T)^A. \quad (\text{B0.41})$$

The reciprocal denominator is independent of w and is estimated **after** inversion.

Proof of the numerator estimate. Appendix B4 proves the following coefficient bounds directly from the six-row table. The factors are polynomials in Q^{-w} : $\deg H_p \leq 2$, $\deg \Delta_p \leq 3$, with degrees one smaller at row primes. Their denominators are independent of w . Off the row, direct expansion of (B0.29) gives

$$|H_{p,0} - 1| + \sum_{j=1}^2 |H_{p,j}| Q^{j(a-1)} \ll Q^{-1-\epsilon_e}, \quad \sum_{j=0}^3 |\Delta_{p,j}| Q^{j(a-1)} \ll Q^{-a}. \quad (\text{B0.42})$$

At a row prime of valuation j , allocate $\min(k, j-1)$ powers of a monomial shift Q^k to $h_r \mid \Lambda_u = q_u/q_{\text{rad}(u)}$; the rest belongs to a free shift h_0 . The exact table yields

$$\sum_k |H_{p,k}| Q^{-k/2} Q^{\delta(k-j+1)+/2} \leq 1 + CQ^{-c_e}, \quad \sum_k |\Delta_{p,k}| Q^{-(k+1)/2} Q^{\delta(k-j+1)+/2} \ll 1. \quad (\text{B0.43})$$

The sharp error at $j \geq 2$ is $\Delta_{p,1} \asymp Q$: central normalization cancels it exactly, and its shift is absorbed by Λ_u . The quadratic error at $j = 2$ contributes $Q^{1-a} Q^{-3/2} Q^{\delta/2} = Q^{-1}$. No valuation is charged twice.

On the original annular support, $W_1(q_n h/Y) \neq 0$ forces $h \leq CY$. Retain this cutoff in the coefficient expansion: it gives $\log_U h_0 \leq m + O(1/\log U)$, so all later moment lengths stay in fixed bounded ranges. Inverting a monomial with shift $h = h_0 h_r$ gives $Y^{-1/2} h^{-1/2} S_{\psi_u}(Y/h)$. Primitive functional-equation reflection preserves central amplitude and gives $\varepsilon_u S_{\overline{\psi_u}}(\mathbf{c}_u h/Y; W_1^\vee)$, where $\mathbf{c}_u = 3q_{\mathfrak{f}_u}/(2\pi)^2 \asymp_S q_{\text{rad}(u)}$ and $\widehat{W}_1^\vee(v) = \widehat{W}_1(1-v)\Gamma(v)/\Gamma(1-v)$. Thus the dual scale is $\ll_S (U/Y)h_0$. Before reflection, delete the fixed Euler factors by their finite expansion; restore their natural zeros afterward. The restoring series has mass bounded by a fixed product of $(1 - Q^{-1/2})^{-1}$ and only shortens scales. The reflected profiles are common and rapidly decreasing at infinity. Moving their Mellin line just left of zero crosses only the simple $\Gamma(v)$ pole there, giving $W_1^\vee(x) = O(1)$ as $x \downarrow 0$. Hence annular decomposition has summable normalized coefficients. The varying conductor changes their scale, not the arithmetic coefficients to which MP is applied. Equations (B0.42)–(B0.43) give the rowwise coefficient mass below. Indeed, after its slot normalizer is removed, an unramified error slot has weighted mass $P^{-1/2} \sum_{q_p \asymp P} q_p^{-a} \ll P^{1/2-a} \leq 1$; ramified choices cost a divisor power. The disjoint slots prevent double counting. Hence

$$\sum_{h_0, h_r} |c_u(h_0, h_r)| h_0^{\delta/2} \ll U^\epsilon. \quad (\text{B0.44})$$

Replace remaining plain values by a common-profile scale supremum **before** applying MP. For $h_0 \asymp U^t$, the allowed capacity is $z(t) = \max\{0, \min(1, 2m-1-2t)/(6\kappa)\}$. The chosen prime product is at least $U^{qz(t)-o(1)}$ throughout its amplitude class. Hölder and MP therefore give $U^{1/4+3R/4-qz(t)/2+\epsilon}$; the fourth root accounts for the gain $qz(t)/2$. Equation (B0.44) pays $U^{-\delta t/2}$. Positive capacity uses MP and zero capacity uses MP0. Reserve small capacity decrements for the reflected-scale enlargement and the two finite meshes before taking the limit. Since $-\delta t/2 - qz(t)/2 \leq -qz_A/2$, summing these dyads proves (B0.41). Subunit reflected scales are handled by the original Schwartz profiles and assigned nonnegative length zero; capacities are capped at $1/(6\kappa)$, never computed from a negative length.

B0.6. Geometry and an exact exponent certificate

Let

$$\kappa_0 = (5 - 3\ell_0)/6, \quad \delta_0 = \frac{25 - 60\ell_0 + 27\ell_0^2}{6(5 + 4\ell_0 - 3\ell_0^2)}.$$

With $C = (6\kappa_0 - 1)/(15\kappa_0 - 2)$, $V = 5/6 - \delta_0$, put

$$R' = -1 + \frac{C(V^2 - \delta_0^2 C)}{2(V + \delta_0 C)^2}, \quad h_0 = -\frac{(1 + \ell_0)/2 + \ell_0/(6\kappa_0)}{3R'/4 - 1/(24\kappa_0)}, \quad x_0 = 1 + \ell_0 - h_0, \quad y_0 = h_0 - 2\ell_0. \quad (\text{B0.45})$$

Numerically

$$(\delta_0, h_0, x_0, y_0) = (.469344858698143, .793947597388136, .373172481118674, .459707440374516).$$

Then $M = 1 - \ell_0$, $\ell_0 < x_0 < y_0 < 2x_0$, $y_0 < 1/2 < h_0 < 2y_0$, and (B0.27)'s physical boundary is $b_0 = 11/12 - \ell_0/4$.

For general nearby geometry with $h = 1 + \ell - x$, the dyadic numerator boundary is

$$F_\kappa(d) = a + h(z_0 - 1/6) - y/2 - \ell/2 + \ell q + d(1/4 + 3R_\kappa/4 - z_0) - \frac{q}{12\kappa} \min(d, 2y - d). \quad (\text{B0.46})$$

It is convex in d ; checking endpoints suffices. At $d = h_0$,

$$F_\kappa^{(4)}(\delta, p) = \frac{1 + \ell_0}{2} + \frac{\delta}{2} + h_0 \left(\frac{3R_\kappa(\delta, p)}{4} - \frac{5}{12} \right) + \delta p \left(\ell_0 - \frac{h_0 - 4\ell_0}{12\kappa} \right). \quad (\text{B0.47})$$

The binding point is $(\delta, p, \kappa) = (\delta_0, 1/2, \kappa_0)$. The equations behind the quartic are

$$(324\kappa^2 - 3)\delta^2 + (-756\kappa^2 + 54\kappa + 5)\delta + 300\kappa^2 - 40\kappa = 0, \quad \delta = \frac{\kappa(5 - 9\ell)}{6\kappa + (6\kappa + 2)\ell}. \quad (\text{B0.48})$$

The first is equivalent to $R_\kappa(\delta, 1/2) = 5/9 + \delta/(18\kappa)$, eliminating the h coefficient; (B0.45) enforces stationarity in δ .

For an exact global certificate, suppress subscripts and let $u = 1/2 - p$. Set

$$\begin{aligned} j_{00} &= 300\kappa - 40, \quad j_{01} = 120\kappa + 80, \quad j_{10} = 24 - 216\kappa, \quad j_{11} = 144\kappa - 96, \\ B_0 &= 24\kappa - 4, \quad B_1 = 48\kappa, \quad B_2 = 16, \quad J = j_{00} + j_{01}u + \delta(j_{10} + j_{11}u + 96u^2), \\ T &= \ell - (h - 4\ell)/(12\kappa), \quad k_0 = b - (1 + \ell)/2 - h/3, \\ k_1 &= -1/2 + 3h/4 - T/2, \quad K_* = 8j_{10}k_1 + 18hB_0. \end{aligned}$$

Then $69 < J \leq 270$, and

$$8J\{b - F_\kappa^{(4)}(\delta, 1/2 - u)\} = K_*(\delta - \delta_0)^2 + uA_1(\delta) + u^2A_2(\delta) + 768Tu^3\delta^2, \quad (\text{B0.49})$$

where

$$\begin{aligned} A_1 &= 8j_{01}k_0 + [8(j_{00}T + j_{01}k_1 + j_{11}k_0) - 15hB_1]\delta + [8(j_{10}T + j_{11}k_1) + 18hB_1]\delta^2, \\ A_2 &= [8(j_{01}T + 96k_0) - 15hB_2]\delta + [8(j_{11}T + 96k_1) + 18hB_2]\delta^2. \end{aligned}$$

This follows by substituting $R = 1 - \delta + \delta(B_0 + B_1u + B_2u^2)(5 - 6\delta)/(2J)$. The zero- u coefficient has a double zero by (B0.48) and stationarity. Rational interval substitution from (B0.1) gives

$$A_1 > 36 - 174\delta + 347\delta^2 > 0, \quad A_2 > 38\delta + 257\delta^2, \quad 179 < K_* < 180, \quad 117 < 768T < 119.$$

Thus $F_{\kappa_0}^{(4)} \leq b_0$, with equality only at the binding point, without relying on a floating-point grid.

All capacities fit: $\ell_0/h_0 > 1/5 > 4/21$, and the physical numerator requires only $dz_A = \min(d, 2y_0 - d)/(6\kappa) \leq \ell_0$, since $y_0 < 4\ell_0$. Also $\partial_\kappa C \leq 1/18$, $\partial_C R \leq 5/12$, hence $\partial_\kappa F^{(4)} < 1/10$. If $\beta_* > b_0$, choose $\kappa = 2\beta_* - 1$; the MP hypothesis holds by equality and $F_\kappa^{(4)} - \beta_* \leq -4(\beta_* - b_0)/5$. The other endpoint in (B0.46) is at most $11/12 - 49h_0/150 + 11\ell_0/12 < b_0 - .06$. The floor $(a, R, q) = (.51, 1, q \leq .01)$ has endpoint at most $.51 + h_0/3 + .51\ell_0 < b_0 - .01$.

B0.7. Completing the zero-free argument

Retain the assumption $\beta_* > b_0$, and put $\mu = \beta_* - b_0 > 0$. The geometry is (B0.45), and k is defined in (B0.36). All real power margins in this section are common to the family; constants, the fixed exclusion S , and sufficiently small analysis-height exponents may depend on the fixed target η .

The principal term

(B0.27), the normalization in (B0.36), and (B0.45) give

$$J_\eta(Z) \ll Z^{b_0 - k + \varepsilon}. \quad (\text{B0.50})$$

We will prove, for some fixed $\Delta > 0$ independent of η ,

$$J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta_* - k - \Delta}. \quad (\text{B0.51})$$

Consider first $u = 1$ in the compensated version of (B0.28). Move to $(\Re s, \Re w, \Re z) = (\beta_* + e, 1 + e, 1/6 + e)$. The global line for s is zero-free by definition and (B0.14). Move w to $3/4$, crossing only the simple numerator pole at one. In its residue, move z to $1/8$, crossing only the simple pole at $1/6$. The local product and replacement are holomorphic on the full principal box following (B0.34). The Mellin transform $\mathcal{M}$ has no pole in these positive strips.

Equations (B0.35)–(B0.36) identify the double residue, including both residue scalars. Substitute the prime-product approximation following (B0.35) on the global s -line before replacing its main integral by the line $\Re s = 2$. The latter move crosses no singularity. Its main term is exactly f_η . The two contour remainders and the error in the residue approximation are respectively

$$O_\eta \left(Z^{\beta_* - k - y/4 + O(e) + \varepsilon} + Z^{\beta_* - k - h/24 + O(e) + \varepsilon} + Z^{\beta_* - k - \mu_P + O(e) + \varepsilon} \right). \quad (\text{B0.52})$$

For the first two bounds use the quotient-free correction on the principal box, with size $O(Z^{\ell \Re z + \varepsilon})$; the remaining geometric exponents are $y(\Re w - 1)$ and $h(\Re z - 1/6)$. The third bound uses the prime-product approximation following (B0.35) only on its stated residue locus. In particular a $Q^{-7/8}$ approximation has not been used below its permissible line.

Retained nonprincipal rows

Fix $d_{\min} = 1/200$ and a sufficiently small positive extension ζ_0 . For $U = Z^d$, with $d_{\min} \leq d \leq h + \zeta_0$, partition nonprincipal rows into the buffered bins of §B0.2.

For each bin move z to $z_0 = 17/50$, keeping $s = 2, w = 3$. Next move s to $\beta_* + 20e$, still keeping $w = 3$. On retained heights move its segment to $a + 16e$. The denominator's zero-free buffer permits this last move for every presentation in the bin; outside those heights leave the reciprocal on the global line. The entire deformation lies in (B0.34). There is no nonprincipal numerator pole, and no z -pole for $\Re z \geq z_0$.

Use the exact expansion (B0.33), fix a main-prime amplitude class, and now perform the full w -inversion on its original absolute line. The reciprocal is independent of w . (B0.40) and (B0.41), and the buffered subpower reciprocal bound, give

$$\sum_{u \text{ in the class}}^{(6)} \left| \frac{q_u^{-z} \mathcal{N}_u(Y; s, z)}{L_{\mathbb{K}}^S(s, \eta \psi_u)} \right| \ll (1 + T)^A Y^{-1/2} Z^{\ell(z_0 - 1/2) + \ell q + \varepsilon_2} U^{1/4 + 3R_\kappa/4 - z_0 - qz_A(d)/2 + \varepsilon_2}, \quad (\text{B0.53})$$

where

$$z_A(d) = \frac{2 \min(y/d, 1) - 1}{6\kappa}, \quad dz_A(d) = \frac{\min(d, 2y - d)}{6\kappa}. \quad (\text{B0.54})$$

For sufficiently small ζ_0 , this capacity is positive and available. The independent heights of the physical prime sums and the detector witnesses are covered by the moment seminorm bounds; no common physical/witness height is asserted.

Multiplying (B0.53) by the remaining factors in (B0.28), and subtracting $\beta_* - k$, gives exactly

$$a - \beta_* + h(z_0 - 1/6) - y/2 - \ell/2 + \ell q + d(1/4 + 3R_\kappa/4 - z_0) - \frac{q}{12\kappa} \min(d, 2y - d). \quad (\text{B0.55})$$

At $d = h$ this is $F_\kappa^{(4)}(\delta, p) - \beta_*$, with $F^{(4)}$ as in (B0.47). Its convexity in d , the endpoint bounds in §B0.6, and the floor estimate therefore supply a fixed negative margin on the whole retained interval. The extension above h costs at most a fixed multiple of ζ_0 . Dyadic sums, amplitude classes, and finite profile-height costs fit inside a reserved fraction of this margin.

All witness capacities are available because $\ell/(h + \zeta_0) > 4/21$. The new numerator capacity is available because $\min(d, 2y - d) \leq y < 4\ell \leq 6\kappa\ell$. If $\ell_{\max}$ is the largest base- Z slot length, its largest relative length on the retained range is at most $\ell_{\max}/d_{\min}$. Finitely many sufficiently short equal slots, on disjoint norm subintervals, therefore meet both the MP mesh and the greedy rounding condition uniformly, including the smallest dyads.

The remaining rows and heights

The remaining units and nonunit rows $q_u \leq Z^{d_{\min}}$ do not need a detector. Use the global lines

$$(\Re s, \Re w, \Re z) = (\beta_* + e, 1/2, z_0).$$

For their nonprincipal numerators the ordinary functional equation and convexity give the more than sufficient bound

$$L_{\mathbb{K}}^S(1/2 + it, \psi_u) \ll U^{3/5+\varepsilon}(1 + |t|)^A.$$

Together with the first of the raw bounds following (B0.34), ideal counting, and $q_u^{-z_0}$, this makes their exponent relative to $\beta_* - k$ at most

$$h(z_0 - 1/6) - y/2 + \frac{63}{50}d_{\min} + O(e) + \varepsilon. \quad (\text{B0.56})$$

For (B0.45) the expression before losses is less than $-1/20$, as follows from the quartic isolating interval (B0.1). The finite unit rows are included by replacing U with a fixed lower bound. No nonprincipal pole is crossed.

For $q_u > Z^{h+\zeta_0}$, take $(\Re s, \Re w, \Re z) = (2, 2, v)$, with fixed $v > 2$. The second of the raw bounds following (B0.34), the absolute L -bounds, and the row count give for a dyad

$$O\left(Z^{B_0+hv+\varepsilon}U^{1+\varepsilon-v}\right), \quad B_0 = x/2 + 1 + y.$$

The dyadic sum is

$$O\left(Z^{B_0+(h+\zeta_0)(1+\varepsilon)-\zeta_0v+\varepsilon}\right). \quad (\text{B0.57})$$

Choose v after ζ_0 , as a sufficiently large fixed number. This term then has any required fixed negative power relative to the principal scale. It does not use the central factorization or a large-row zero count.

We spell out the height control because it is needed for a common-family argument. The three independent frequency coordinates can be taken to be

$$\Im(s+z), \quad \Im z, \quad \Im w.$$

The Gaussian decays in the first, while the two Mellin transforms have arbitrary polynomial decay in the latter two on every fixed positive real strip used above. A fixed fraction of the bin's height buffer is allocated to each coordinate. The joining segments leave every reciprocal either within its retained zero-free buffer or on a global zero-free line. The unseparated correction has raw bound Z^{BU^ε} on all these fixed real boxes, by (B0.31); B is fixed before the integration-by-parts order. Off the retained region the elementary conductor-height bounds and this raw estimate thus give

$$O_N(Z^{B'}T^{-N}) \tag{B0.58}$$

with B' fixed before N . For the retained numerator estimate the w -integral was performed exactly on $\Re w = 3$; its common reflected-profile tails were handled before MP in the proof of (B0.41). The remaining deformations use only the buffered s coordinate and the external z weight. The same bound also applies to the tails in Lemma 1W, after the finite smooth orders required by the moments have been fixed. In particular rapid decay is applied before any tail-dependent separation that could alter B' .

Here is a noncircular order of choices. First fix the contradiction gap μ , a fraction of the high margin, and a small row extension ζ_0 . Fix the moment loss, capacity decrement, and amplitude width in terms of this margin. Obtain the uniform MP mesh, choose a finite slot count satisfying that mesh on $d \geq d_{\min}$, and freeze the slot lengths. This fixes a positive μ_P in the prime-product approximation following (B0.35). Now choose the real contour and detector buffers smaller than the high margin and the principal margins $y/4, h/24, \mu_P$. The moment loss was already fixed; shrinking these buffers does not demand a new mesh. Losses used just for the principal product or the physical estimate can be chosen independently of the marked-moment loss.

Next fix the target, its finite character group and cutoff, and all finite internal Sobolev and moment orders. Choose a positive, sufficiently small τ , and take $T = 3Z^\tau$. It is small enough both in terms of $d_{\min}$ and every positive ℓ_i to meet all retained-height restrictions and to absorb $(1+T)^A$ into the reserved margin. It is also bounded below by a fixed positive power of U on the retained row interval. Finally take the external tail order N sufficiently large in (B0.58). Enlarging this last order affects only the external Schwartz weights. No slot count, real margin, or raw power is selected after the asymptotic limit.

All exponent gaps before the final choice of τ are common to the family. A target-dependent τ can absorb its target-dependent polynomial height costs into the same reserved gap. Taking a fixed sufficiently small $\Delta > 0$, less than fixed fractions of $\mu, y/4, h/24, \mu_P$ and the gap in (B0.56), proves (B0.51).

Mellin continuation

Choose the ε in (B0.50) smaller than $\mu/4$, and reduce Δ , if necessary, so that $\Delta < \mu/4$. Equations (B0.50)–(B0.51) imply

$$f_\eta(Z) \ll_\eta Z^{\beta_* - k - \Delta} \quad (Z \rightarrow \infty). \tag{B0.59}$$

For $0 < Z \leq 1$, move the defining line in (B0.36) to any fixed $A > 2$. Absolute Euler convergence, the Gaussian, and (B0.35) give

$$f_\eta(Z) = O_{\eta,A}(Z^{A-k}). \tag{B0.60}$$

Consequently

$$\mathfrak{F}_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s+k} \frac{dZ}{Z} \quad (\text{B0.61})$$

is holomorphic on $\Re s > \beta_* - \Delta$. On $\Re s > 1$, Fourier inversion in $\log Z$, first on a fixed vertical line, identifies it with

$$\mathfrak{F}_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_\mathbb{K}^S(s, \eta)}. \quad (\text{B0.62})$$

The Fourier density on that line is integrable by Gaussian decay. Its inverse transform is integrable by (B0.59)–(B0.60), which justifies the identification as an ordinary Mellin integral. Identity on the full half-plane $\Re s > 1$ also follows by analytic continuation from any intervening strip of absolute integrability.

Choose Δ additionally so that $\beta_* - \Delta > b_0 > 87/100$. Formula (B0.35) and the nonvanishing Gaussian show that the right side of (B0.62) would have a pole at every zero of $L_\mathbb{K}^S(s, \eta)$ in this half-plane. Its holomorphic continuation (B0.61) excludes all such zeros. This conclusion holds for each primitive finite-order target, with the same real boundary $\beta_* - \Delta$. Constants and thresholds may depend on the target; the excluded strip does not. It contradicts the definition of β_* . Thus $\beta_* \leq b_0$.

The classical nonvanishing on $\Re s = 1$ and the Euler product on $\Re s > 1$ complete the asserted half-plane statement. Finally

$$\zeta_\mathbb{K}(s) = \zeta(s) L(s, \chi_{-3}). \quad (\text{B0.63})$$

The second factor is holomorphic in the relevant strip. A zeta zero there would be a zero of the principal Hecke function, which has just been excluded. Reflection by the zeta functional equation gives the stated symmetric location of nontrivial zeros. This proves Theorem B0.

Appendix B1. Finite arithmetic used by both expansions

This appendix fixes the phases and proves the finite identities used in the probe, reflection and moment arguments. All moduli are primary and prime to the fixed set S . Character powers are extended by zero on nonunits, including the zeroth and sixth powers.

B1.1. The fixed ray phase and sextic reciprocity

For an odd $c = a + b\omega$, put

$$\Gamma_q(c) = \frac{1 + i^{-b} + i^a + i^{b-a}}{2}, \quad G(c) = \overline{\chi_c(4)} \Gamma_q(c), \quad R(v, w) = \frac{\Gamma_q(vw)}{\Gamma_q(v) \Gamma_q(w)}. \quad (\text{B1.1})$$

These formulas apply to arbitrary odd moduli, not just squarefree ones. In fact

$$\Gamma_q(c) = q_c^{-1/2} \sum_{x \bmod c} e(x^2/c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4). \quad (\text{B1.2})$$

To prove the second equality without a squarefreeness restriction, apply lattice Poisson summation to $e(z^2/c) e^{-\pi\delta|z|^2}$. With the self-dual measure, its transform is

$$\frac{|c|}{2\sqrt{r}} \exp\left(-\frac{\pi\delta q_c |y|^2}{4r}\right) e(-cy^2/(4r)), \quad r = 1 + 3q_c\delta^2/16.$$

Divide the identity by $2/(\sqrt{3}\delta)$ and let $\delta \downarrow 0$. The original periodic mean is $q_c^{-1} \sum_{x \bmod c} e(x^2/c)$; the transformed mean is $(2|c|)^{-1} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4)$. Replacing r by one produces an $O_c(1)$

error before division: its phase error is bounded by $O_c(\delta^2 \sum_y |y|^2 e^{-c_1 \delta |y|^2})$, and the amplitude and damping errors are smaller. The four representatives $0, 1, \omega, 1 + \omega$ give (B1.1).

On $(\mathcal{O}/4\mathcal{O})^\times$ modulo squares, the representatives $1, -1, \lambda, -\lambda$ have values $1, 1, i, -i$. Thus R is a symmetric sign-valued bicharacter, explicitly

$$R((-1)^e \lambda^f, (-1)^g \lambda^h) = (-1)^{eh+fg+fh}. \quad (\text{B1.3})$$

Cubic reciprocity gives $\chi_c(4) = (c/(-2))_3$. Hence G and R depend only on the fixed ray class modulo $36\mathcal{O}$, and

$$G(vw) = G(v)G(w)R(v, w), \quad G(p^3) = \gamma_3(p), \quad R(p, p) = \gamma_3(p)^2 = \chi_p(-1). \quad (\text{B1.4})$$

Indeed p^2 is a square unit modulo four, so $\Gamma_q(p^3) = \Gamma_q(p)$, while $\chi_{p^3}(4) = 1$. At a prime, the quadratic exponential sum equals the quadratic-character Gauss sum; its square is $\chi_p(-1)$.

For distinct primary primes v, w , cubic reciprocity makes $(\chi_v(w)/\chi_w(v))^2 = 1$. The quadratic Gauss sum and CRT identify its cube with $R(v, w)$. A sign equals its cube. Multiplicativity and (B1.3) therefore give the full sextic reciprocity law

$$\chi_v(w) = R(v, w)\chi_w(v), \quad (v, w) = 1, \quad (\text{B1.5})$$

including nonsquarefree v, w .

B1.2. Prime Gauss sums and squarefree coefficients

Write $g_\chi(p, h) = \sum_{x \bmod p} \chi(x)e(hx/p)$, and $Q = q_p$. The finite Gauss–Jacobi identity, obtained by grouping the double Gauss sum according to $x + y$, is

$$g_A(p, 1)g_B(p, 1) = J(A, B)g_{AB}(p, 1), \quad J(A, B) = \sum_x A(x)B(1-x), \quad (\text{B1.6})$$

when A, B, AB are nonprincipal. Also $g_A g_{\bar{A}} = A(-1)Q$.

For $\psi = \chi_p^2$, reduction of $J(\psi, \psi)$ modulo p gives the sum of $x^{(Q-1)/3}(1-x)^{(Q-1)/3}$, a polynomial of degree less than $Q-1$, so $p \mid J$. Its absolute value is $\sqrt{Q}$. The order-three map $x \mapsto 1/(1-x)$ preserves the summand; its two fixed points have $x(1-x) = 1$, and every other orbit has size three. Therefore $J \equiv -1 \pmod{3\mathcal{O}}$. Since p is primary, the unit J/p is -1 , and $J = -p$. Applying (B1.6) gives

$$\gamma_2(p)^3 = -\alpha(p). \quad (\text{B1.7})$$

The changes of variable $x = (1+y)/2$ and $y \mapsto y/(y-1)$, with zero extensions at the omitted points, give

$$J(\chi^2, \chi^2) = \overline{\chi(4)}^2 J(\chi^3, \chi^2) = \overline{\chi(4)}^2 J(\chi, \chi^2), \quad \chi = \chi_p.$$

Since $\chi(4)^3 = 1$, (B1.6) now yields

$$\gamma_1(p)\gamma_2(p) = -\alpha(p)G(p). \quad (\text{B1.8})$$

CRT gives, for coprime squarefree a, b ,

$$\gamma_j(ab) = \gamma_j(a)\gamma_j(b)\chi_a(b)^j\chi_b(a)^j. \quad (\text{B1.9})$$

Cubing this formula, and using (B1.4)–(B1.8), proves

$$\begin{aligned} \gamma_2(c)^3 &= \mu(c)\alpha(c), & \gamma_1(c)\gamma_2(c) &= \mu(c)\alpha(c)G(c), \\ \mu(c)\gamma_{-1}(c) &= \chi_c(-1)G(c)^{-1}\overline{\alpha(c)}\gamma_2(c). \end{aligned} \quad (\text{B1.10})$$

The last formula also follows directly from $\gamma_1(c)\gamma_{-1}(c) = \chi_c(-1)$. Finally, for $a_\nu(c) = \overline{\alpha(c)}\gamma_2(c)\nu(c)$, reciprocity in (B1.9) gives

$$a_\nu(ab) = a_\nu(a)a_\nu(b)\chi_b(a)^4. \quad (\text{B1.11})$$

B1.3. Imprimitive Gauss sums and the complete correlation

The two-variable notation

$$\mathcal{G}(a, h) = q_a^{-1/2} \sum_{x \bmod a} \chi_a(x) e(hx/a), \quad \mathcal{G}(1, h) = 1, \quad (\text{B1.12})$$

is distinct from the fixed ray function $G(a)$. At a prime power p^j , with $j \geq 1$, lifting $x = x_0 + py$ first forces $p^{j-1} \mid h$. The remaining prime sum is nonprincipal if $6 \nmid j$, and is a Ramanujan sum if $6 \mid j$. Consequently

$$\begin{aligned} |\mathcal{G}(p^j, h)| &= Q^{(j-1)/2} \mathbf{1}_{v_p(h)=j-1} \quad (6 \nmid j), \\ \mathcal{G}(p^j, h) &= Q^{j/2} \mathbf{1}_{p^j \mid h} - Q^{j/2-1} \mathbf{1}_{p^{j-1} \mid h} \quad (6 \mid j). \end{aligned} \quad (\text{B1.13})$$

In particular the zeros at nonunits survive every imprimitive reduction.

For arbitrary u, v , define

$$F(u, v; j) = \sum_{\substack{x \bmod u, y \bmod v \\ vx - uy \equiv j \pmod{uv}}} \chi_u(x) \overline{\chi_v(y)}. \quad (\text{B1.14})$$

Expanding the Gauss sums and using additive orthogonality gives

$$F(u, v; j) = \frac{1}{\sqrt{q_u q_v}} \sum_{h \bmod uv} \mathcal{G}(u, h) \overline{\mathcal{G}(v, h)} e(-jh/(uv)). \quad (\text{B1.15})$$

At $j = 0$, a nonzero summand requires $u \mid v$ and $v \mid u$, since x, y must be units. Thus

$$F(u, v; 0) = \mathbf{1}_{u=v} \phi(u). \quad (\text{B1.16})$$

To expose the common support, write $u = Cn_1, v = Cn_2$, where $C = (u, v)$ and $(n_1, n_2) = 1$. The congruence forces $C \mid j$. For $j = Ck$, solve it first modulo n_1, n_2 and use (B1.5). The exact result is

$$F(Cn_1, Cn_2; Ck) = \chi_{n_1}(k) \overline{\chi_{n_2}(-k)} R(n_1, n_2) L_C(n_1, n_2; k), \quad (\text{B1.17})$$

where

$$L_C(n_1, n_2; k) = \sum_{\substack{x, y \bmod C \\ n_2 x - n_1 y \equiv k \pmod{C}}} \chi_C(x) \overline{\chi_C(y)}.$$

Here C may meet either residual column. The full local table, for $p^c \parallel C$, is

Residual columns at p	$L_{p^c}(n_1, n_2; k)$
$p \nmid n_1 n_2, p \mid k$	$Q^{c-1} \chi_p(n_1/n_2)^c (Q - 1)$
$p \nmid n_1 n_2 k, 6 \nmid c$	$-Q^{c-1} \chi_p(n_1/n_2)^c$
$p \nmid n_1 n_2 k, 6 \mid c$	$Q^{c-1} (Q - 2)$
p divides exactly one of n_1, n_2	$Q^{c-1} (Q - 1) \mathbf{1}_{6 \mid c} \mathbf{1}_{p \nmid k}$

For the first three rows, each solution modulo p has Q^{c-1} lifts. If $p \mid k$, then $x/y = n_1/n_2$. Otherwise put $y = (k/n_1)t$, $x = (k/n_2)(1 + t)$; the ratio $(1 + t)/t$ runs through $\mathbf{F}_Q^\times \setminus \{1\}$. Its character sum is -1 for a nonprincipal power and $Q - 2$ for a principal power. If exactly one residual column is divisible by p , the other variable is a fixed unit and the free unit-character sum gives the last row. This proves the table and $|L_C| \leq q_C$.

For separation, extend this table to all residual pairs by setting the local value to zero whenever $p \mid (n_1, n_2)$. This produces a periodic coefficient modulo $\text{rad } C$ in each column. **It equals the original correlation only on the coprime locus.** Accordingly insert

$$\mathbf{1}_{(n_1, n_2)=1} = \sum_{d \mid n_1, d \mid n_2} \mu(d) \quad (\text{B1.18})$$

before separating columns; no common prime can be omitted from this projector.

A second form is useful when a moment induction has already extracted the complete common prime support. If

$$u = Da, \quad v = Eb, \quad (a, b) = (ab, DE) = 1,$$

then, without any squarefreeness condition on D, E ,

$$F(Da, Eb; j) = F(D, E; j)R(a, E)R(b, D)R(a, b)\chi_a(j)\overline{\chi_b(-j)}. \quad (\text{B1.19})$$

Indeed the congruence fixes the residues modulo a, b . In the remaining congruence set $x' = bx \pmod{D}$, $y' = ay \pmod{E}$; the unit factors cancel by (B1.5), leaving precisely the displayed signs. One may extend the right side to all (a, b) with $(ab, DE) = 1$, retaining all character zeros, but must again use the full projector (B1.18). This is the common-support identity used in both moment inductions.

Appendix B2. Completed reflection and marked row energy

This appendix proves the reflected-energy estimate used in the moment argument. A whole-index prime mark changes the reflection conductor when its local frequency vanishes. The proof therefore keeps the compatible conductor branches until the row and marked-prime phases have separated; a quadratic–cubic large sieve then bounds the remaining norm.

B2.1. The completed family and fixed conventions

Write $\mathcal{O} = \mathbb{Z}[\omega]$, $\lambda = 1 + 2\omega$, $q_a = |a|^2$, and $\alpha(a) = a/|a|$. Throughout $Z \geq 2$. Ideals prime to λ are represented by their unique primary generators $a \equiv 1 \pmod{3}$. Let S be a fixed finite set containing the primes above 6; “good” means prime to S . We use the sextic characters and normalized Gauss coefficients of the arithmetic section. Every integer power of a residue character is zero on a nonunit, including exponent zero:

$$\chi_p(A)^0 = 1_{p \nmid A}. \quad (\text{B2.1})$$

In particular, exponent reduction modulo six never removes a coprimality condition.

Fix a finite family of finite-ray characters ν , with conductors supported on S , extended by zero outside the primary ideals coprime to S . A squarefree good label f and a squarefree puncture radical r_ρ may vary between invocations. Within a row norm they are fixed, and

$$\rho(A) = 1_{(A, r_\rho)=1}, \quad \Psi_k(A) = \nu(A)\rho(A)\chi_A(k)\chi_A(f)^4. \quad (\text{B2.2})$$

The puncture may depend on previously frozen outer labels; it is independent of the current row and, when f is averaged later, of that averaging variable.

For a fixed number of pairwise disjoint prime lists $\mathcal{P}_i$, with $q_p \asymp Z^{z_i}$ on the i -th list and $|a_i(p)| \leq 1$, put

$$d(A) = \sum_{(p_i) \in \prod_i \mathcal{P}_i} \prod_i a_i(p_i) 1_{\prod_i p_i | A}, \quad z = \sum_i z_i. \quad (\text{B2.3})$$

The completed sum at length X is

$$\mathcal{T}_d(X; k; V) = \sum_{\substack{n \text{ sf}, b \\ (nb, S)=1}} \frac{\overline{\alpha(n)}\gamma_2(n)\Psi_k(n)\overline{\alpha(b)}^3\Psi_k(b)^3}{\sqrt{q_n}q_b} d(nb^3)V(q_nq_b^3/X). \quad (\text{B2.4})$$

The two primal ideals may share primes. An annular smooth test is allowed, as is any test rapidly decreasing with all Euler derivatives at both endpoints. The factor $y^{1/2}$ converting an annular uncompleted sum to (B2.4) is included in V .

We also allow one common smooth profile in the completed norm and the individual marked-prime norms, supported in fixed logarithmic boxes in the latter variables: expand (B2.3) in (B2.4), then put this profile inside the tuple sum. Logarithmic Fourier inversion in those variables reduces it to (B2.4), with an absolutely integrable common family of tests V . Bounds below depend on finitely many weighted Euler seminorms of that profile. They are uniform in the prime lists, their bounded coefficients, k, f, r_ρ , and all norm labels whose logarithms to base Z lie in fixed bounded ranges. Imaginary norm powers incur only a fixed polynomial in their heights. Product form in (B2.3) is required when slots are assigned to extracted factors; after that assignment, the remaining tuple coefficient may be any bounded function independent of the row and dual indices.

The only imported analytic results are the unconditional theta expansions in Dunn–Radziwiłł [DR, §5 and Appendix A], Goldmakher–Louvel’s quadratic Hecke large sieve [GL, Definition 1 and Theorem 1.1], and Heath-Brown’s cubic large sieve [HB, Theorem 2]. All moving-conductor and marked-index reductions are proved below.

B2.2. Exact reflection, with the local zeros retained

B2.2.1. The finite theta source

Put $\check{e}(z) = \exp(2\pi i(z + \bar{z}))$ and $e(z) = \check{e}(z/\lambda)$. The theta input [DR, (5.4), (5.7), (5.9), (5.12)–(5.17), Appendix A] gives a smooth cubic theta function on $\mathbb{H}^3$, automorphic on

$$\Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma(3) \rangle, \quad \theta(gw) = \kappa(g)\theta(w), \quad \kappa \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c \\ a \end{pmatrix}_3 \quad (g \in \Gamma(3), c \neq 0). \quad (\text{B2.5})$$

The multiplier is one on $\mathrm{SL}_2(\mathbb{Z})$ and on the $c = 0$ elements of $\Gamma(3)$. Let

$$L(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \quad \Theta_j(w) = \theta(L(t_j)w), \quad (t_0, t_1, t_2) = (0, \omega^2, \omega).$$

Translations by $\mathbb{Z} + 3\mathcal{O}$ have multiplier one. Thus $L(\lambda)$ and $L(-\lambda)$ give Θ_1, Θ_2 . Also, for $u_0 = s + t\omega$, $0 \leq s, t < 3$, $\theta \begin{pmatrix} u_0 & -1 \\ 1 & 0 \end{pmatrix} w = \Theta_t(w)$.

For each of these three sources define d_H by

$$\overline{\theta(H(z, v))} = C_H(v) + \sum_{\mu \neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z). \quad (\text{B2.6})$$

Their support and coefficient bound are

$$\mu = u\lambda^h n b^3, \quad h \geq -4, \quad n, b \equiv 1 \pmod{3}, \quad n \text{ sf}, \quad |d_H(\mu)| \leq 27 \cdot 3^{h/6} |b|. \quad (\text{B2.7})$$

Here u is an Eisenstein unit. The dual n, b may share primes and may contain primes of S other than λ . This is the full source support; the primal S -mask is not imposed on it.

For clarity, the bound follows directly from the cited coefficient formulas. Their ordinary cubic Gauss sums

$$\tilde{g}_3(n) = q_n^{-1/2} \sum_{v \bmod n} \left(\frac{v}{n} \right)_3 \check{e}(v/n)$$

have modulus one for every primary squarefree n , by the finite-field Gauss identity and CRT. This includes the prime -2 of norm 4, where only the cubic symbol is used. The possible magnitudes in the three sources are $3^{h/6+8/3}|b|$, $3^{h/6+3}|b|$, and $9|b|$ at $h = -4$, all bounded by (B2.7). Valuations make the displayed n, b unique when the representation exists. At infinity the condition $\lambda^3 \mu \equiv 1 \pmod{3}$ selects exactly

$$d_I(\lambda^{-3} n b^3) = 3^{5/2} |b| \tilde{g}_3(n), \quad \tilde{g}_3(n) = \chi_n(\lambda)^{-2} \gamma_2(n) \quad ((n, S) = 1). \quad (\text{B2.8})$$

No coprimality between n and b is required.

B2.2.2. All nonzero element rows use one fixed modulus

Write a nonzero row uniquely as $k = uk_S k_{\text{good}}$, using fixed generators at S and primary generators elsewhere. The established reciprocity identity gives, on primary S -units,

$$\chi_A(k) = \chi_A(uk_S) R(A, k_{\text{good}}) \prod_{p|k_{\text{good}}} \chi_p(A)^{v_p(k)}. \quad (\text{B2.9})$$

It holds at collisions too: both sides are zero, including when $6 \mid v_p(k)$. Fix u , the S -valuations modulo six, and the class of k_{good} in the fixed reciprocity ray group. The first two factors, multiplied by ν , then belong to one fixed finite family of characters Ψ_0 supported on S . The supplementary characters $A \mapsto \chi_A(uk_S)$ have conductor supported there; only the finitely many exponents modulo six occur. The fourth-power label introduces no reciprocity sign, since $R(A, f)^4 = 1$.

Choose once a modulus E supported exactly on S , divisible by the conductors of this finite family. Choose a sufficiently divisible L , with the same prime support and $18 \mid L$, $E \mid L$, such that

$$\varphi(x) = \begin{cases} \chi_x(\lambda)^2 \Psi_0(x), & x \equiv 1 \pmod{3}, (x, E) = 1, \\ 0, & \text{otherwise} \end{cases}$$

is L -periodic. Such an L exists because the cubic supplementary law makes $(\lambda/x)_3$ periodic modulo 9, and the other characters and zero mask have fixed moduli. The zero branch is evaluated before either character. Set

$$\widehat{\varphi}(h_0) = q_L^{-1} \sum_{x \bmod L} \varphi(x) e(-h_0 x/L), \quad \mathfrak{M} = \lambda^{12} L^4. \quad (\text{B2.10})$$

Every good prime of k, f, r_p stays in a moving local set $\mathcal{P}$, with its exponents combined modulo six. No moving good prime enters L . Thus it is enough to reflect $\Psi(A) = \Psi_0(A) \prod_{p \in \mathcal{P}} \chi_p(A)^{j_p}$, $0 \leq j_p < 6$.

B2.2.3. The reflection identity

For $m \not\equiv 0 \pmod{6}$, let

$$\tau_{p,m}^{\pm} = q_p^{-1/2} \sum_{y \bmod p} \chi_p(y)^m e(\pm y/p), \quad |\tau_{p,m}^{\pm}| = 1.$$

An active branch consists of $h_0 \bmod L$ and a subset of $\mathcal{P}$ containing every prime with $j_p \neq 0$. Write $r = \prod_{p \text{ active}} p$. Reduce $\lambda^2 h_0/L = a_F/c_F$, normalizing its numerator to be primary if $\lambda \mid c_F$, and its denominator otherwise. Put $c = c_F r$. For active p define, modulo p ,

$$\sigma_p = \lambda^2 c/p, \quad \epsilon_p = -((\lambda^3 c/p) \sigma_p)^{-1} = -\lambda^{-5} (c/p)^{-2},$$

$$B_{p,j}(x) = \begin{cases} \chi_p(x)^{-j-2}, & j \neq 0, 4, \\ q_p^{-1/2} (-1 + q_p 1_{p|x}), & j = 4, \\ q_p^{-1/2} \chi_p(x)^{-2}, & j = 0, \end{cases} \quad \omega_{p,j} = \begin{cases} \tau_{p,j}^- \tau_{p,j+2}^+ \chi_p(\epsilon_p)^{-j-2}, & j \neq 0, 4, \\ \tau_{p,4}^-, & j = 4, \\ -\tau_{p,2}^+ \chi_p(\epsilon_p)^{-2}, & j = 0. \end{cases} \quad (\text{B2.11})$$

Indices on τ are reduced modulo six.

Define

$$K_* = \frac{(2\pi)^4}{27}, \quad \mathcal{R}(t) = \prod_{\pm} \frac{\Gamma(1+t \pm 1/6)}{\Gamma(1-t \pm 1/6)}, \quad V^{\sharp}(x) = \frac{1}{2\pi i} \int_{(\sigma)} \widehat{V}(-t) \mathcal{R}(t) (K_* x)^{-t} dt, \quad \sigma \geq 0. \quad (\text{B2.12})$$

The integral is independent of the nonnegative line. Write $\mathcal{T}_1(X; \Psi; V)$ for (B2.4) with no mark and with Ψ_k replaced by Ψ . For each branch, there are a source d from (B2.6), a fixed-modulus additive character ϑ on $\mathcal{O}$, and a unit scalar κ_F , all determined by h_0 and $r \bmod \mathfrak{M}^2$, such that

$$\begin{aligned} \mathcal{T}_1(X; \Psi; V) &= \sum_{\text{branches}} \zeta \sum_{0 \neq \mu \in \lambda^{-4}\mathcal{O}} \frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} \prod_{p \text{ active}} B_{p,j_p}(\lambda^4\mu) V^\sharp(q_\mu X/q_c^2), \\ \zeta &= -\frac{i}{81} \overline{\alpha(c)}^2 \widehat{\varphi}(h_0) \overline{\kappa_F} \prod_{p \text{ inactive}} (1 - q_p^{-1}) \prod_{p \text{ active}} \chi_p(\sigma_p)^{-2} \omega_{p,j_p}. \end{aligned} \quad (\text{B2.13})$$

All dual sums converge absolutely and $|\zeta| \leq 1/81$. There are $O_L(2^{|\mathcal{P}|})$ branches, or $O_L(4^{|\mathcal{P}|})$ after splitting the two terms of every $j = 4$ factor. The branch choices are simultaneous across local sets: adding an inactive $j = 0$ prime changes only the scalar, by $1 - q_p^{-1}$. It leaves the conductor, source, additive character, other local phases, and kernel unchanged.

Proof of the conductor and phase assertions. Finite Fourier inversion gives

$$C_{p,j}(h) = q_p^{-1} \sum_{x \bmod p} \chi_p(x)^j e(-hx/p) = \begin{cases} q_p^{-1/2} \tau_{p,j}^- \chi_p(h)^{-j}, & j \neq 0, h \neq 0, \\ 0, & j \neq 0, h = 0, \\ -q_p^{-1}, & j = 0, h \neq 0, \\ 1 - q_p^{-1}, & j = 0, h = 0. \end{cases} \quad (\text{B2.14})$$

Hence the masked theta function is

$$F(z, v) = \sum_{h_0, (h_p)} \widehat{\varphi}(h_0) \prod_p C_{p,j_p}(h_p) \overline{\theta(z + z_h, v)}, \quad z_h = \lambda^2 \left(h_0/L + \sum_p h_p/p \right). \quad (\text{B2.15})$$

At an infinity index $\lambda^3\mu = nb^3$, the whole-index mask first removes $(nb, E) > 1$. On the remaining domain its coefficient multiplier is $\chi_n(\lambda)^2 \Psi(n) \Psi(b)^3$, also at a shared moving prime. Equations (B2.8) therefore identify its direct series with the completed sum.

Fix h_0 and the active frequencies. Then

$$z_h = a/c, \quad c = c_F r, \quad a = a_F r + \lambda^2 c_F \sum_{p|r} (r/p) h_p. \quad (\text{B2.16})$$

At a prime of c_F , $a \equiv a_F r$ is a unit; at $p \nmid r$, $a \equiv \sigma_p h_p$ is a unit. Thus the fraction is reduced even for nonunit h_0 . Lift every h_p to be 0 mod $\mathfrak{M}^2$ while retaining its class modulo p , and fix $r \bmod \mathfrak{M}^2$. Then $a \bmod \mathfrak{M} c_F$ is fixed. Choose δ' by CRT:

$$\begin{aligned} \delta' &\equiv a^{-1} \pmod{\mathfrak{M} c_F} \text{ at primes of } c_F, \\ \delta' &\equiv 0 \pmod{\mathfrak{M}} \text{ at other primes of } \mathfrak{M}, \\ \delta' &\equiv a^{-1} \pmod{p} \text{ } (p \mid r). \end{aligned}$$

Set $b = (a\delta' - 1)/c$ and $g = \begin{pmatrix} a & b \\ c & \delta' \end{pmatrix}$. Then $b \equiv 0 \bmod \mathfrak{M}$ at primes of c_F , and $b \equiv -c^{-1} \bmod \mathfrak{M}$ elsewhere. All required fixed-prime residues of the matrix are therefore independent of the active frequencies. Periodic changes of lifts, followed by adding a multiple of the CRT modulus to δ' , avoid zero matrix entries if a symbol calculation requires this.

There are three cases. If $v_\lambda(c) \geq 2$, take $H = I$; then $g \in \Gamma(3)$ and $\kappa(g) = (c_F/a)_3 (a/r)_3$. If $v_\lambda(c) = 1$, choose $u_0 \in \{\lambda, -\lambda\}$ with $u_0 \equiv c \pmod{3}$, set $H = L(u_0)$, and write $A = a - u_0 b$. Then $G = gH^{-1} \in \Gamma(3)$, a, A are coprime primary elements, and

$$a(c - u_0 \delta') = cA - u_0, \quad bc \equiv -1 \pmod{a}.$$

Cubic reciprocity gives

$$\kappa(G) = \left(\frac{-u_0}{A} \right)_3 \left(\frac{a}{A} \right)_3^{-1} = \left(\frac{-u_0}{A} \right)_3 \left(\frac{c_F/u_0}{a} \right)_3 \left(\frac{a}{r} \right)_3.$$

If $\lambda \nmid c$, choose $u_0 = s + t\omega \equiv a \pmod{3}$, $0 \leq s, t < 3$, and $H = \begin{pmatrix} u_0 & -1 \\ 1 & 0 \end{pmatrix}$. Then

$$G = gH^{-1} = \begin{pmatrix} -b & a + bu_0 \\ -\delta' & c + \delta'u_0 \end{pmatrix} \in \Gamma(3).$$

Both $-b, c$ are primary and $-bc = 1 - a\delta' \equiv 1 \pmod{9}$. Factoring $\delta' = u\lambda^h\delta_0$, the supplementary laws and reciprocity give $(\delta'/(-bc))_3 = 1$: the unit and λ -factors use the congruence modulo 9, and after reciprocity the remaining numerator is $1 \pmod{\delta_0}$. Consequently $\kappa(G) = (a/c_F)_3(a/r)_3$. In every case,

$$\kappa(G) = \kappa_F \prod_{p|r} \chi_p(a)^2, \quad (\text{B2.17})$$

where κ_F and the choice of d_H are fixed by the displayed sector. This proves the needed conductor uniformity; it does not assume a unit-frequency support condition on $\widehat{\varphi}$.

For $x = \lambda^4\mu$ and $D_F = \lambda^3c_F$, additive CRT gives

$$\check{e}(-\delta'\mu/c) = \vartheta(x) \prod_{p|r} e(\epsilon_p h_p^{-1}x/p), \quad \vartheta(x) = e(-\delta'_F r^{-1}x/D_F). \quad (\text{B2.18})$$

The fixed congruences determine $\delta'_F \pmod{D_F}$, and the sector determines r^{-1} . Conjugating (B2.17) supplies $\overline{\kappa_F} \prod_p \chi_p(\sigma_p)^{-2} \chi_p(h_p)^{-2}$. Multiply by (B2.14) and replace h_p^{-1} by y . For $j \neq 0, 4$ the resulting sum is a Gauss sum of exponent $j + 2$; for $j = 4$ it is $\sum_{y \neq 0} e(\epsilon_p xy/p) = -1 + q_p 1_{p|x}$; an active $j = 0$ gives the last line of (B2.11). These are exactly $\omega_{p,j} B_{p,j}(x)$, including their zeros. An absent prime and an inactive zero-exponent prime have identical active radical, so their only difference is the coefficient $1 - q_p^{-1}$. This also proves simultaneous branch compatibility.

Proof of the Mellin normalization and continuation. Put

$$D(s, \Psi) = \sum_{n \text{ sf}} \overline{\alpha(n)} \gamma_2(n) \Psi(n) q_n^{-s}, \quad L_0(s, \Psi) = \sum_b \overline{\alpha(b)}^3 \Psi(b)^3 q_b^{-3s+1/2}, \quad T = L_0 D.$$

The sums are over primary S -units and converge absolutely for $\Re s > 1$. For $J(s) = \int_0^\infty \partial_{\bar{z}} F(0, v) v^{2s-1} dv$, use

$$\int_0^\infty K_{1/3}(y) y^{w-1} dy = 2^{w-2} \Gamma((w-1/3)/2) \Gamma((w+1/3)/2).$$

Since $\partial_{\bar{z}} \check{e}(\mu z) = 2\pi i \bar{\mu} \check{e}(\mu z)$ and $\overline{\alpha(\lambda^{-3})} = -i$, (B2.8) yields

$$J(s) = \frac{3^{5/2}}{4} \left(\frac{27}{(2\pi)^2} \right)^s \Gamma(s+1/3) \Gamma(s+2/3) T(s, \Psi). \quad (\text{B2.19})$$

For one translated cusp $g = GH$, its inverse action is

$$g^{-1}(z + a/c, v) = \left(-\delta'/c - \frac{\bar{z}}{c^2(v^2 + |z|^2)}, \frac{v}{q_c(v^2 + |z|^2)} \right).$$

At $z = 0$, the chain rule changes $\partial_{\bar{z}}$ into $-\kappa(G)(cv)^{-2} \partial_z$ at $(-\delta'/c, 1/(q_c v))$. The derivative removes the constant Fourier term at every cusp. Substituting $v \mapsto 1/(q_c v)$ and the Bessel integral gives, for $\Re s < 0$,

$$-\frac{i}{4} (2\pi)^{2s-2} \overline{\alpha(c)}^2 q_c^{1-2s} \overline{\kappa(G)} \prod_{\pm} \Gamma(3/2 - s \pm 1/6) \sum_{\mu \neq 0} d_H(\mu) \alpha(\mu) \check{e}(-\delta'\mu/c) q_\mu^{-(1-s)}. \quad (\text{B2.20})$$

The series converges absolutely there by (B2.7). The differentiated theta function decays exponentially at both height endpoints, using the direct expansion at infinity and this inverse-cusp formula at zero. Repeated integration by parts in $\log v$ makes J entire and rapidly decreasing on fixed vertical strips. Equation (B2.19) continues T to an entire function. The direct series bounds it on a line right of 1, and (B2.20), divided by (B2.19), gives polynomial growth left of 0. Within the

intervening strip, the preliminary bound from J is $O((1 + |s|)^C e^{\pi|\Im s|})$. Apply the maximum principle to $T(s)(B + s)^{-C'} e^{\epsilon s^2}$, choosing $-B$ outside the strip; then let $\epsilon \downarrow 0$. This gives polynomial strip growth.

Mellin inversion expresses (B2.4), with no marks, as $(2\pi i)^{-1} \int_{(a)} \widehat{V}(s - 1/2) X^{s-1/2} T(s, \Psi) ds$, $a > 1$. Shift to $\Re s < 0$, insert (B2.20)/(B2.19), and put $t = 1/2 - s$. The gamma quotient is $\mathcal{R}(t)$; the remaining factor is

$$-\frac{i}{81} \overline{\alpha(c)}^2 q_\mu^{-1/2} (K_* q_\mu X / q_c^2)^{-t}.$$

Absolute convergence and rapid Mellin decay justify both interchanges and the contour shift. Combining the finite local factors proves (B2.13).

B2.3. Common profiles and whole-series tails

For $A, B \geq 0$, with B an integer, set

$$\mathcal{M}_{A,B}(V) = \sup_{-A \leq \eta \leq 1/4} \int_{\mathbb{R}} (1 + |t|)^B |\widehat{V}(\eta + it)| dt.$$

The first negative numerator pole of $\mathcal{R}$ is $-5/6$, and on $-1/4 \leq \sigma \leq A$, $|\mathcal{R}(\sigma + it)| \ll_A (1 + |t|)^{4\sigma}$. Shifting (B2.12) to $-1/4, 0, A$ therefore proves

$$|(x\partial_x)^j V^\sharp(x)| \ll_{A,j} \min\{x^{1/4}, (1+x)^{-A}\} \mathcal{M}_{A, [4A]+j+2}(V). \quad (\text{B2.21})$$

These are finite seminorms: $B + 2$ integrations by parts bound $\mathcal{M}_{A,B}$ by

$$\ll_{A,B} \int_0^\infty (1 + x^{-A} + x^{1/4}) (|V(x)| + |(x\partial_x)^{B+2} V(x)|) \frac{dx}{x}.$$

Replacing V by $x^{it_0} V$ costs at most $(1 + |t_0|)^B$.

On a fixed compact logarithmic box, let $F_Y(\mathbf{y}) = W(\mathbf{y}) V^\sharp(Y \prod_i y_i^{a_i})$, where Y is constant within the block. Euler differentiation and (B2.21) bound every fixed derivative of F_Y by

$$m_A(Y) = \min\{Y^{1/4}, (1+Y)^{-A}\}$$

times fixed seminorms. Fourier integration by parts in all logarithmic variables shows that the common separating density has weighted L^1 -norm bounded by the same scalar. For a row vector linear in this whole profile, write $F_Y = m_A(Y) \widehat{F}_Y$, apply its single common Fourier inversion, and then Minkowski:

$$\|U(F_Y)\|_{\ell^2(I)} \leq m_A(Y) \int |\widehat{F}_Y(\mathbf{t})| \|B(\mathbf{t})\|_{\ell^2(I)} \frac{d\mathbf{t}}{(2\pi)^d}. \quad (\text{B2.22})$$

Only now may the positive row set be enlarged. The separated formula defines B on the added rows; no kernel estimate is asserted there. Squaring gives $m_A(Y)^2$. In particular the small-argument gain for a linear reflected sum is $Y^{1/2}$ in its energy. Choosing separate Fourier densities for different rows would not justify (B2.22).

For either lattice $\Lambda = \mathcal{O}, \lambda^{-4}\mathcal{O}$, if $|K(x)| \leq C_A x^{-A}$ for $x \geq 1$, $A > 1$, then

$$\sum_{\substack{0 \neq \mu \in \Lambda \\ aq_\mu > U}} |K(aq_\mu)| \ll_A C_A (1 + a^{-1}) U^{1-A} \quad (a > 0, U \geq 1). \quad (\text{B2.23})$$

Indeed the shell $2^j U < aq_\mu \leq 2^{j+1} U$ has $O(1 + 2^j U/a)$ lattice points; sum the two resulting geometric series. This estimate controls the entire discarded dual series, without presupposing a bound on its retained indices.

B2.4. The quadratic–cubic norm, including both collision masks

B2.4.1. Imported sieves and their domains

For squarefree primary good ideals, the quadratic sieve used here is

$$\sum_{\substack{a \text{ sf} \\ q_a \leq U}} \left| \sum_{\substack{b \text{ sf} \\ q_b \leq V}} c_b \chi_a(b)^3 \right|^2 \ll_{\epsilon} (UV)^{\epsilon} (U+V) \sum_b |c_b|^2. \quad (\text{B2.24})$$

Here is the family check needed to apply [GL]. The quadratic symbol at the primary generator of an ideal (x) , initially on λ -units, is

$$\vartheta_a((x)) = \epsilon_{\lambda}(x)^{e(a)}(x/a)_2, \quad e(a) = (q_a - 1)/2 \bmod 2,$$

where ϵ_{λ} is the nontrivial character modulo λ . The powers of ω used to make a generator primary are squares; its sign contributes exactly the displayed λ -character. CRT gives primitive conductor $a\lambda^{e(a)}$ and trivial infinite type. When $e(a) = 0$, take the primitive inducing character with the λ -factor absent. In a fixed primary square class modulo 4, two indices have the same $e(a)$; for coprime such indices the primitive product character has conductor exactly their product. The established reciprocity factor is a bicharacter of a fixed ray group. These are the three conditions of [GL, Definition 1]; its theorem and transpose duality give (B2.24).

Heath-Brown’s cubic sieve gives

$$\sum_{\substack{a \text{ sf}, a \equiv 1(3) \\ q_a \leq U}} \left| \sum_{\substack{b \text{ sf}, b \equiv 1(3) \\ q_b \leq V}} c_b(b/a)_3 \right|^2 \ll_{\epsilon} (UV)^{\epsilon} \{U + V + (UV)^{2/3}\} \sum_b |c_b|^2. \quad (\text{B2.25})$$

Squarefreeness is in $\mathcal{O}$; squarefree rational norms and an exclusion of rational prime factors are not hypotheses. Conjugation and cubic reciprocity permit the opposite orientation. In either sieve, fixed row restrictions decrease the positive outer sum, and fixed column restrictions are implemented by zero coefficients. A row–column collision mask must first be resolved.

B2.4.2. Quadratic reduction for a squarefree index times a cube

Let k be squarefree good with $q_k \ll K$, n squarefree primary with $q_n \asymp N$, and b primary with $q_b \asymp B$. The dual n, b may contain primes of S other than λ . Let $\beta(n, b)$ be arbitrary, independent of k ; fixed support restrictions are allowed.

Write uniquely

$$b = gt^2, \quad c = (n, g), \quad n = cm, \quad g = ch,$$

where g is squarefree and c, m, h are pairwise coprime squarefree ideals. There is no restriction $(g, t) = 1$. On dyads $q_c \asymp C$, $q_g \asymp G$, $q_t \asymp T$, we have $B \asymp GT^2$ and the exact identity

$$\chi_k(nb)^3 = \chi_k(mh)^3 1_{(k, ct)=1}. \quad (\text{B2.26})$$

Cauchy in t costs $O(T)$. Fix t , retain $(k, t) = 1$ as a row restriction, and expand the entire remaining mask: $1_{(k, c)=1} = \sum_{r|(k, c)} \mu(r)$. Rowwise divisor Cauchy costs $(KNB)^{\epsilon}$. For fixed r , write $k = rk'$, $c = rc'$. Squarefreeness gives fixed restrictions $(k', r) = (c', r) = 1$, and terms with $(r, t) > 1$ vanish. There is no remaining (k', c') -mask in an individual r -summand.

The factor $\chi_r(mh)^3$ is column-only. Split the finitely many squarefree S -parts of m, h , whose characters are row scalars. Their good product $j = m_{\text{good}} h_{\text{good}}$ is squarefree with norm $O(NG/C^2)$. Group by j , use (B2.24) with row length K/q_r , then Cauchy over the divisor-bounded factorizations

of j and the $O(C/q_r)$ choices of c' . If $\mathcal{D}_r(t)$ retains all original conditions on (c', m, h) , the resulting bound is

$$Q_{C,G,T}(\beta) \ll_{\epsilon} (KNB)^{\epsilon} T \sum_{qt \asymp T} \sum_{\substack{r \text{ sf, good} \\ q_r \ll \min(K, C)}} \left(\frac{K}{q_r} + \frac{NG}{C^2} \right) \frac{C}{q_r} \sum_{(c', m, h) \in \mathcal{D}_r(t)} |\beta(rc'm, rc'ht^2)|^2. \quad (\text{B2.27})$$

This more precise positive bound will preserve the tuple independence in the next step. In particular, for $|\beta| \leq 1$, the triple count is $O(NG/(q_r C))$. In a dyad $q_r \asymp r_0$, counting its $O(r_0)$ divisors and the $O(T)$ values of t gives

$$\ll (KNB)^{\epsilon} \frac{NB}{r_0} \left(\frac{K}{r_0} + \frac{NG}{C^2} \right) \ll (KNB)^{\epsilon} NB(K + NB).$$

Triangle inequality over the logarithmically many C, G, T -dyads proves

$$\sum_k \left| \sum_{n,b} \beta(n, b) \chi_k(nb)^3 \right|^2 \ll_{\epsilon} (KNB)^{\epsilon} (K + NB) NB. \quad (\text{B2.28})$$

Subunit quotient ranges are empty; bounded ranges, including the unit ideal, are covered by fixed comparison constants.

B2.4.3. Hybrid norm

Let P be squarefree good with $q_P \asymp L$. Suppose $|a(P)|, |\beta(n, b)| \leq 1$, with a independent of k, n, b and β independent of k, P . Then

$$\begin{aligned} \mathcal{H} &= \sum_k \left| \sum_P \frac{a(P)}{\sqrt{q_P}} 1_{(P,k)=1} \sum_{n,b} \beta(n, b) \chi_k(nb)^3 \chi_P(n)^{-2} 1_{(P,b)=1} \right|^2 \\ &\ll_{\epsilon} (KNBL)^{\epsilon} (K + NB) B \{N + L + (NL)^{2/3}\}. \end{aligned} \quad (\text{B2.29})$$

Fixed support restrictions and finite ray sectors are allowed if they preserve those two independence conditions.

To prove this, expand the complete mask $1_{(P,k)=1}$ over $D \mid (P, k)$, and apply rowwise divisor Cauchy. Put $P = DP'$, $k = Dk'$, and

$$a_D(P') = a(DP') 1_{(D,P')=1}, \quad \beta_D(n, b) = \beta(n, b) \chi_D(nb)^3 \chi_D(n)^{-2} 1_{(D,b)=1}.$$

The energy obtains a factor q_D^{-1} ; the new row and tuple lengths are $K/q_D, L/q_D$. No (P', k') -mask is reinstated. Apply (B2.27) to k' , treating

$$\beta_D(n, b) \sum_{P'} a_D(P') q_{P'}^{-1/2} \chi_{P'}(n)^{-2} 1_{(P',b)=1}$$

as its row-independent coefficient. In its notation the positive output for one C, G, T -block is bounded by

$$\begin{aligned} &(KNBL)^{\epsilon} \sum_D \frac{1}{q_D} T \sum_{qt \asymp T} \sum_r \left(\frac{K}{q_D q_r} + \frac{NG}{C^2} \right) \frac{C}{q_r} \sum_{c', m, h} |\beta_D(rc'm, rc'ht^2)|^2 \\ &\cdot \left| \sum_{P'} \frac{a_{D,r}(P')}{\sqrt{q_{P'}}} \chi_{P'}(c'm)^{-2} 1_{(P',ht)=1} \right|^2, \quad a_{D,r}(P') = a_D(P') \chi_{P'}(r)^{-2}. \end{aligned} \quad (\text{B2.30})$$

The identity used to obtain it, with zeros included, is

$$\chi_{P'}(rc'm)^{-2} 1_{(P',rc'ht^2)=1} = \chi_{P'}(r)^{-2} \chi_{P'}(c'm)^{-2} 1_{(P',ht)=1}.$$

The character supplies the zeros at r, c', m .

Only in this positive sum discard $|\beta_D|^2 \leq 1$. For fixed h, t, D, r , the P' -coefficients are independent of $j = c'm = n/r$, and their squared mass is $\sum_{P'} q_{P'}^{-1} |a_{D,r}(P')|^2 1_{(P',ht)=1} \ll (KNBL)^\epsilon$. Group c', m by the squarefree primary j , with norm $O(N/q_r)$. Its allowed S -parts remain. The cubic sieve (B2.25), then the $O(G/C)$ choices of h , give, on $q_D \asymp D_0$, $q_r \asymp r_0$,

$$\ll (KNBL)^\epsilon \frac{G}{C} \left\{ \frac{N}{r_0} + \frac{L}{D_0} + \left(\frac{NL}{r_0 D_0} \right)^{2/3} \right\}.$$

The t -sum and the factor C/q_r contribute $T^2(C/r_0)(G/C) \asymp B/r_0$. There are $O(r_0)$ choices of r , and $\sum_{q_D \asymp D_0} q_D^{-1} \ll 1$. Replacing D_0, r_0 by 1 only increases the sieve factors; $NG/C^2 \ll NB$. Summing the dyads proves (B2.29), without introducing either $(g, t) = 1$ or a new pair-dependent mask.

B2.5. Whole-index marks and separation of their phases

For a set $\mathcal{Q}$ of marked primes disjoint from the base local set, expand $\prod_{q \in \mathcal{Q}} 1_{q|A} = \prod_q (1 - \chi_q(A)^0)$ and apply (B2.13). Its simultaneous branch compatibility makes the inactive contribution at q

$$1 - (1 - q_q^{-1}) = q_q^{-1}.$$

The active contribution occurs only in the subtracted zero-mask transform. It contributes $q_q^{-1/2} \chi_q(\lambda^4 \mu)^{-2}$, with scalar $-\omega_{q,0}$. Thus the marked reflection is

$$\begin{aligned} \mathcal{T}_{\mathcal{Q}}(X; \Psi; V) &= \sum_{\mathcal{B}} \zeta_{\mathcal{Q}} \sum_{\mu \neq 0} \frac{d(\mu) \alpha(\mu) \vartheta(\lambda^4 \mu)}{\sqrt{q_\mu}} \prod_{p \in \mathcal{A}_0} B_{p,j_p}(\lambda^4 \mu) \prod_{q \in \mathcal{A}_{\mathcal{Q}}} q_q^{-1/2} \chi_q(\lambda^4 \mu)^{-2} V^\#(q_\mu X / q_c^2), \\ \zeta_{\mathcal{Q}} &= -\frac{i}{81} \overline{\alpha(c)}^2 \widehat{\varphi}(h_0) \overline{\kappa_F} \prod_{p \in \mathcal{I}_0} (1 - q_p^{-1}) \prod_{q \in \mathcal{I}_{\mathcal{Q}}} q_q^{-1} \prod_{p \in \mathcal{A}_0} \chi_p(\sigma_p)^{-2} \omega_{p,j_p} \prod_{q \in \mathcal{A}_{\mathcal{Q}}} \chi_q(\sigma_q)^{-2} (-\omega_{q,0}). \end{aligned} \quad (\text{B2.31})$$

Here $\mathcal{A}_0, \mathcal{I}_0$ are the base active and inactive sets, $\mathcal{Q} = \mathcal{A}_{\mathcal{Q}} \sqcup \mathcal{I}_{\mathcal{Q}}$, and $c = c_F \prod_{p \in \mathcal{A}_0 \cup \mathcal{A}_{\mathcal{Q}}} p$. An inactive marked prime is absent from that branch's conductor and kernel. If a marked prime belongs to the base local set, its primal term is identically zero and is removed first: its whole-index mark and the base zero extension cannot both be nonzero.

The phase at an active prime has the exact form

$$\chi_p(\sigma_p)^{-2} \omega_{p,j} = \xi_{p,j} \chi_p(c/p)^{2j+2}, \quad |\xi_{p,j}| = 1, \quad (\text{B2.32})$$

where $\xi_{p,j}$ depends only on p, j . This follows by inserting $\epsilon_p = -\lambda^{-5}(c/p)^{-2}$ into (B2.11); for $j = 4$, use $2j + 2 \equiv -2 \pmod{6}$. Consequently, two distinct active primes contribute

$$\chi_p(q)^{2j_p+2} \chi_q(p)^{2j_q+2} = \left(\frac{q}{p} \right)_3^{j_p+j_q+2}. \quad (\text{B2.33})$$

A residual row prime has $j_p = 1$, and an active mark has $j_q = 0$. Their phase is therefore one. In contrast, a $j = 4$ prime would leave a nontrivial cubic phase with the residual row, so every nonresidual base prime is fixed before the row norm.

Let R be the residual squarefree row, P the product of active marked primes, and F_{act} the frozen active base product. Fix separate classes of R and P modulo the full $\mathfrak{M}^2$. This fixes d, ϑ, κ_F without imposing a pair-dependent sector restriction. Equation (B2.33) assigns row-row phases to a row scalar, mark-mark phases to a tuple scalar, and frozen interactions to the relevant one of these; $\overline{\alpha(c)}^2$ separates multiplicatively. The removed primal collisions remain as $1_{(P,R)=1}$.

After fixing the dual unit, ramified power, and extracted frozen factors, the remaining dual part is nb^3 . Its moving columns are exactly

$$\chi_R(nb^3)^3 = \chi_R(nb)^3, \quad \chi_P(nb^3)^{-2} = \chi_P(n)^{-2} 1_{(P,b)=1}. \quad (\text{B2.34})$$

The source coefficient, fixed additive character, and frozen local columns depend only on the full extracted dual index. A collision with an extracted frozen prime gives a fixed row, tuple, or dual restriction. Hence after common smooth separation the tuple coefficient is independent of R, n, b , the dual coefficient is independent of R, P , and the remaining mixed masks are exactly those in (B2.29).

B2.6. Reflected energy and its two consequences

B2.6.1. Block parameters and estimate

Fix $q_f \asymp Z^V$, $Q = \log_Z q_{r_\rho}$, $X = Z^{N_*}$, and rows $0 < q_k \ll Z^M$. Split the ideal of k as

$$(k) = k_{\text{pow}} k_{\text{sup}} R.$$

The maximal powerful part contains every prime of valuation at least two; k_{sup} consists of the valuation-one primes in $S \cup \text{supp}(fr_\rho)$; and R is the remaining squarefree part. Their supports are disjoint. Freeze the actual $k_{\text{pow}}, k_{\text{sup}}$, the unit, and the fixed sectors above. On dyads $q_{k_{\text{pow}}} \asymp Z^O$, $q_R \asymp Z^H$, retain the actual row cutoff. Necessarily

$$0 \leq H \leq M - O + O(1/\log Z). \quad (\text{B2.35})$$

It need not equal $M - O$.

Choose the active zero-exponent base primes and split every $j = 4$ factor. For a large $j = 4$ term use the disjoint assignment

$$1_{p|nb^3} = 1_{p|n} + 1_{p \nmid n} 1_{p|b}.$$

Let A_0 be the logarithmic norm of all active non-slot primes outside R ; let S_0 count the small $j = 4$ terms and active base zero masks; and let N_0, B_0 count the large terms assigned respectively to the squarefree and cube factors. Let $z_a = \sum_{i \text{ active}} z_i \leq z$. The active marked product satisfies $q_P \asymp Z^{z_a}$. All these parameters are nonnegative.

For actual frozen products F_{act}, N_F, B_F of logarithmic norms A_0, N_0, B_0 , write a dual index as

$$\mu = u\lambda^h N_F n (B_F b)^3, \quad h \geq -4, \quad q_n \asymp Z^v, \quad q_b \asymp Z^{\ell_b}, \quad e_\lambda = h \log_Z 3,$$

with $v, \ell_b \geq 0$. All original source restrictions and permitted S -parts remain. Define

$$T_d = 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0, \quad u_h = \min\{v, z_a, (v + z_a)/3\}. \quad (\text{B2.36})$$

For any fixed $\tau_{\text{ref}} > 0$, all dyads outside

$$v + 3\ell_b + e_\lambda \leq T_d + \tau_{\text{ref}} \quad (\text{B2.37})$$

have arbitrarily small total size. A retained dyad, including the sum over active tuples, satisfies

$$\begin{aligned} E_{\text{ref}} &= \frac{O}{2} + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u_h - \ell_b - \frac{2}{3}e_\lambda \\ &\quad - \frac{1}{2}(T_d - v - 3\ell_b - e_\lambda)_+. \end{aligned} \quad (\text{B2.38})$$

Precisely, for the fixed $k_{\text{pow}}, k_{\text{sup}}$ its energy over R is $\ll Z^{E_{\text{ref}} - O/2 + \epsilon}$; summing these fixed parts in their O -dyad and all local choices gives $\ll Z^{E_{\text{ref}} + \epsilon}$. Constants are uniform under the scope in §B2.1. Current row restrictions must be independent of the active tuple and dual indices apart from $1_{(P,R)=1}$; the character, puncture, and original product mark are also independent of a subsequently averaged f . The thresholds depend only on the bounded log-length ranges, fixed profiles and moduli, $\epsilon, \tau_{\text{ref}}$. Negative e_λ has size $O(1/\log Z)$.

Proof. An inactive slot has absolute mass $\sum_{p \in \mathcal{P}_i} q_p^{-1} \ll Z^\epsilon$, including bounded lists. Triangle inequality in the row Hilbert space therefore freezes those primes at this cost; their collision exclusions become fixed row restrictions. Keep all active tuples inside the norm.

Equation (B2.7), before the local factors, gives

$$\frac{|d(\mu)|}{\sqrt{q_\mu}} \leq 27 q_{\lambda^h}^{-1/3} q_{N_F n}^{-1/2} q_{B_F b}^{-1}.$$

A small $j = 4$ or active base zero term contributes $q_p^{-1/2}$. A large $j = 4$ term contributes $q_p^{1/2}$: assigning p to N_F cancels that square-root denominator; assigning it to B_F leaves $q_p^{-1/2}$. If N_F meets b , its entire cube occurrence remains in b . Thus the common coefficient, per active tuple, has exponent

$$-v/2 - \ell_b - e_\lambda/3 - (S_0 + B_0 + z_a)/2. \quad (\text{B2.39})$$

The denominator is $c = c_F F_{\text{act}} R P$. With actual annular coordinates

$$y_R = q_R/Z^H, \quad y_n = q_n/Z^v, \quad y_b = q_b/Z^{\ell_b}, \quad y_i = q_{p_i}/Z^{z_i},$$

the kernel argument is exactly

$$\frac{q_\mu X}{q_c^2} = Y y_n y_b^3 y_R^{-2} \prod_{i \text{ active}} y_i^{-2}, \quad Y = q_{c_F}^{-2} Z^{v+3\ell_b+e_\lambda-T_d}. \quad (\text{B2.40})$$

This identity remains true at shared prime factors. On the original annuli the monomial is bounded above and below by fixed constants. Equation (B2.22) therefore permits the scalar

$$m_{\text{ker}} = Z^{-(T_d-v-3\ell_b-e_\lambda)_+/4}$$

to be factored from the whole linear row vector, with bounded normalized seminorms.

Remove tails before this Fourier separation. For sufficiently large Z , a dyad violating (B2.37) has actual argument $> Z^{\tau_{\text{ref}}/2}$. The full source satisfies $|d(\mu)|/\sqrt{q_\mu} \leq 27 \cdot 3^{4/3}$, and every local column factor is at most $q_p^{1/2}$. Row and tuple counts, branch counts, and $1 + q_c^2/X$ cost a fixed power $Z^{B_{\text{tail}}}$, chosen independently of the kernel order. Apply (B2.23) with $a = X/q_c^2$, $U = Z^{\tau_{\text{ref}}/2}$ and

$$A > 1 + 2(B_{\text{tail}} + D)/\tau_{\text{ref}}.$$

The entire discarded sum is $O(Z^{-D})$ times a finite test seminorm, with its corresponding fixed polynomial height cost. This counts all discarded μ .

For a retained block, apply the single common separation to (B2.40), then enlarge only the separated positive row norm to $q_R \ll Z^H$. By §B2.5 the separated arithmetic expression is the one in (B2.29), with

$$K = Z^H, \quad N = Z^v, \quad B = Z^{\ell_b}, \quad L = Z^{z_a}.$$

Leave $q_p^{-1/2}$ inside that norm. The outside squared coefficient is $Z^{-v-2\ell_b-2e_\lambda/3-S_0-B_0+\epsilon}$. Since

$$\max\{v, z_a, 2(v + z_a)/3\} = v + z_a - u_h,$$

(B2.29) gives the first line of (B2.38), without $O/2$. Multiplication by m_{ker}^2 supplies its last line.

Every powerful ideal is uniquely $x^2 y^3$ with y squarefree. Ideal counting and $\sum_y q_y^{-3/2} < \infty$ give $O(Z^{O/2})$ choices in its dyad. The supported part divides the radical of Sfr_ρ and has $O(Z^\epsilon)$ choices; the local branches have the same divisor bound. Rowwise divisor Cauchy and the positive sum over frozen parts add $O/2 + \epsilon$ once. Active tuples were already bounded by the hybrid norm and are not counted again. This proves (B2.38).

B2.6.2. General label and puncture width

At an active non-slot prime, the coefficient of its log-norm in $2A_0 - N_0 - S_0 - 4B_0$ is

Local choice	Coefficient
j odd or $j = 2$	2
$j = 0$, active	1
$j = 4$, small or assigned to n	1
$j = 4$, assigned to b	-2
$j = 0$, inactive	0

A residual row prime contributes two. These coefficients are bounded prime by prime by the valuations in $(q_k^2/q_{k_{\text{pow}}})q_fq_{r_\rho}$: a powerful row valuation $e \geq 2$ supplies e , a simple row prime supplies two, and a prime occurring only in f or r_ρ supplies one. Intersections only increase the available valuation. Hence

$$q_R^2 Z^{2A_0 - N_0 - S_0 - 4B_0} \leq C_S \frac{q_k^2}{q_{k_{\text{pow}}}} q_f q_{r_\rho},$$

and therefore

$$T_d - S_0 - B_0 \leq 2M - O + Q + V - N_* + 2z_a + O(1/\log Z). \quad (\text{B2.41})$$

This is the general width bound used when completion shortens the column length to $N_* = N - 3\ell_h$. It is uniform in the actual f and puncture, without replacing H by $M - O$.

With no slots, $z_a = u_h = 0$. On a retained dyad, (B2.38) gives

$$E_{\text{ref}} \leq O/2 + \max\{H, T_d - S_0 - B_0\} + O(1/\log Z) + \tau_{\text{ref}}.$$

For the row branch drop the nonpositive terms; for the column branch use $v \leq T_d - 3\ell_b - e_\lambda + \tau_{\text{ref}}$. Apply (B2.35), (B2.41), take τ_{ref} and the small-power losses sufficiently small, and sum the logarithmically many blocks:

$$\sum_{\substack{0 < q_k \ll Z^M \\ q_{k_{\text{pow}}} \asymp Z^O}} |\mathcal{T}_1(Z^{N_*}; k; V)|^2 \ll_\epsilon Z^{O/2 + \max\{M - O, 2M - O + Q + V - N_*\} + \epsilon}. \quad (\text{B2.42})$$

In particular, with $f = r_\rho = 1$, summing O gives the unmarked completed bound

$$\sum_{0 < q_k \ll Z^M} |\mathcal{T}_1(Z^{N_*}; k; V)|^2 \ll_\epsilon Z^{\max(M, 2M - N_*) + \epsilon}. \quad (\text{B2.43})$$

This includes the balanced energy $Z^{1+\epsilon}$ at $M = N_* = 1$.

B2.6.3. Short marked energy without label or puncture

When $f = r_\rho = 1$, every active non-slot prime outside R divides the powerful part. Thus

$$2A_0 \leq O + O(1/\log Z), \quad 0 \leq N_0 \leq A_0.$$

If $M + 2z \leq N_*$, then for every $\epsilon > 0$,

$$\sum_{0 < q_k \ll Z^M} |\mathcal{T}_d(Z^{N_*}; k; V)|^2 \ll_\epsilon Z^{M + (N_* + 2z - 2M)_+ / 4 + \epsilon}. \quad (\text{B2.44})$$

To see this directly from (B2.38), first ignore fixed $O(1/\log Z)$ errors and choose τ_{ref} arbitrarily small. Since

$$T_d - H \leq M + 2z - N_* \leq 0,$$

retention forces $\max(H, v + \ell_b) = H + O(\tau_{\text{ref}} + 1/\log Z)$. If $u_h = z_a$, (B2.38) is at most M , up to those errors. Otherwise $u_h \geq v/2$. Put $y_d = v + 3\ell_b + e_\lambda$; because $\ell_b \geq 0$ and $e_\lambda \geq -O(1/\log Z)$,

$$u_h + \ell_b + \frac{2}{3}e_\lambda + \frac{1}{2}(T_d - y_d)_+ \geq \frac{1}{4}y_d + \frac{1}{2}(T_d - y_d)_+ - O(1/\log Z) \geq \frac{1}{4}T_d - O(1/\log Z).$$

With $\Delta_H = M - O - H \geq -O(1/\log Z)$, substitution into (B2.38) gives

$$\begin{aligned} 4(E_{\text{ref}} - M) &\leq N_* + 2z_a - 2M - 2\Delta_H - (2A_0 - N_0 + B_0 + 4S_0) + O(\tau_{\text{ref}} + 1/\log Z) \\ &\leq N_* + 2z - 2M + O(\tau_{\text{ref}} + 1/\log Z). \end{aligned}$$

Restoring the row branch, tails, and small-power losses proves (B2.44). No moment estimate or zero-free conclusion enters this proof.

Appendix B3. The inverse and plain moment estimates

This appendix proves UM and MI, the unconditional zero-slot moment, and the canonical common-profile positive-slot moment in the range stated in B3.4. Its arithmetic inputs are the finite Gauss identities (B1.1)–(B1.19) and the completed reflected-energy estimate (B2.35)–(B2.41), proved in the preceding appendices. In particular, the inverse-moment induction terminates at RE; the plain-moment induction is independent of the inverse moment.

B3.1. Common conventions and analytic bookkeeping

Write $q_a = Na$, sum ideals through their primary generators, and retain every zero of every displayed residue symbol, including powers divisible by six. Element rows include units and arbitrary prime valuations. All column ideals avoid the fixed set S . A fixed ray character has its entire defining modulus supported in S ; a moving coprimality condition is displayed separately. Put

$$a_0(n) = \overline{\alpha(n)}\gamma_2(n) \quad (n \text{ squarefree}).$$

For disjoint prime lists $\mathcal{P}_i$, fixed throughout the current row and label averages, define a **product mark**

$$d_I(A) = \sum_{(p_i) \in \prod_{i \in I} \mathcal{P}_i} \prod_{i \in I} a_i(p_i) \mathbf{1}_{p_i | A}, \quad q_{p_i} \asymp Z^{z_i}, \quad |a_i(p_i)| \leq 1. \quad (\text{B3.1})$$

The number of slots is fixed; their coefficients need not be multiplicative. Smooth slot weights are included in the individual a_i . The useful property is the product form, not just the consequent divisor bound. In particular,

$$\mathbf{1}_{p|D_1 \cdots D_h n} = \sum_{j=1}^h \mathbf{1}_{p|D_j} \prod_{a < j} \mathbf{1}_{p \nmid D_a} + \mathbf{1}_{p|n} \prod_{a \leq h} \mathbf{1}_{p \nmid D_a} \quad (\text{B3.2})$$

assigns each slot first to one extracted factor or leaves it on the remaining column. Assigned coefficients are divisor bounded; surviving coefficients retain their original product form. Each copy of a squared mark is allocated independently.

Common-profile lemma. Suppose finitely many operations use smooth functions on fixed compact intervals of positive normalized norms, radial Schwartz Fourier kernels, and finite Fourier expansions on fixed ray groups. All arithmetic restrictions that are not smooth remain in the outer weights. For a coupled profile $H(y_1, \dots, y_d)$, multiply by larger individual cutoffs equal to one on the full current supports and use

$$H(y) = (2\pi)^{-d} \int_{\mathbf{R}^d} \widehat{H}(t) \prod_j y_j^{it_j} dt, \quad \int |\widehat{H}(t)| |\langle t \rangle|^J dt \ll_{J,d} \max_{|\beta| \leq 2r} \|D^\beta H\|_\infty, \quad 2r > J + d, \quad (\text{B3.3})$$

where $D_j = y_j \partial_{y_j}$ and $\langle t \rangle = 1 + |t|$. The coefficient measure is defined on the ambient box before any current row or live label is chosen. Relations among those norms restrict evaluation points, not the measure. Real inverse roots belong to this coupled profile once; the final column tests receive only their individual cutoffs and Fourier powers.

The estimate follows by integration by parts with $(1 - \Delta)^r$ in logarithmic coordinates. If a kernel is $K(A \prod_j y_j^{e_j})$, its Euler derivatives are fixed multiples of $(x \partial_x)^r K(x)$, uniformly for every $A > 0$. Consequently arbitrarily high fixed derivative orders do not introduce a power of Z . Norm twists introduce only fixed polynomials in their heights. For any $Y \geq 1$, radial decay and lattice counting also give

$$\sum_{h \neq 0: a q_h > Y} |K(a q_h)| \ll_B (1 + a^{-1}) Y^{1-B} \quad (a > 0, B > 1). \quad (\text{B3.4})$$

Indeed the shell $2^j Y < a q_h \leq 2^{j+1} Y$ has $O(1 + 2^j Y/a)$ points. This proves the displayed bound by two geometric series.

Here and below a tail is discarded on the genuine arithmetic expression before extending a coprime sum or taking an absolute Fourier integral. To do so, enclose the retained frequencies in an outer ball, fixed for the current norm sector, whose complement forces the actual kernel argument above Z^τ . Keep the full kernel inside that ball. If the remaining raw factors and $1 + a^{-1}$ cost Z^{B_0} , choose $B > 1 + (B_0 + T)/\tau$ in (B3.4) to obtain $O(Z^{-T})$. All raw scales below are polynomially bounded, so B_0 is fixed before B is chosen.

For completeness, the finite-order uniformity used in both inductions follows from this rule. At a terminal vertex choose its required profile and height orders. Along an edge, (B3.3) integrates the polynomial demanded by its child after increasing finitely many parent derivative orders. Work backwards through a bounded-depth acyclic graph. For a discrete family with weighted count $\ll Z^b$, extract Z^b once and normalize its counting measure before this argument. Cauchy-Schwarz averages two child exponents; it does not add them. Minkowski takes their maximum after the coefficient mass is accounted for. Thus the path length bounds the total numerical exponent loss, while all derivative orders remain finite. Fresh quotient cutoffs, larger separation cutoffs and bounded scale-clipping families are fixed through this finite depth before the final threshold in Z . A nonempty window at a nominal negative length n has $1 \leq Z^{-n} \leq B$, where B is its fixed upper endpoint; replacing it by length zero is therefore a bounded dilation, not a growing support enlargement.

Masked primitive Poisson. If ψ is primitive modulo $\mathfrak{m}$, zero on nonunits, and Φ is radial Schwartz, then

$$\sum_{k \in \mathcal{O}} \psi(k) \mathbf{1}_{(k, R)=1} \Phi(q_k/K) = \frac{K \gamma(\psi; \mathfrak{m})}{\sqrt{q_{\mathfrak{m}}}} \sum_{d | \text{rad } R} \frac{\mu(d) \psi(d)}{q d} \sum_h \overline{\psi(h)} \mathcal{F} \Phi \left(\frac{K q_h}{q d q_{\mathfrak{m}}} \right). \quad (\text{B3.5})$$

Insert the full Möbius expansion of the mask, write $k = dk'$, and apply lattice Poisson in residue classes modulo $\mathfrak{m}$. The finite transform is $\sqrt{q_{\mathfrak{m}}} \gamma(\psi; \mathfrak{m}) \overline{\psi}(h)$; it vanishes at nonunits by primitivity. Finite Parseval gives $|\gamma| = 1$. For the primitive principal character use $\mathfrak{m} = 1$, $\gamma = 1$, including $h = 0$. Its nonzero frequencies have absolute sum $O_{\Phi}(d(\text{rad } R))$, since $A \sum_{h \neq 0} |\mathcal{F} \Phi(A q_h)| \ll 1$. Thus principal nonzero terms may be restored with the same retained outer mask at their original diagonal cost. Fixed ray restrictions are first expanded into characters; deleted conductor zeros are kept in R .

B3.2. A canonical marked energy

Fix a squarefree puncture $\rho(n) = \mathbf{1}_{(n, r_{\rho})=1}$, independent of the current row k and squarefree label f , and put $Q = \log_Z q_{r_{\rho}}$. Let $q_f \asymp Z^V$, $0 \leq w(f) \leq C d(f)^C$, with w depending only on f . For

$F = N + V$, define

$$\mathcal{E}(N, V, M) = Z^{-V} \sum_f w(f) \sum_{0 < q_k \ll Z^M} \left| Z^{-N/2} \sum_{n \text{ sf}} a_0(n) \nu(n) \rho(n) \chi_n(k) \chi_n(f)^4 d_I(n) W(q_n/Z^N) \right|^2. \quad (\text{B3.6})$$

Here ν is a fixed finite ray character, all real lengths range over fixed bounded sets, the slot sum is at most a fixed cap z_0 , and no further coefficient on n is permitted.

Canonical estimate. For any fixed $c_* > 0$,

$$F - M - Q - z_0 \geq c_*, \quad 4F - 3M - 6z_0 \geq c_* \implies \mathcal{E}(N, V, M) \ll Z^{F+\epsilon}. \quad (\text{B3.7})$$

The bound is uniform over the moving punctures and all label-only weights in the displayed class. It holds simultaneously for every surviving slot subcollection with the original cap z_0 .

Proof: terminal completion. Choose $0 < d \leq c_*/200$. Put $\Psi_k(n) = \nu(n) \rho(n) \chi_n(k) \chi_n(f)^4$, multiplicatively on all ideals. For $V_*(y) = y^{1/2} W(y)$ and a mark function d_* , write the completed marked sum as

$$T_{d_*}(X; k) = \sum_{n \text{ sf}, b} \frac{a_0(n) \Psi_k(n) \overline{\alpha(b)}^3 \Psi_k(b)^3}{\sqrt{q_n} q_b} d_*(nb^3) V_*(q_n q_b^3 / X).$$

The exact cube inversion is

$$Z^{-N/2} \sum_{n \text{ sf}} a_0(n) \Psi_k(n) d_I(n) W(q_n/Z^N) = \sum_h \frac{\mu(h) \overline{\alpha(h)}^3 \Psi_k(h)^3}{q_h} T_{d_h}(Z^N/q_h^3; k), \quad d_h(A) = d_I(h^3 A). \quad (\text{B3.8})$$

After expansion put $c = hb$. The mark is $d_I(nc^3)$, independent of the divisor h , and its coefficient contains $\sum_{h|c} \mu(h)$; only $c = 1$ remains. This also verifies the central normalization and every mask.

Use fixed dyadic windows, with the unit window centered at zero. If $V < d$ and the center ℓ_1 of h is less than d , freeze h , assign its slots by (B3.2), and apply RE at length $N_* = N - 3 \log_Z q_h$, with surviving cap $z_a \leq z_0$. The coefficient $\Psi_k(h)^3$ is a contraction in the joint (f, k) Hilbert space, and $\sum_{q_h \asymp Z^{\ell_1}} q_h^{-1} \ll 1$. To record the terminal exponent check, let c be the current margin in (B3.7), and let η bound fixed-window logarithmic errors. The RE support inequalities give

$$H \leq M - O + \eta, \quad T_d - S_0 - B_0 \leq M - O + 2z_a - z_0 - c + 5d + 4\eta. \quad (\text{B3.9})$$

The second inequality follows from $T_d - S_0 - B_0 \leq 2M - O + Q + V - N_* + 2z_a + \eta$: use the first margin and $2V + 3 \log_Z q_h < 5d + 3\eta$.

On retained reflected blocks $v + 3b + e_\lambda \leq T_d + \tau_{\text{ref}}$ and $e_\lambda \geq -\eta$. In the row branch of RE, drop its nonpositive terms to obtain $E \leq M + z_a + 5\eta/3 \leq F - c + 5\eta/3$. In the column branch, $u = z_a$ gives $E \leq O/2 + T_d - S_0 - B_0 + 5\eta/3 + \tau_{\text{ref}}$; if $u \neq z_a$, its definition gives $u \geq v/2$ and $E \leq O/2 + (T_d - S_0 - B_0)/2 + z_a + 7\eta/6 + \tau_{\text{ref}}/2$. Substitution of (B3.9), followed respectively by the first and second margins, yields

$$E \leq F - 2c + 5d + 17\eta/3 + \tau_{\text{ref}}, \quad E \leq F - M/4 - 3c/4 + 5d/2 + 19\eta/6 + \tau_{\text{ref}}/2. \quad (\text{B3.10})$$

All are strictly below F when $c \geq c_*/2$ and η, τ_{ref} , together with the reflected divisor loss, are at most $c_*/1000$. Summing the fixed-label RE bound costs $Z^{V+o(1)}$, canceled by Z^{-V} .

Nonterminal block. If $V < d$ and $\ell_1 \geq d$, reopen the completion in (B3.8), combine its cube variable with h , and call the result b . If its center is ℓ , put $r = N - 3\ell$. If $V \geq d$, instead take $b = 1, \ell = 0, r = N$. Triangle inequality in the joint Hilbert space and the common-profile lemma

reduce every remaining term to

$$Z^{-r-2\ell-V} \sum_f w(f) \sum_k \Phi(q_k/Z^M) \times \left| \sum_{q_b \asymp Z^\ell} \beta(b, f) \chi_b(k)^3 \times \sum_{n \text{ sf}} a_0(n) \nu(n) \rho(n) \chi_n(k) \chi_n(f)^4 d_I(nb^3) W(q_n/Z^r) \right|^2, \quad (\text{B3.11})$$

where $V + \ell \geq d$, β is divisor bounded and independent of k, n , and includes the original $(b, f) = 1$ mask. The variables b, n need not be coprime. We now reduce (B3.11) to (B3.6) with strictly shorter rows.

First Poisson transformation. In the square write

$$\mathfrak{B} = \text{rad}(b_1 b_2), \quad n_i = A_i C u_i, \quad A_i = (n_i, \mathfrak{B}), \quad C = (n_1/A_1, n_2/A_2).$$

Then u_1, u_2 are individually squarefree, coprime to one another and to $C\mathfrak{B}$. At $p \mid \mathfrak{B}$, set $a_i = \mathbf{1}_{p \mid A_i}$, $\pi = v_p(b_1 b_2) \bmod 2$, and $t_p = a_1 - a_2 + 3\pi \bmod 6$. Put $R_1 = \prod_{t_p \neq 0} p$. The primitive row character and its residual zero mask are

$$\chi_{u_1} \overline{\chi_{u_2}} \prod_{t_p \neq 0} \chi_p^{t_p}, \quad \mathbf{m}_1 = u_1 u_2 R_1, \quad R_{\text{mask}} = C\mathfrak{B}/R_1. \quad (\text{B3.12})$$

Apply (B3.5), introducing $d_k \mid C\mathfrak{B}/R_1$ and frequency h . Write A_i, B, R, δ also for the centers of A_i, C, R_1, d_k . The actual conductor length is $\log_Z(q_{n_1} q_{n_2} / (q_{A_1} q_{A_2} q_C^2)) + \log_Z q_{R_1}$. The principal case forces $u_1 = u_2 = 1$, $\pi = 0$, and $a_1 = a_2$ at every prime, hence $n_1 = n_2$ and $b_1 b_2$ a square. Even counting all cube pairs, its normalized contribution is $O(Z^{M+o(1)})$. The same bound covers restoring principal nonzero frequencies with the mask below.

CRT, $\gamma_1(u) \gamma_2(u) = \mu(u) \alpha(u) G(u)$, and reciprocity change the two residual columns to inverse coefficients. Their remaining joint ray factor is $G([u_1 u_2^{-1}]) R([u_1 u_2^{-1}], [\prod p^{t_p}])$; expand it on the fixed ray group before extending coprimality. Put $E = \prod_{\pi=1} p^{a_1+a_2}$, $y = h f^2 E$. The residual exponent at each prime of $\mathfrak{B}$ is $4a_i + \mathbf{1}_{t_p \neq 0}(1 \pm t_p) + \pi(a_1 + a_2) \bmod 6$. The following exhaustive table gives the same exponent on both sides:

π	a_1	a_2	t_p	Residual exponent
0	0	0	0	0
0	1	0	1	0
0	0	1	5	0
0	1	1	0	4
1	0	0	3	4
1	1	0	4	4
1	0	1	2	4
1	1	1	3	4

An exponent zero still means a puncture. Write $b_1 b_2 = q^2 s$, s squarefree, and put

$$J_2 = \prod_{\pi=0, a_1=a_2=1} p, \quad J = s J_2, \quad q = J_2 q_0.$$

Then J is squarefree and the residual factor is exactly $\chi_n(J)^4 \mathbf{1}_{(n, \text{rad } q_0)=1}$. If j centers J , the actual identities

$$\log_Z q_{q_0} = \frac{1}{2} \log_Z(q_{b_1} q_{b_2}) - \frac{1}{2} \log_Z q_s - \log_Z q_{J_2} \geq 0, \quad \log_Z \frac{q_{R_1} q_E}{q_{A_1} q_{A_2}} = \log_Z q_s - 2 \log_Z q_{J_2} \quad (\text{B3.13})$$

give the retained row enclosure

$$H_c = 2r - 2B - M + 4\ell + 2V + \delta - j, \quad H_+ = H_c + 12\eta + \tau, \quad 0 < q_y \leq Z^{H_+}. \quad (\text{B3.14})$$

Indeed the actual first kernel argument is $Z^M q_h / (q_{d_k} q_{m_1})$. If it is at most Z^τ , (B3.13) bounds $\log_Z q_y$ by $H_c + 12\eta + \tau$. Discard only the complement of this outer ball by (B3.4). A negative H_+ leaves no nonzero row.

Now define the quotient-free expression on all individually allowed u_1, u_2 , keeping its full kernel, formal product conductor and outer ball, and insert the complete identity $\mathbf{1}_{(u_1, u_2)=1} = \sum_{t'|(u_1, u_2)} \mu(t')$. Write $u_i = t'x_i$ and let t center t' . No Poisson formula is asserted on nonco-prime conductors: the full Möbius sum annihilates precisely those artificial terms. Squarefreeness retains $(x_i, t') = 1$. With

$$s_i = r - A_i - B - t, \quad \varkappa_i = M - 2r - 2\ell - V - \delta + A_i + B - R/2, \quad (\text{B3.15})$$

the separated inverse polynomial on side i is

$$P_i(y) = \sum_{x \text{ sf}} \mu(x) \nu_i(x) \rho(x) \overline{\chi_x(y)} \chi_x(CJ)^4 \chi_x(d_k) \mathbf{1}_{(x, \text{rad}(q_0 t'))=1} d_{I_i}(x) W_i(q_x / Z^{s_i}). \quad (\text{B3.16})$$

Here ν_i is fixed in its ray summand. The fourth-power factor at f became $\overline{\chi_x(f^2)}$, which explains both the row y and its conjugation. Priority allocation to $(b_i, A_i, C, t'; x_i)$ leaves an original slot subcollection: all earlier survival exclusions are already supplied by CJ , q_0 , and t' .

For precision, the first coupled profile uses the independent normalized norms of $(A_1, A_2, C, d_k, R_1, t', h, x_1, x_2)$. Apart from its larger cutoffs, its real root and kernel are

$$\begin{aligned} & y_d^{-1} y_R^{-1/2} y_t^{-1} (y_1 y_2)^{-1/2} W(y_{A_1} y_C y_t y_1) \overline{W(y_{A_2} y_C y_t y_2)} \\ & \times K_1 \left(A_1^{\text{ker}} \frac{y_h}{y_d y_R y_t^2 y_1 y_2} \right), \\ & A_1^{\text{ker}} = Z^{M+h_1-\delta-R-2t-s_1-s_2}, \end{aligned} \quad (\text{B3.17})$$

where h_1 centers the bare frequency. Multiply this profile by $Z^{-9\eta/2}$; its outside side factors are $Z^{(\varkappa_i+9\eta/2)/2}$. This is an exact separation of the full prefactor and kernel, with full fresh x_i -supports satisfying $|\log_Z q_{x_i} - s_i| \leq 4\eta$. The common-profile lemma applies uniformly to $A_1^{\text{ker}} > 0$.

Apply weighted Cauchy–Schwarz over the **whole** outer tuple before bounding its weights. At fixed old reconstruction and $y \neq 0$,

$$\sum_{f^2 E|(y)} w(f) \ll d((y))^{C+1} \ll Z^{o(1)}. \quad (\text{B3.18})$$

Thus f, h disappear together, since $h = y/(f^2 E)$, while the original $-V$ remains in $\varkappa_i$. Only at this positive-norm stage replace the retained ball by a fixed smooth majorant at scale Z^{H_+} . Keep J averaged and q_0 fixed. From the table, if r_{diff} is the actual radical length of even-parity primes with $a_1 \neq a_2$, $\log_Z q_{R_1} = \log_Z q_s + r_{\text{diff}}$. Hence

$$\log_Z q_{q_0} \leq \ell + R/2 - j + 5\eta/2. \quad (\text{B3.19})$$

This bounds the count of q_0 ; counting J as well costs $Z^{\ell+R/2+7\eta/2+o(1)}$. Its pieces are not counted independently later.

Second Poisson transformation. Fix a first Cauchy side. Expand $|P_i(y)|^2$, write $x_a = g' z_a$, $g' = (x_1, x_2)$, and center g' at g . Apply (B3.5) with primitive character $\overline{\chi_{z_1}} \chi_{z_2}$, modulus $z_1 z_2$, residual mask g' , divisor $d_2 \mid g'$ of center θ , and frequency k'' . Put

$$\begin{aligned} \lambda_c &= H_c - \theta - (s_i - g), \\ M_c &= 2(s_i - g) - H_c + \delta + 2\theta \\ &= M - 4\ell - 2A_i - 2t - 2g + 2\theta - 2V + j, \\ M' &= M_c + \eta. \end{aligned} \quad (\text{B3.20})$$

Retain $0 < q_{d_k d_2 k''} \leq Z^{M'}$. The actual conductor differs from $2(s_i - g)$ by at most 10η . The two appearances of d_2 and one of d_k add 3η ; subtracting $H_+ = H_c + 12\eta + \tau$ cancels the same tail tolerance τ . Thus its complement is an actual-ratio tail, exactly as required by (B3.4).

The principal case $z_1 = z_2 = 1$ has nominal total exponent

$$\varkappa_i + H_c + s_i + B + t + \ell + R/2 = F - B - j. \quad (\text{B3.21})$$

The first prefactor, row enclosure, diagonal column window and outer counts add at most $26\eta + \tau + o(1)$. It is therefore within the target bound, including any principal nonzero terms restored with the identical second mask.

The Gauss signal is $\mu(z)\gamma_{-1}(z) = a_0(z)\chi_z(-1)\overline{G(z)}$. After the second side is placed inside its conjugate, the relative ray factor is $G([z_2 z_1^{-1}])$. It is unchanged on removing a common factor. Extend the separate columns and full kernel off coprimality and insert the full Möbius sum $z_a = v'n_a$, with v' squarefree of center v . The common extracted factor and the old gcd restriction retain $(v', g' CJ) = 1$, $(g', CJ) = 1$, and the frequency zero $(v', k'') = 1$. Set

$$\begin{aligned} \gamma &= (q_0, t', r_g), \\ r_g &= g'/d_2, \\ k_{\text{new}} &= d_k d_2 k'', \\ f_{\text{new}} &= JC d_2 v', \\ \rho_\gamma(n) &= \rho(n) \mathbf{1}_{(n, \text{rad}(q_0 t' r_g))=1}. \end{aligned} \quad (\text{B3.22})$$

The retained common zeros make J, C, d_2, v' pairwise coprime and squarefree. The new label is therefore squarefree, and

$$\chi_n(d_k) \chi_n(k'') \overline{\chi_n(d_2)} \chi_n(CJ)^4 \chi_n(v')^4 = \chi_n(k_{\text{new}}) \chi_n(f_{\text{new}})^4. \quad (\text{B3.23})$$

This uses $\overline{\chi_n(d_2)} = \chi_n(d_2) \chi_n(d_2)^4$, valid also at nonunits, and $a_0(v'n) = a_0(v') a_0(n) \chi_n(v')^4$. Expanding $G([n_2 n_1^{-1}])$ yields the same fixed character ν' on both child sides. Priority allocation leaves original slot subcollections; their old survival zeros are supplied by f_{new} and ρ_γ . Freeze only γ ; the four factors of f_{new} remain inside its average.

The formal child lengths and their actual enclosing windows are

$$\begin{aligned} N_c &= r - A_i - B - t - g - v, \\ V_c &= B + \theta + v + j, \\ F_c &= N_c + V_c, \\ N' &= \max(0, N_c), \\ F' &= N' + V_c, \\ 0 &\leq \delta_N = N' - N_c \leq 6\eta. \end{aligned} \quad (\text{B3.24})$$

The full fresh column window has logarithmic error 6η , the full label window 4η . An empty window is omitted; otherwise the clipping is a bounded dilation as in the common-profile lemma. With $D_c = \ell + A_i + t + g - \theta + V$, direct substitution gives

$$\begin{aligned} F_c &= F - D_c - 2\ell + j, & M_c &= M - 2D_c - 2\ell + j, \\ F_c - M_c &= F - M + D_c, & 4F_c - 3M_c &= 4F - 3M + 2(A_i + t + g - \theta + V) + j. \end{aligned} \quad (\text{B3.25})$$

The fixed triples range over the projection of the genuine outer reconstruction set, before positive row or label enlargement. Every such triple has a source witness, giving $g - \theta \geq -2\eta$, $j \leq 2\ell + 3\eta$, and $\log_Z q_{q_0} \leq \ell + \eta$. Consequently its actual puncture satisfies $Q' \leq Q + \ell + t + g - \theta + 4\eta \leq Q + D_c + 4\eta$. Therefore

$$F' - M' - Q' - z_0 \geq c - 5\eta, \quad 4F' - 3M' - 6z_0 \geq c - 7\eta, \quad M - M' \geq 2(V + \ell) - 8\eta, \quad F' \leq F + 11\eta. \quad (\text{B3.26})$$

These inequalities hold for the structural triples themselves, not for every point in a containing count ball.

To finish the normalization, the second common profile uses normalized norms of $(g', d_2, v', k'', n_1, n_2)$, with real root $y_d^{-1} y_v^{-1} (y_1 y_2)^{-1/2}$, old windows $W_i(y_g y_v y_1) \overline{W_i(y_g y_v y_2)}$, and kernel $K_2(A_2^{\ker} y_k / (y_d y_v^2 y_1 y_2))$, $A_2^{\ker} = Z^{H_+ + h_2 - \theta - 2v - 2N_c}$. Multiplying this profile by $Z^{-6\eta}$ leaves the exact outside factor $Z^{\lambda_c + 18\eta + \tau}$. Fresh tests are $\omega_a(Z^{\delta_N} y) y^{\pm it_a}$; they and the density are independent of the current $k_{\text{new}}, f_{\text{new}}, \gamma$. The signed outer weight retains both common extracted squared magnitudes, every priority mask, one $\mu(d_2)$, one $\mu(v')$, and the second frequency mask. Expanding the separated columns and inverting (B3.3) restores the full expression term by term. Only then apply weighted Cauchy over all outer tuples and bound their common zeros by one.

Here is an explicit sufficient reconstruction bound, which prevents a hidden label cost. Let K_0 bound the original slot count. At fixed $\gamma, f_{\text{new}}, k_{\text{new}}$, splitting f_{new} into J, C, d_2, v' , splitting $J = sJ_2$, reconstructing $b_1 b_2 = q_0^2 J_2^2 s$, and choosing A_i, d_k gives at most

$$D(f_{\text{new}})C(\gamma), \quad D(f) = d(f)^{9+4K_0}, \quad C(\gamma) = d(q_0)^{5+2K_0} d(t')^{2K_0} d(r_g)^{2K_0}. \quad (\text{B3.27})$$

Indeed $d_k \mid f_{\text{new}} \text{ rad } q_0$; all extracted slot primes divide these fixed reconstructed factors and contribute only the indicated additional divisor powers. Once these factors are chosen, the row frequency is determined. No old f, h remains. By (B3.19), the distinct triples have weighted count

$$\ll Z^{C_c + 11\eta/2 + o(1)}, \quad C_c = \ell + R/2 - j + t + g - \theta. \quad (\text{B3.28})$$

Extract this factor once and normalize the triple measure. The unnormalized squared child norm equals $Z^{F'} \mathcal{E}(N', V_c, M')$. If the child estimate is $\mathcal{E} \ll Z^{F' + \epsilon_{\text{child}}}$, its contribution is $Z^{2F' + \epsilon_{\text{child}}}$. The exact balance

$$\varkappa_i + \lambda_c + C_c + 2F_c = F \quad (\text{B3.29})$$

then gives $Z^{F+40\eta+\tau+\pi+\epsilon_{\text{child}}}$: the two profile allowances and triple count contribute $9\eta/2 + 18\eta + 11\eta/2 = 28\eta$, and $2\delta_N \leq 12\eta$. Here π is the aggregate freely chosen divisor, ray-expansion, dyadic and coefficient loss. Both Cauchy inequalities take geometric means, so neither this exponent nor the triple mass is doubled.

Closure. Set $D = \lceil (M_{\max} + 2)/d \rceil + 1$. At depth h , use margins $c_h = c_* - 7h\eta$ and row cap $M \leq M_{\max} - hd$. Choose

$$\eta \leq \min\{d/16, c_*/(14D), \epsilon/(160D), c_*/1000\}, \quad \tau, \pi \leq \epsilon/(4D), \quad \tau \leq c_*/1000, \quad (\text{B3.30})$$

and the reflected losses as above. Then (B3.26) gives a decrease of at least d , both margins remain at least $c_*/2$, and F stays in a bounded range through depth D . Nonnegative rows preclude a strict call at depth D . Choose all local small-power shares within π , then the raw tail orders by (B3.4), then the required finite smooth and height orders backwards by (B3.3), and finally Z large enough for all fixed-window errors. The accumulated path loss is less than ϵ . The terminal completion, principal counts and this strict descent prove (B3.7). $\square$

B3.3. The marked inverse estimate and inverse amplification

Let $\varphi_u(n) = \nu(n) \chi_n(u)^{\varepsilon_\chi}$, with one common sign $\varepsilon_\chi \in \{\pm 1\}$, and define the centrally normalized inverse and prime polynomials as in the main text. We prove the full-width version of MI:

$$r + 2z \leq m - c_1, \quad 2r + 8z \leq 3m - c_2 \implies \sum_{0 < q_u \ll Z^m} |M_u(Z^r) Q_u|^2 \ll Z^{m+\epsilon}. \quad (\text{B3.31})$$

The prime coefficients are arbitrary bounded row-independent coefficients, and no mesh is required.

Let $P = \prod_i p_i$, $j = (n, P)$, and fix the subset of slot primes dividing j . Write $n = jn_0$, $P = jP_0$; then $(n_0, P_0) = 1$ and $c = n_0 P_0$ is squarefree and prime to j . The exact identities $\mu(n_0) = \mu(c)\mu(P_0)$ and $\varphi_u(n)\varphi_u(P) = \varphi_u(j)^2 \varphi_u(c)$ retain all zeros. If G is the sum of assigned slot lengths, put

$$z_0 = z - G, \quad D_0 = r + z - 2G, \quad Z^{-(r+z)/2} = Z^{-G} Z^{-D_0/2}. \quad (\text{B3.32})$$

There are $O(Z^{G+o(1)})$ assigned tuples. Their Hilbert-space triangle mass is canceled by Z^{-G} , while $\varphi_u(j)^2$ is a contraction. For a fixed j , the old inverse profile has argument $q_n/Z^r = (q_j/Z^G)(q_c/Z^{D_0})/\prod_{i \text{ surviving}}(q_{p_i}/Z^{z_i})$. Separate this one common profile by (B3.3), before choosing surviving primes. Absorb $\mu(p_i)$ in their coefficients. Conjugate the whole residual polynomial when necessary; it remains to bound

$$R_0(u) = Z^{-D_0/2} \sum_{c \text{ sf}} \mu(c) \nu_0(c) \mathbf{1}_{(c,j)=1} \overline{\chi_c(u)} d_0(c) W_0(q_c/Z^{D_0}). \quad (\text{B3.33})$$

The prior Fourier measure is uniform over the fixed q_j/Z^G interval.

Expand its square in a smooth row ball, write $c_i = Cz_i$, $(z_1, z_2) = 1$, and apply (B3.5) to $\overline{\chi_{z_1}} \chi_{z_2}$, with divisor $d' \mid C$ and frequency h' . Let B, θ center C, d' , put $P_1 = B - \theta \geq -2\eta$, and retain

$$M_c = 2D_0 - m - 2P_1, \quad M_{\text{init}} = M_c + 6\eta + \tau_{\text{init}}, \quad 0 < q_{d'h'} \leq Z^{M_{\text{init}}}. \quad (\text{B3.34})$$

The actual conductor differs from $2(D_0 - B)$ by at most 4η , and the two appearances of d' add 2η . Thus the omitted complement is an actual-ratio tail. The principal diagonal is $O(Z^{m+o(1)})$, also for a bounded nonempty negative D_0 -window.

The same Gauss calculation as the second canonical transform produces $a_0(z) \nu_*(z) \overline{\chi_z(d')} \chi_z(h')$, with the same ν_* on both sides after expanding $G([z_2 z_1^{-1}])$. Insert the full mutual-gcd Möbius expansion $z_a = sn_a$, and retain the common zeros $(C, j) = 1$, $(s, Cj) = 1$, $(s, h') = 1$. Set

$$t = C/d', \quad k = d'h', \quad f = d's, \quad \rho_{t,j}(n) = \mathbf{1}_{(n, \text{rad}(tj))=1}. \quad (\text{B3.35})$$

The zero-preserving identity

$$a_0(sn) \nu_*(sn) \overline{\chi_{sn}(d')} \chi_{sn}(h') = a_0(s) \nu_*(s) \overline{\chi_s(d')} \chi_s(h') a_0(n) \nu_*(n) \chi_n(k) \chi_n(f)^4 \quad (\text{B3.36})$$

holds on $(s, n) = 1$, using $\overline{\chi_n(d')} = \chi_n(d') \chi_n(d')^4$. The new label f is squarefree. Slot priority allocation to $(t, d', s; n)$ leaves an original subcollection; its survival zeros are exactly supplied by $\rho_{t,j}$ and $\chi_n(f)^4$.

If v centers s , the formal child parameters are

$$\begin{aligned} N_c &= D_0 - B - v, \\ V_c &= \theta + v, \\ F_c &= D_0 - P_1, \\ N' &= \max(0, N_c), \\ 0 \leq \delta_N &= N' - N_c \leq 3\eta, \\ F' &= F_c + \delta_N. \end{aligned} \quad (\text{B3.37})$$

For the six normalized norms (C, d', s, h', n_1, n_2) , the exact root is $y_d^{-1} y_s^{-1} (y_1 y_2)^{-1/2}$, the old windows are $W_0(y_C y_s y_1) \overline{W_0(y_C y_s y_2)}$, and the kernel is $K_{\text{init}}(A^{\text{ker}} y_h / (y_d y_s^2 y_1 y_2))$, with $A^{\text{ker}} = Z^{m+H-\theta-2(D_0-B)}$ and H the bare-frequency center. Multiply the profile by $Z^{-3\eta}$; the exact outside factor is $Z^{\kappa_c+3\eta}$, where $\kappa_c = m - 2D_0 + P_1$. Fresh tests are bounded clipped cutoffs times their Fourier powers. Inversion restores the full kernel before Cauchy, with the mask (B3.34) and all three common zeros outside it.

At fixed t, f, k , a reconstruction is determined by $d' \mid f$, since $s = f/d'$, $C = td'$, $h' = k/d'$. With at most K_0 original slots, its multiplicity is bounded by $d(f)^{1+2K_0} d(t)^{2K_0}$. The distinct t 's are projected from the genuine outer set once; each has $\log_Z q_t \leq P_1 + 2\eta$, so their weighted count is $Z^{P_1+2\eta+o(1)}$. Normalize this measure after extracting that factor. The new actual puncture obeys $Q' \leq P_1 + G + 3\eta$. Direct substitution gives

$$\begin{aligned} F' - M_{\text{init}} - Q' - z_0 &\geq c_1 - 9\eta - \tau_{\text{init}}, \\ 4F' - 3M_{\text{init}} - 6z_0 &\geq c_2 - 22\eta - 3\tau_{\text{init}}, \\ \kappa_c + P_1 + 2F_c &= m. \end{aligned} \quad (\text{B3.38})$$

The first two follow respectively from the exact formal differences $m - r - 2z + 2G$ and $3m - 2r - 8z + 10G + 2P_1$. Apply the canonical estimate with margin $\min(c_1, c_2)/2$, after choosing η, τ_{init} sufficiently small. The last identity and $2\delta_N \leq 6\eta$ give total exponent at most $m + 11\eta + \pi_{\text{init}} + \epsilon_{\text{canonical}}$. Choose these shares below the requested ϵ , then fix raw tail and backwards seminorm orders as in the common-profile lemma. This proves (B3.31), hence MI at $m = 1$. $\square$

Unmarked consequence (UM). Take no slots in (B3.31), $Z = H$, $m = 1$, and $r = \log_H D$. If $H \geq D^{1+\vartheta}$, then $1 - r \geq \vartheta/(1 + \vartheta)$ and $3 - 2r \geq 1$. Bounded scales are direct; otherwise the lengths lie in a fixed bounded range. Thus for every fixed $\vartheta, \epsilon > 0$,

$$\sum_{0 < q_u \leq H} \left| D^{-1/2} \sum_n \mu(n) \nu(n) \chi_n(u)^{\epsilon_x} W(q_n/D) \right|^2 \ll H(HD)^\epsilon, \quad H \geq D^{1+\vartheta}. \quad (\text{B3.39})$$

All element rows and all original zero extensions have survived. The same finite-seminorm and polynomial-height bounds follow from the proof above.

B3.4. The enlarged plain-moment family

The induction requires a larger family than the physical application. Let $\psi_k(n) = \tau(n) \chi_n(k)$, where $0 < q_k \ll Z^m$. The displayed twist τ is a product of a fixed finite ray character and moving residue-symbol factors. The union of the primes in those moving factors, **before** canceling phases or reducing exponents modulo six, has norm at most Z^q . Put $M = m + q$. An additional mask $\mathbf{1}_{(n,R)=1}$, with R squarefree and $q_R \leq Z^B$, may be applied to every plain and prime factor. It is fixed throughout the current row sum, although it may depend on previously frozen labels. A cancellation between τ and $\chi_n(k)$ retains its natural row zero and is never recast as an independently chosen row mask.

For a positive-slot family fix a nonzero integral ray modulus $\mathfrak{M}$, a ray subgroup containing the global units, and one common smooth profile W supported in a fixed compact annulus. Fix an anchor character η_0 and an ideal $Q \leq \mathfrak{M}$. Write $Q_0 = \text{internalQ}(Q, \eta_0)$, and require $Q_0 \neq 0$, $Q_0 \neq \mathcal{O}$, and $Q_0 \subseteq (72)$. Each slot coefficient is the individual relative ray character $\nu_i = \text{relativeCharacter}(\mathfrak{M}, H, \eta_0, \theta_i)$, where θ_i is a character of the fixed ray quotient. With bounded real powers σ_i and height parameters v_i , write

$$Q_{k,i} = Z^{-z_i/2} \sum_{p \in \mathcal{P}_i^{\text{can}}} \psi_k(p) \nu_i(p) W(q_p/Z^{z_i}) (q_p/Z^{z_i})^{-\sigma_i + iv_i}.$$

Here $\mathcal{P}_i^{\text{can}}$ is the genuine canonical good-prime pool at the scale Z^{z_i} . Every natural zero and the same common puncture remain in the summand. Arbitrary prime subsets are allowed in MI, but are not asserted for MP. The physical application has separated annular scales. There is no assertion here for arbitrary slot-specific W_i or for a finite-character-expansion extension.

Write $z = \sum_i z_i$, $A = n_1 + n_2 + z$, and let $S_k(Z^{n_i})$ be the centrally normalized plain sums with two independent smooth profiles. Their norm-twist heights may be independent and have polynomial costs.

Plain-moment theorem. The zero-slot assertion is unconditional. For positive slots assume

$$7/10 \leq \kappa \leq 3/4, \quad \beta_* \geq 51/100, \quad 2\beta_* - 1 \leq \kappa.$$

Then

$$\begin{aligned} z = 0 : \quad & \sum_{\psi_k^* \neq 1} |S_k(Z^{n_1}) S_k(Z^{n_2})|^2 \ll Z^{M+\epsilon} \quad \text{for all bounded } n_1, n_2 \geq 0; \\ z > 0 : \quad & n_1 + n_2 + 6\kappa z \leq M \implies \sum_{k \text{ eligible}} |S_k(Z^{n_1}) S_k(Z^{n_2}) \prod_i Q_{k,i}|^2 \ll Z^{M+\epsilon}. \end{aligned} \quad (\text{B3.40})$$

The positive assertion is the original-family `Cycle25.Energy.terminal_positive_at`, with `PositiveAt` defined in `OAI.NumberTheory.DirichletL.Energy.Bands`. Its state is `CenteredMomentEnergyState`.

NaturalState: its width is the row width plus the character width; the presented character modulus has norm at most Z^q ; and the common nonzero puncture has norm at most Z^B . Its radial weight and every original coefficient zero are retained. The kept rows exclude **all FixedInducingRow** presentations supported by the fixed modulus and common mask. This is an explicit eligibility condition, not an identification with $\psi_k^* \notin \Theta$ for an arbitrary finite group.

For each requested loss and bounded width, puncture and length ranges, the common mesh is chosen before the arithmetic family, slot count, profiles and target height, uniformly for $\kappa \in [7/10, 3/4]$. After fixing a finite slot index type and the common W , one finite seminorm set and one height order precede the fixed character and fixed ideal. Their constant and eventual threshold precede the live slot subset, slot characters, lengths, real powers, heights, radial state and the two plain profiles. For the positive theorem a common whole-product norm power is allowed explicitly; independent twists of the two plain profiles are controlled by their seminorms. The zero-slot theorem separately retains independent plain heights, unrestricted bounded lengths, and all nonprincipal inducing characters, including fixed-conductor rows.

Fixed conductor constants are absorbed by an arbitrarily small positive width padding before the eventual threshold, and that padding is charged to the requested loss. Moving support remains part of the character width; it is not absorbed into a profile seminorm.

The physical translation is the proved adapter `OAI.SevenEighths.Cycle25WeightedNumeratorPhysical.finite_physical_pair_positiveAt` in `Cycle25.Energy.Physical.PhysicalMomentInput`. It constructs the original `initialState`, identifies the exact coefficients, and uses `weighted_source_keep_iff` and `weighted_source_exceptional_norm` from `Cycle25.Energy.Physical.SourceState`. The exceptional sixth-power-free physical rows have a fixed norm bound and disappear from the growing lower row range. In that adapter $W = \overline{F.W}$, each ray-quotient character θ_i is one, so the relative slot coefficient retains η^{-1} , the real power is $33/50$, and the slot height is $-\Im z_i$. Thus the physical application uses precisely the proved common-profile endpoint.

The zero-slot endpoint is `ZetaZeroFree.Analytic.Moments.Unconditional.all_nonprincipal_at`. Fixed nonprincipal rows are restored by masked lattice cancellation. The common moving-support norm is counted before cancellation, the puncture is excluded from the width, and independent profile and height controls remain explicit. For the child analysis, fix a finite conjugation-stable group Θ of characters supported on S , containing the fixed phases, their inverses and $n \mapsto \chi_n(-1)$. This group is bookkeeping for the exceptional child terms; it is not substituted for the original positive-slot eligibility condition. Taking $m = 1, q = 0, n_1 = n_2$ gives MP in that positive family and MP0 in the full nonprincipal zero-slot family.

Removing fixed extra masks. Call a presentation natural if its only zeros are those from its primitive conductor, redundant displayed moving factors and row radical, and S . Let ψ_k^0 be obtained by deleting the additional fixed R -mask. Multiplicativity gives the exact expansions

$$\begin{aligned} S_{k,R}(X) &= \sum_{d|R} \frac{\mu(d)\psi_k^0(d)}{\sqrt{qd}} S_k^0(X/q_d), \\ Q_{k,i,R} &= Q_{k,i}^0 - Z^{-z_i/2} \sum_{p|R} \psi_k^0(p) \nu_i(p) \\ &\quad \times W(q_p/Z^{z_i})(q_p/Z^{z_i})^{-\sigma_i + i\nu_i}. \end{aligned} \tag{B3.41}$$

The total absolute coefficient mass is $Z^{o(1)}$, because $\prod_{p|R} (1 - q_p^{-1/2})^{-1} \ll_\epsilon Z^\epsilon$; on a frozen prime's annulus $Z^{-z_i/2} \ll q_p^{-1/2}$. Apply both identities simultaneously and use Minkowski. The remaining affine length $A + (6\kappa - 1)z$ only decreases. If all slots disappear, the zero-slot rows include the original eligible rows. Nonempty subunit plain scales are clipped to one by bounded dilation and still do not exceed their old nonnegative lengths. Thus it suffices to prove the natural assertion uniformly, keeping this deletion available at each child.

Reflection of a whole plain factor. Let ψ_k^* induce ψ_k^0 , let $R_{0,k}$ be its redundant natural radical, and let $C_k = 3Q_k/(2\pi)^2$, where Q_k is its primitive conductor norm. Each good prime

occurs once in either that conductor or the redundant radical, so

$$C_k q_{R_{0,k}} \ll Z^M. \quad (\text{B3.42})$$

Unit and S -supported row factors, with valuations reduced modulo six, give only finitely many fixed ray sectors. For a nonprincipal inducing character, the ordinary Hecke functional equation [GZ, equation (1.1)] and Mellin inversion give the centrally normalized primitive reflection at scale C_k/X , with root number of modulus one and profile

$$\widehat{W}^\#(s) = \widehat{W}(1-s) \frac{\Gamma(s)}{\Gamma(1-s)}. \quad (\text{B3.43})$$

This identity follows by substituting the functional equation in the Mellin integral and replacing s by $1-s$. There is no L -pole. Moving the inverse Mellin contour left shows that $W^\#$ and its Euler derivatives are bounded at zero; moving it right gives arbitrary decay at infinity. Stirling's formula and $\widehat{W}y^{it}(s) = \widehat{W}(s+it)$ give finite-seminorm and polynomial-height bounds.

Delete the redundant Euler factors before reflection and restore them geometrically afterward. This gives the exact natural-character identity

$$S_k^0(X; W) = \varepsilon_k \sum_{d_0 | R_{0,k}} \sum_{\text{rad}(h_0) | R_{0,k}} \frac{\mu(d_0) \psi_k^*(d_0) \overline{\psi_k^*(h_0)}}{\sqrt{q_{d_0} q_{h_0}}} S_{\psi_k^0} \left(\frac{C_k q_{d_0}}{X q_{h_0}}; W^\# \right). \quad (\text{B3.44})$$

Its coefficient mass is $Z^{o(1)}$. Decompose $W^\#$ into annuli. Lower annuli carry summable central coefficients $O(2^{-j/2})$; upper annuli have arbitrary summable decay. Discard the upper tail after relative enlargement $Z^{\xi/2}$. A main scale $Y < Z^{-\xi}$ is also negligible: absolute counting and rapid decay give $O_B(Y^{B-1/2})$ for its normalized sum, for arbitrarily large fixed B . Row-dependent coefficients and retained scales are handled first by their absolute mass and then by a rowwise scale supremum. The latter is bounded by integrating the function and its logarithmic scale derivative over $O(\log Z)$ unit boxes; for two scales use the two-variable version. The derivative replaces a profile by $-W/2 - yW'$, so the profile lemma applies. Conjugating a whole reflected factor inside a product's absolute value restores the original row character. By (B3.42), every retained nonempty reflected scale has declared length

$$n_{\text{ref}} \leq M - n + \xi. \quad (\text{B3.45})$$

All fixed endpoint and scale-box factors are included by increasing the final Z -threshold. A negative right side means the retained range is empty. Reflecting each plain longer than $M/2$ once reduces the zero-slot problem to the padded core

$$n_i \leq M/2 + \xi, \quad A \leq M + 2\xi \leq M + \delta. \quad (\text{B3.46})$$

Prime cost. On the original eligible rows, each individual slot character is nonprincipal. The hypothesis $2\beta_* - 1 \leq \kappa$ permits moving the smooth von Mangoldt Mellin contour to $s_\kappa + e$, $s_\kappa = (1 + \kappa)/2$. Here $|L'/L(s_\kappa + e + it)| \ll_e \log(Q(3 + |t|)^2)$: take the holomorphic Euler branch of $g = \log L$ on the disk centered at $2 + it$ with radius $2 - s_\kappa - e/4$. It is zero free. Polynomial strip growth from the ordinary functional equation, Stirling and the strip principle bounds $\Re g$ by $O(\log(Q(3 + |t|)^2))$, while $g(2 + it) = O(1)$ by the Euler product. Borel–Carathéodory on a slightly smaller disk and Cauchy's derivative estimate on a further disk containing $s_\kappa + e + it$ give the claim. All gaps are fixed multiples of e . Prime powers of degree at least two cost $O(P^{1/2+\epsilon})$. Dividing the annular weight by $\log(Py)$ removes the von Mangoldt factor. Redundant natural zeros delete only $O(\log Z)$ primes. After central normalization and multiplication over the fixed slot list,

$$\left| \prod_i Q_{k,i} \right|^2 \ll Z^{\kappa z + \epsilon_1} p_J(W)^b \langle \mathbf{t} \rangle^h. \quad (\text{B3.47})$$

The contour displacement costs $2ez$, depending on total length rather than slot count; height twists merely translate rapidly decreasing Mellin transforms. Thus the exponent is independent of the required derivative and height orders.

B3.5. Induction order and the centered comparison

Choose a width floor $\rho > 0$, step $\sigma > 0$, and padding $\delta \ll \min(\rho, \sigma)$. Divide the bounded M -range into bands of length $\sigma/4$. Prove all zero-slot bands in increasing width, then all positive-slot bands. Within each band prove the uncentered range $A \leq 5M/6$ first, then the remaining allowed range using a centered difference. Reflection then supplies unrestricted zero-slot lengths. At $M \leq \rho$, absolute counting gives the padded core with excess at most $\rho + \delta$. The positive-slot case follows from the completed zero-slot estimate and (B3.47), since

$$z \leq M/(6\kappa), \quad \kappa z \leq M/6. \quad (\text{B3.48})$$

Its additional terminal excess is at most $\rho/6$.

For $A > 5M/6$, set $L = M/4$. If an original plain has length $b < L$, reflect the other. Otherwise compare with plain lengths $L, A - z - L$ and the same profiles, reflecting the longer comparison factor only when estimating that comparison separately. In the positive-slot range, with $q_\kappa = 6\kappa - 1 > 2$, $z < (M/6)/q_\kappa < M/12$; hence both comparison lengths are at least L . Whole-factor reflection gives

$$\begin{aligned} A_{\text{comp}} &\leq 3M/2 - A + 2z + \xi, \\ \min\{5M/6 - A_{\text{comp}}, M - A_{\text{comp}} - q_\kappa z\} &\geq \mathfrak{m}(7/10)M - \xi, \\ \mathfrak{m}(\kappa) &= \frac{2\kappa - 1}{2(6\kappa - 1)}. \end{aligned} \quad (\text{B3.49})$$

For example, write $t = M - A < M/6$ and use $z \leq t/q_\kappa$: the first margin is at least $M/3 - t(1 + 2/q_\kappa) - \xi$, and the second at least $M/2 - t(2 + 2/q_\kappa) - \xi$. Both have the displayed lower bound. For zero slots each comparison has margin at least $M/6 - \xi$ to the uncentered boundary. Choose $\xi < \rho \mathfrak{m}(7/10)/2$. Every comparison then calls the already proved uncentered stage in its own band.

It remains to treat originals with both plain lengths at least L . Put $X_i = Z^{n_i}$, $Y_1 = Z^L$, $Y_2 = X_1 X_2 / Y_1$. Subtract the unreflected comparison, keeping the two equal-product rectangles together. After plain powers b_i are extracted, its coefficient is

$$D_b(l_1, l_2) = W_1(q_{b_1 l_1} / X_1) W_2(q_{b_2 l_2} / X_2) - W_1(q_{b_1 l_1} / Y_1) W_2(q_{b_2 l_2} / Y_2), \quad X_1 X_2 = Y_1 Y_2. \quad (\text{B3.50})$$

The same b_i is used in both terms even if an allocation makes one term zero. The comparison has already been bounded on the permissible original rows. Enlarge the nonnegative norm of the **whole difference** to all rows of a smooth ball. In the uncentered range enlarge the original squared norm instead. No Poisson formula is applied to a character-eligibility indicator.

The invariant coefficient carried through the transformations is

$$\prod_{i \in I'} \nu_i(p_i) W(q_{p_i} / Z^{z_i}) (q_{p_i} / Z^{z_i})^{-\sigma_i + i v_i} D_b(l_1, l_2) \tau_1(u) q_u^{it} \mathbf{1}_{(u, R)=1} \mathbf{1}_{s|u}, \quad u = \left(\prod_{i \in I'} p_i \right) l_1 l_2, \quad (\text{B3.51})$$

where s is squarefree and fixed. For an uncentered input omit the second rectangle. The character, mask and norm power act on the whole product; this is valid even when its factors share primes, because all zero powers retain their zeros.

Use one aggregate support tolerance. For the eventual fixed slot count N , choose H_N containing the sum of absolute logarithmic endpoints of all its slot windows, the two plain windows and the finitely many later support enlargements. Put $\theta_N = H_N / \log Z$. Every slot subset has total logarithmic ratio error at most θ_N . Extracted plain powers always use exact norms. Thus one stage's boundary errors are $O(\theta_N)$, with a numerical coefficient independent of N ; there is no tolerance charged once per slot.

B3.6. First Poisson: from plain products to Gauss polynomials

Set $H = Z^m$, $X = Z^A$. At frequency zero the mean of the row character can be nonzero only when all net prime exponents are zero modulo six. No prime then has total multiplicity one in the product of the two full column products, so that product is powerful. A powerful ideal is a square times a cube of a squarefree ideal; summing over the latter counts $O(Y^{1/2+\epsilon})$ powerful ideals up to norm Y . With $Y \asymp X^2$, divisor-bounded allocations, central factor X^{-1} , and row mean at most H , the zero term is $O(Z^{m+o(1)})$.

For nonzero frequencies write the two full products as Ca, Db , where C, D contain their **complete** common prime support. Thus $(a, b) = (ab, CD) = 1$. At every common prime fix its full valuation in each plain variable and its occurrence in a slot. Extract all those powers, freeze any slot supplying the prime, and puncture every remaining factor there. The same allocation acts on both rectangles. A scalar zero forced by an old character or mask remains zero before any enlargement.

Let c, d, p be the logarithmic norms of $C, D, \text{rad}(CD)$. Let r contain common primes of nonzero net exponent modulo six, let $R = \log_Z q_r$, and let ξ_r be their primitive character. The remaining common row mask introduces a divisor e , $E = \log_Z q_e \leq p - R$. If A_C, A_D are the allocated convolution coefficients, omitting the whole-product twist, put $\tau_C(a) = \tau(a)\chi_a(er)\overline{\xi_r(a)}$ and similarly for τ_D . Residue-class Poisson and CRT give, up to a bounded frozen scalar,

$$\frac{H}{Xq_e\sqrt{q_r}} \sum_{h \neq 0} \mathcal{G}_{\xi_r}(r, h) \sum_{(a,b)=1} \frac{A_C(a)\overline{A_D(b)}}{\sqrt{q_a q_b}} \tau_C(a)\overline{\tau_D(b)} R(a, b) \mathcal{G}(a, h) \overline{\mathcal{G}(b, -h)} K_1 \left(\frac{Hq_h}{q_e q_r q_a q_b} \right). \quad (\text{B3.52})$$

Here $\mathcal{G}$ is the normalized two-variable Gauss sum of (B1.12); $|\mathcal{G}_{\xi_r}| \leq 1$. To check (B3.52), Poisson modulo rab after $k = ek'$ contributes $H/(q_e q_r q_a q_b)$; the three unnormalized Gauss sums restore $\sqrt{q_r q_a q_b}$. CRT contributes $\chi_a(br)\overline{\chi_b(ar)}\overline{\xi_r(ab)}$, and reciprocity replaces the a, b cross ratio by $R(a, b)$. These are exactly the displayed factors. Every new moving prime occurs in the whole-product character and punctures all residual factors, including when phases cancel. Its full support length is bounded by

$$\tilde{q} = q + R + E, \quad K_0 = 2A - c - d + R + E - m. \quad (\text{B3.53})$$

The first kernel's nominal frequency scale is Z^{K_0} .

To localize both Poisson formulas used here, choose one nonnegative dyadic partition of frequency norms with fixed relative supports. In a sector of aggregate outer norms let T_{sec} be the maximum nominal frequency scale. Retain whole partition weights meeting $q_h \leq T_{\text{sec}} Z^{\xi/2}$. Their union has upper length $K \leq K_0 + \delta_1$, $0 \leq \delta_1 < \xi$, after Z exceeds a fixed endpoint threshold. Each discarded weight has actual kernel argument at least $Z^{\xi/4}$, because the full column-product support costs at most e^{2H_N} . Formula (B3.4), $|\mathcal{G}(a, h)| \leq \sqrt{q_a}$, and the finite bound $|F(u, v; j)| \leq q_u q_v$ justify arbitrary tail savings in the two formulas respectively. The original columns have bounded polynomial norms and divisor-bounded coefficients, so these raw bounds cost a fixed power before the decay order is chosen. Retained weights and the full kernel are common to both rectangles and independent of live columns.

On each retained dyad define the displayed bridge off $(a, b) = 1$ by its separate Gauss sums, fixed bicharacter, masks and formal norms. It is an artificial coefficient there, not a formula for the original Gauss correlation. Insert the **complete** projector $\sum_{s|(a,b)} \mu(s)$ before taking absolute values. Let $s_0 = \log_Z q_s$. Separate the kernel and inverse roots in the normalized row and two whole-product norms by (B3.3), adding one aggregate outer-ratio coordinate if necessary. This introduces one norm power on each entire column, never different powers on its two plains or rectangles. Only now use Cauchy–Schwarz and $|\mathcal{G}_{\xi_r}| \leq 1$. The positive C-side norm is

$$\mathcal{N}_C = \sum_{h \neq 0} \omega_K(q_h/Z^K) \left| Z^{-(A-c)/2} \sum_{s|a} A_C(a) \tau'_C(a) q_a^{it} \mathcal{G}(a, h) \right|^2. \quad (\text{B3.54})$$

It has coefficient (B3.51), and similarly for D. The outside exponent is $m - A - R/2 - E$. The common-support and Möbius labels cost $Z^{p+s_0+o(1)}$: count the radical first, then its bounded-range powers by $\#\{v : \text{rad } v \mid c, q_v \leq Z^T\} \leq Z^{aT} \prod_{p|c} (1 - q_p^{-a})^{-1}$, choosing $a > 0$ small; all allocations are divisor bounded.

It suffices to prove

$$\mathcal{N}_C \ll Z^{A-c+\tilde{q}+B_c-s_0+\epsilon_G}, \quad B_c = (3c - 5d - R)_+/6, \quad (\text{B3.55})$$

and its exchanged counterpart. Indeed their geometric mean, with the outside factors, has exponent $M + \{B_c + B_d - (c + d - 2p - R)\}/2$. Primewise,

$$B_c + B_d \leq c + d - 2p - R. \quad (\text{B3.56})$$

For multiplicities $i \geq j \geq 1$ and $r_p = \mathbf{1}_{6 \nmid i-j}$, only the larger side can have a positive local numerator. Six times the difference between the right side and $(3i - 5j - r_p)_+/6$, when the latter is positive, is $3i + 11j - 12 - 5r_p \geq 0$; otherwise the right side is already nonnegative. Positive parts of sums are bounded by sums of positive parts. This proves (B3.56).

B3.7. Enlarging a Gauss-row norm

Let w be a column length extracted from (B3.54), and $w_o \leq w$ the additional moving-radical length, whose squared scalar coefficient is Z^{-w_o} . Put $a_0 = A - c - w$. The allowance (B3.55) for the unweighted remaining norm is

$$\Lambda_c = a_0 + \tilde{q} + w_o + w + B_c - s_0 + \epsilon_G. \quad (\text{B3.57})$$

Define $J = d - c + (K_0 - K) - 2w + w_o$, $J_+ = \max(J, 0)$. With zero slots take $w = w_o = 0$, $\ell = 0$, and enlarge the row norm to a smooth ball of length $K + g$, $g = J_+ + \sigma$.

With positive slots, put $\ell_* = \sigma/3$, choose the mesh below $\sigma/6$, and use the fixed pool $\mathcal{P} = \{p : Z^{\ell_*}/2 < q_p \leq Z^{\ell_*}, p \notin S\}$. The fixed-field prime ideal theorem [TZ, Theorem 1.1, specialized to the identity extension of $\mathbb{K}$] gives $|\mathcal{P}| = Z^{\ell_*+o(1)}$. The pool is disjoint from every live slot window once Z is large. For each $h \neq 0$, omit primes dividing h, s or frozen supports; only $O(\log Z)$ are omitted.

For $p \nmid uh$, the prime-power Gauss identities (B1.13) and CRT give

$$\mathcal{G}(p^i u, h) = R(p, u)^i \chi_p(u)^{2i} \mathcal{G}(p^i, h) \mathcal{G}(u, h), \quad \mathcal{G}(u, hp^6) = \mathcal{G}(u, h). \quad (\text{B3.58})$$

Replacing h by hp^6 changes only column valuations $i = 1, 6, 7$. Allocate their powers to the two plain variables, since the pool is disjoint from the slots, and puncture the residual product at p . The same allocation updates both rectangles. Jensen in the eligible primes and this finite allocation list bounds the original squared value by the average of the amplified squared value and these extracted squared errors. A fixed output hp^6 has boundedly many representations, since all pool primes have norm $\asymp Z^{\ell_*}$ and the output norm has bounded logarithmic length. Thus the amplified norm gets one density factor $Z^{-\ell_*+o(1)}$. For it take $\ell = \ell_*$, $g = J_+ + 2\sigma$, which contains the new rows because $6\ell_* = 2\sigma$.

The extracted alternatives are exactly:

Column valuation i	Squared extraction		w	w_o
		coefficient		
1		q_p^{-1}	ℓ_p	ℓ_p
6		$(1 - q_p^{-1})^2$	$6\ell_p$	0
7		q_p^{-1}	$7\ell_p$	ℓ_p

Here $\ell_p = \log_Z q_p \in [\sigma/6, \sigma/3]$. At valuation six the exact zero power $\chi_p(u)^{12}$ is its already common puncture; retain the fixed phase $R(p, u)^6$. At valuations one and seven the local character remains displayed and its prime is counted in w_o . Freeze p for each error, set $\ell = 0, g = J_+ + \sigma$, and apply the second transform directly, with no further amplification. The normalized average $|\mathcal{P}|^{-1} \sum_p$ costs no second prime count. Always

$$0 \leq w_o \leq w \leq 7\sigma/3. \quad (\text{B3.59})$$

The coefficient remains (B3.51) times $\mathcal{G}(u, h)$, including its whole difference, common puncture and s -divisibility.

Add row zero only when passing to the final smooth ball. Formula (B1.13) gives $\mathcal{G}(u, 0) = 0$ unless u is a sixth power and $|\mathcal{G}(u, 0)| \leq \sqrt{q_u}$. If $Y = e^{H_N} Z^{a_0}$ is a nonempty column upper bound and $s \mid u$, then $q_s^6 \leq Y$. Counting $O(Y^{1/6})$ sixth powers bounds the normalized squared zero row by $Z^{-a_0} Y^{4/3+o(1)} \leq Z^{a_0-s_0+2\theta_N+o(1)}$, within (B3.57). Zero is never amplified.

B3.8. Second transform and the actual children

Restore live slot sums before squaring. If $B(u)$ is the complete convolution coefficient, with $s \mid u$, (B1.14) gives

$$\begin{aligned} & Z^{-\ell} \sum_h \Phi_2(q_h/Z^{K+g}) \left| Z^{-a_0/2} \sum_u B(u) \tau_1(u) q_u^{it} \mathcal{G}(u, h) \right|^2 \\ &= Z^{K+g-\ell-a_0} \sum_{u,v} \frac{B(u) \overline{B(v)} \tau_1(u) \overline{\tau_1(v)} q_u^{it} q_v^{-it}}{\sqrt{q_u q_v}} \\ &\quad \times \sum_j F(u, v; j) K_2 \left(\frac{Z^{K+g} q_j}{q_u q_v} \right). \end{aligned} \quad (\text{B3.60})$$

At $j = 0$, (B1.16) leaves $u = v$, $F(u, u; 0) = \phi(u) \leq q_u$. There are $O(Z^{a_0-s_0+\theta_N})$ supported multiples of s , so its exponent is $K + g - \ell - s_0 + \theta_N + o(1)$. Its nominal excess above (B3.57) is

$$(A - M) - d - (K_0 - K) - w_o - B_c + g - \ell \leq A - M + 5\sigma/3 + \delta_1. \quad (\text{B3.61})$$

For the inequality use J_+ : when $J \geq 0$, the part $-d - (K_0 - K) - w_o + J_+$ equals $-c - 2w$; otherwise it is at most δ_1 . The largest remaining increment is $2\sigma - \ell_* = 5\sigma/3$. This is a terminal loss, with an additional δ in the padded zero-slot core.

Localize the genuine nonzero j -sum at its nominal scale Z^{2a_0-K-g} by the whole-dyad rule of §B3.6, before extracting common support. Now write $u = D_2 a, v = E_2 b$, extracting their complete common support. Put

$$\begin{aligned} c_2 &= \log_Z q_{D_2}, \\ d_2 &= \log_Z q_{E_2}, \\ b_2 &= (c_2 + d_2)/2, \\ p_2 &= \log_Z q_{\text{rad}(D_2 E_2)}, \\ G_c &= (D_2, E_2), \\ g_2 &= \log_Z q_{G_c}. \end{aligned}$$

Use the exact complete-support factorization (B1.19)

$$F(D_2 a, E_2 b; j) = F(D_2, E_2; j) R(a, E_2) R(b, D_2) R(a, b) \chi_a(j) \overline{\chi_b(-j)} \quad (\text{B3.62})$$

only on $(a, b) = (ab, D_2 E_2) = 1$. It permits arbitrary powers and all slot/plain coincidences. All extracted primes puncture all remaining factors, and any slot supplying them is frozen. In particular a plain p^5 and slot p on one side leave the zero-extended factor $\chi_p(j)^6$, not the constant one.

The local correlation table (B1.17) implies $G_c \mid j$. At a common prime of equal multiplicity $i \not\equiv 0 \pmod{6}$, split according as j/G_c is a unit or nonunit. Let t_2 be the radical length of the unit primes and V that of the nonunit primes; let V_{id} be the latter radical. Write $j = G_c V_{\text{id}} h'$. The partitioned scalar $\mathbf{1}_{\text{part}} F(D_2, E_2; G_c V_{\text{id}} h') / Z^{g_2 - t_2}$, extended by zero off its partition, has modulus at most one. The necessary local bounds are

Common multiplicities and divided frequency	Bound
$i = i, 6 \nmid i, \text{ unit}$	P^{i-1}
$i = i, 6 \nmid i, \text{ nonunit}$	P^i
$i = i, 6 \mid i, \text{ either}$	P^i
$i > j_0, 6 \mid j_0, \text{ unit}$	P^{j_0}

Every other unequal case is zero. At the fixed factor $v = G_c V_{\text{id}}$, write each valuation modulo six. A nonzero residue retains its moving character; a zero residue is exactly the already present common puncture. Active fixed primes are contained in those counted by $t_2 + V$. This operation factors only $\chi_n(v)$; it keeps the natural varying-row zeros of $\chi_n(h')$. Hence the nominal child widths are

$$m' = 2a_0 - K - g - g_2 - V, \quad q' = \tilde{q} + w_o + t_2 + V, \quad M' = m' + q' = M + J - g - g_2 + t_2 \leq M - \sigma. \quad (\text{B3.63})$$

The actual retained row weight, pulled back under $j = G_c V_{\text{id}} h'$, lies in $q_{h'} \leq Z^{m' + \delta_2}$, $0 \leq \delta_2 < \xi$. A nonempty range has $m' + \delta_2 \geq 0$; declare $m'_{\text{act}} = m' + \delta_2$, $M'_{\text{act}} = M' + \delta_2$. Extend the pulled-back weight and partitioned scalar by zero on the rest of this ball. They are still part of the signed expression.

The condition $s \mid u, v$ forces $s \mid \text{rad}(D_2 E_2)$. Thus common-support labels cost $Z^{p_2 - s_0 + o(1)}$, not Z^{p_2} . The second-transform exponent ledger is

Factor	Exponent
Original squared central normalization	$-a_0$
Poisson factor and inverse roots	$K + g - a_0$
Amplifier density	$-\ell$
Common-support count with fixed s	$p_2 - s_0$
Common correlation	$g_2 - t_2$
Conversion to normalized residual products	$a_0 - b_2$

Subtracting this sum from (B3.57) leaves the inner allowance

$$M' + \Delta, \quad \Delta = b_2 - p_2 + w + B_c + \ell \geq w + \ell. \quad (\text{B3.64})$$

The last inequality uses $b_2 \geq p_2$.

Before separating child factors, define the right side of (B3.62) artificially on all individually allowed residual pairs, keeping the original common-support masks, formal full-product norms, inverse roots, kernel and pulled-back row weight. Insert the full residual coprimality projector $\sum_{t \mid (a,b)} \mu(t)$. Only its complete sum recovers the genuine correlation; the extra common primes in an individual t -term are not added to D_2, E_2 or its frequency restrictions. Separate the full kernel and roots in the row and two whole-product norms as before. Then allocate each prime of t with

$$\mathbf{1}_{p \mid \prod_i n_i} = \sum_{\emptyset \neq J} (-1)^{|J|+1} \prod_{i \in J} \mathbf{1}_{p \mid n_i}, \quad (\text{B3.65})$$

where the factors are the two plains and live slots. A selected plain is $n_i = p l_i$ with no restriction on l_i ; divide both rectangle scales for it by q_p . A selected slot is frozen. Selecting two different

primes in one prime slot is zero. Quotients may retain p , and no new one-variable puncture is inserted.

This proves the needed coefficient invariance: each separated child side has one whole-product character, common mask and norm power in both plains and both rectangles; surviving slots retain their original ν_i , without Gauss factors. The two sides may have different whole-product norm powers. Their fixed-inducing predicates agree under the paired reflection, with the same fixed modulus and retained natural masks; this is `Family.paired_inducing_iff` in `OAI.NumberTheory.DirichletL.Moments.SecondExceptionalFamily`, under its fixed-modulus condition $Q \leq (72)$. At each fixed-modulus stage enlarge Θ to contain every primitive character whose conductor divides that modulus and the finitely many fixed ray twists used here. This is still a finite group. On an exceptional row with primitive inducing character ϑ , fixing the live slots leaves precisely

$$\vartheta(l_1)\vartheta(l_2)\mathbf{1}_{(l_1l_2, R_*)=1}q_{l_1l_2}^{it}D_b(l_1, l_2) \quad (\text{B3.66})$$

for one squarefree polynomial-size R_* , common to the two variables and rectangles. Every extracted row scalar, including zero scalars, is retained until the following positive or absolute estimate.

Exact extraction normalization. Put $\alpha_1 = a_0 - c_2$, $\alpha_2 = a_0 - d_2$. For side j , let F_j be the entrance slots frozen by the genuine extractions, and I_j the live slots before (B3.65). If T_j is the common product of its two pre-allocation plain scales, then $T_j \prod_{i \in I_j} Z^{z_i} = Z^{\alpha_j + e_j}$, where $e_j = \sum_{i \in F_j} \log_Z(q_{p_i}/Z^{z_i})$. For the final t -allocation, let $d_{j,i}$ be the product selected in plain i , and $J_j \subset I_j$ the newly frozen slots. Put

$$a_{j,i} = \log_Z q_{d_{j,i}}, \quad r_j = a_{j,1} + a_{j,2} + \sum_{i \in J_j} z_i, \quad \omega_j = \sum_{i \in J_j} \log_Z(q_{p_i}/Z^{z_i}).$$

Use lower-endpoint dyads $Z^{t-} \leq q_t < 2Z^{t-}$. Every selected product is divisible by t , so

$$r_j + \omega_j \geq t_-, \quad |\omega_j| \leq \theta_N, \quad |e_j + \omega_j| \leq \theta_N. \quad (\text{B3.67})$$

The last bound uses disjoint subsets of the current entrance slots. Ancestor slots are not counted again. The raw child normalizer is $Z^{-(\alpha_j - r_j)/2}$, and its exact extraction scalar relative to $Z^{-\alpha_j/2}$ is $Z^{-r_j/2}$. The formal post product of scales is $Z^{\alpha_j + e_j - r_j}$. These signed raw rectangles need not have nonnegative individual lengths.

Exact child eligibility. Split each actual child by the literal `FixedInducingRow` predicate, retaining its fixed modulus and source masks. For the retained family, `Family.nonexceptional_iff` in `OAI.NumberTheory.DirichletL.Moments.SecondExceptionalFamily` identifies the nonexceptional part with

$$z \neq 0 \quad \text{and} \quad \neg \text{FixedInducingRow}(\tau_\chi, Q, \text{fixedBadMask}, 1, z).$$

The actual radial norm is the canonical radial norm by `sourceRadial_eq_canonical` in `OAI.NumberTheory.DirichletL.Moments.SecondNonexceptionalChosenBlock`. In `OAI.NumberTheory.DirichletL.Energy.ChildState`, `canonical_mask_nonexceptional` restores the common nonzero puncture without changing the exclusion, and `canonicalState` supplies the required natural state. The smaller-width moment estimates therefore apply to precisely these rows: `actual_canonical_child_bands` in `OAI.NumberTheory.DirichletL.Energy.CanonicalChildBound` is the proved source adapter. The finite group Θ records the exceptional characters for lattice cancellation; membership in that group is not used in place of the child's eligibility predicate.

Nonexceptional children. On this exact nonexceptional part, use Cauchy to form positive child norms, then split a centered child into its two rectangles. Only now bound the common row weights and extracted scalars by one and enlarge to the declared ball. For a retained rectangle with pre-allocation plain log scales x_i , post scales $y_i = x_i - a_{j,i}$, set $\pi_0 = \sum_i (-x_i)_+$, $\pi = \sum_i (-y_i)_+$. A rectangle with a plain upper support endpoint below the first ideal is identically zero and is omitted; otherwise $0 \leq \pi_0 \leq \pi \leq \theta_N$. Its clipped reduction and total are

$$r_{\text{clip},j} = r_j - (\pi - \pi_0) \geq 0, \quad A_{\text{clip},j} = \alpha_j + e_j - r_j + \pi, \quad (\text{B3.68})$$

and its exact scalar relative to the pre-normalizer is $Z^{(-r_{\text{clip},j}+e_j+\pi_0)/2}$. For a paired divisor dyad, the count and both scalars satisfy

$$t_- - \frac{r_{\text{clip},1} + r_{\text{clip},2}}{2} + \frac{e_1 + \pi_{0,1} + e_2 + \pi_{0,2}}{2} \leq 2\theta_N. \quad (\text{B3.69})$$

Substitute (B3.68) and use (B3.67); there are only four rectangle pairs. No common clipped reduction is assigned to a signed difference.

When no slot survives, (B3.69) and the already completed smaller-width zero-slot assertion fit (B3.64), since $\Delta \geq 0$. For positive slots, before the final allocation the child total on side one is $A' = A - c - w - c_2$. Direct substitution in (B3.63) gives

$$(A' - M') - (A - M) = g + w - d - c_2 - (K_0 - K) - w_o + g_2 - t_2 \leq 6(w + \ell) + \delta_1. \quad (\text{B3.70})$$

Use $g_2 - t_2 \leq c_2$. In the amplified main norm $g \leq d + (K_0 - K) + 6\ell + \delta_1$; in an error $g \leq d + (K_0 - K) + \sigma + \delta_1$, and the three rows of the extraction table give $w - w_o + \sigma \leq 6w$. The other side is identical with c_2, d_2 exchanged.

Delete the extra mask by (B3.41). For each resulting actual natural child with surviving slots, let $F_{\text{act}} = (A_{\text{act}} - M'_{\text{act}} + (6\kappa - 1)z_{\text{act}})_+$. Equations (B3.68)–(B3.70), the parent affine condition, and the aggregate endpoint bound imply

$$F_{\text{act}} \leq 6(w + \ell) + \delta_1 + 2\theta_N. \quad (\text{B3.71})$$

Remove whole slots using (B3.47) until the affine defect disappears or no slot remains. One slot of length d reduces that defect by $6\kappa d$ and costs κd in squared norm. A single greedy prefix overshoots its required total length by at most one mesh, so

$$\kappa d_z \leq F_{\text{act}}/6 + \kappa\eta_{\text{mesh}} \leq \Delta + \delta_1/6 + \theta_N/3 + \eta_{\text{mesh}}. \quad (\text{B3.72})$$

If slots remain, apply the smaller-width positive-slot assertion. Otherwise apply the already completed unrestricted zero-slot assertion; its plain lengths may now be arbitrary. Combining (B3.69), (B3.72), and $M'_{\text{act}} - M' = \delta_2$, the strict edge costs at most $\delta_2 + \delta_1/6 + 7\theta_N/3 + \eta_{\text{mesh}}$, besides its allocated small-power share. Cauchy averages the two costs; it does not double the mesh. This handles all nonexceptional children at smaller width.

B3.9. Fixed-character rows and rectangle cancellation

Call a child exceptional if it satisfies the complementary fixed-inducing predicate above. Before taking absolute values, a nonzero genuine second allocation is disjoint from every old moving support: those primes are common column zeros. Thus no old character at such a prime can cancel a factor supplied by $G_c V_{\text{id}}$. Let v_1 be the radical length of nonunit-partition primes whose common multiplicity is one, and set $f = 2v_1$. At each such prime the fixed row factor $G_c V_{\text{id}}$ has valuation two. Its primitive inducing character has conductor dividing the fixed modulus and is therefore unramified at these good primes. Exceptionality forces $v_p(h') \equiv 4 \pmod{6}$. More generally, at every good prime it specifies one residue class modulo six, with residue zero outside the frozen moving supports. Unit and S -valuations contribute finitely many sectors, while their sixth powers remain in the free sixth-power factor. Therefore $(h') = h_0 v^6$ for a fixed sixth-power-free h_0 , and $q_{h_0} \geq Z^{4v_1} = Z^{2f}$. Ideal counting gives the stronger exponent $(m'_{\text{act}} - 2f)/6$; we need only

$$\#\{h' \text{ exceptional} : q_{h'} \ll Z^{m'_{\text{act}}}\} \ll Z^{(m'-f)/6+\delta_2/6+o(1)}. \quad (\text{B3.73})$$

Below the forced initial norm the range is empty, apart from a fixed endpoint factor. This counts the full fixed-inducing part, including the principal character, and also rows admitted when a former partition restriction is bounded by one. The corresponding source estimates are `chosen_exceptional_block` in `OAI.NumberTheory.DirichletL.Moments.SecondExceptionalFixed`

QChosenBlock and original_exceptional_energy in OAI.NumberTheory.DirichletL.Moments.SecondExceptionalFixedQSource. No saving from the unit set t_2 is used.

After the full Möbius allocation, take the pointwise absolute product on these rows. Absolute volume on side j , with its raw normalizer and live slots, is $O(Z^{(\alpha_j - r_j)/2 + e_j})$. Together with both extraction scalars and the divisor count, (B3.67) gives paired volume at most $Z^{a_0 - b_2 + 2\theta_N + o(1)}$; indeed $t_- - r_1 - r_2 \leq \omega_2$, and $r_1 \geq 0$. Its nominal excess over the inner allowance (B3.64), including (B3.73), is exactly

$$\frac{m' - f}{6} + a_0 - b_2 - (M' + \Delta) = A - \frac{5}{6}M - F_1 - F_2, \quad (\text{B3.74})$$

where

$$\begin{aligned} F_1 &= c/6 + 5\{d + (K_0 - K)\}/6 + w/3 + \tilde{q}/6 + w_o + B_c - 5g/6 + \ell, \\ F_2 &= 2b_2 - 5g_2/6 - p_2 + t_2 + V/6 + f/6. \end{aligned} \quad (\text{B3.75})$$

These expressions obey

$$F_1 \geq \frac{2}{3}(c + w) - 3\sigma - 5\delta_1/6, \quad F_2 \geq \frac{2}{3}b_2. \quad (\text{B3.76})$$

For the first, initially take $g = J_+$, $\ell = 0$, and put $D_0 = d + K_0 - K$. If $J \geq 0$, substitution gives $F_1 = c + 2w + \tilde{q}/6 + w_o/6 + B_c$. If $J < 0$, use $B_c + \tilde{q}/6 \geq (3c - 5D_0)_+/6 - 5\delta_1/6$, since $\tilde{q} \geq R$. Hence $F_1 \geq [c + 5D_0 + (3c - 5D_0)_+]/6 + w/3 - 5\delta_1/6 \geq 2(c + w)/3 - w/3 - 5\delta_1/6$. Restoring $g \leq J_+ + 2\sigma$, $\ell \geq 0$, and $w \leq 7\sigma/3$ loses less than 3σ . For the second, the local contributions in units of $\log_Z P$ are:

Common case	F_2	b_2
$i = i, 6 \nmid i$, unit	$7i/6$	i
$i = i, 6 \nmid i$, nonunit	$(7i - 5)/6 + \mathbf{1}_{i=1}/3$	i
$i = i, 6 \mid i$	$7i/6 - 1$	i
$i > j_0, 6 \mid j_0$, unit	$i + j_0/6 - 1$	$(i + j_0)/2$

Each first entry is at least two thirds of the second: the only smallest case is $i = 1$ in the second row, where the $f/6$ contribution makes equality. This proves (B3.76). Consequently (B3.74) is at most $3\sigma + O(\xi + \theta_N)$ when $A \leq 5M/6$. This completes the current uncentered stage and validates its comparisons from §B3.5.

Masked lattice cancellation. For a fixed $\vartheta \in \Theta$, a squarefree mask R_* of polynomial norm, a fixed annular W_i , and real t , put $L_i(X, t) = \sum_l \vartheta(l) \mathbf{1}_{(l, R_*)=1} q_l^{it} W_i(q_l/X)$. Then, uniformly for $X > 0$,

$$L_i(X, t) = c_{\vartheta, R_*} X^{1+it} I_i(t) + O(Z^{\epsilon_1} \langle t \rangle^J), \quad |L_i(X, t)| \ll X, \quad (\text{B3.77})$$

where c_{ϑ, R_*} is independent of X, t , and $I_i(t)$ is the fixed lattice measure constant times $\int_0^\infty W_i(y) y^{it} dy$.

To prove it first omit the mask. The primary ideals outside S , with fixed weight ϑ , form a finite weighted set of classes in a fixed lattice. Poisson for $q_z^{it} W_i(q_z/X) = X^{it} f_t(z/\sqrt{X})$ gives zero-frequency main term $c_\vartheta X^{1+it} I_i(t)$. Integration by parts on the fixed annulus gives $|\hat{f}_t(\xi)| \ll_J \langle t \rangle^J (1 + |\xi|)^{-J}$. For $X \geq 1$, summing nonzero frequencies bounds the error by $O(\langle t \rangle^J)$ for fixed sufficiently large J ; for $0 < X < 1$, both the finite lattice sum and main term are $O(1)$. Insert the mask with $L_i(X, t) = \sum_{d|R_*} \mu(d) \vartheta(d) q_d^{it} L_i^0(X/q_d, t)$, omitting already excluded primes. Its main coefficient is $c_{\vartheta, R_*} = c_\vartheta \sum_{d|R_*} \mu(d) \vartheta(d)/q_d$; the two norm powers of d cancel. The divisor bound gives the stated error. Finally absolute counting on a nonempty annulus is $O(X)$; below its first possible ideal it is empty. This proves (B3.77).

For $U_i = X_i/q_{b_i}$, $V_i = Y_i/q_{b_i}$ and $T = U_1U_2 = V_1V_2$, the sum of (B3.66) is $L_1(U_1, t)L_2(U_2, t) - L_1(V_1, t)L_2(V_2, t)$. Its two main terms agree. If all four scales are at least Z^r , $r \geq 0$, (B3.77) bounds the centrally normalized difference by

$$O(Z^{\epsilon_1} \langle t \rangle^{2J} T^{1/2} Z^{-r}). \quad (\text{B3.78})$$

Each one-error term uses one scale at most TZ^{-r} ; the two-error term fits because $1 \leq TZ^{-r}$. Without a nonnegative lower length the all-scale volume fallback is $O(T^{1/2})$. This is an equality-of-main-terms argument for the same row, mask and norm power; it would fail for independently modified rectangles.

Apply (B3.78) to the centered exceptional children. Initially all four plain lengths are at least L . Before the final t -allocation, the two sides have lower lengths at least $L - c - w - c_2$, $L - c - w - d_2$. Let $r = (L - c - w - \min(c_2, d_2))_+$, and call the side with this saving side one. Its final allocation removes at most r_1 from either plain. If $r_1 < r$, use (B3.78) with saving $r - r_1$; otherwise use volume. Keep the raw equal-product scales, with no clipping of a signed rectangle. The divisor dyad and both reductions give exactly

$$t_- - r_1 - r_2 - (r - r_1)_+ = (t_- - r_2) - \max(r, r_1) \leq \omega_2 - r. \quad (\text{B3.79})$$

Thus the paired absolute volume improves to $Z^{a_0 - b_2 - r + 2\theta_N + o(1)}$, uniformly in the live slots. Their labels were absent from the Fourier density; sum them only now by absolute volume, using the exact normalization (B3.67).

Put $v = c + w + \min(c_2, d_2)$. From (B3.76), $F_1 + F_2 \geq 2v/3 - 3\sigma - 5\delta_1/6$. The remaining centered exceptional deficit is bounded by

$$A - 5M/6 - 2v/3 - (L - v)_+ \leq (A - M)_+, \quad L = M/4. \quad (\text{B3.80})$$

The left side increases for $v \leq L$, decreases for $v \geq L$, and equals $A - M$ at $v = L$. In the positive-slot range $A \leq M$; in the padded zero-slot core $A \leq M + \delta$. The exceptional terminal cost is therefore at most $3\sigma + \delta$, with its numerical support and frequency errors already reserved. This finishes the centered stage.

B3.10. Closure and uniform common mesh

Every strict child has nominal width at most $M - \sigma$ by (B3.63). The actual enclosing width and support corrections cost at most $C_*\xi$, for one numerical C_* independent of the eventual slot count. This follows from the single aggregate θ_N , the two frequency enclosures, the at most two plain clippings on a rectangle, and the fixed list of aggregate lengths in (B3.53)–(B3.80). Every ledger is affine or a positive part of an affine expression in those lengths. In particular the positive-slot boundary changes by at most $|\Delta M| + |\Delta A| + 5|\Delta z|$, because $0 < 6\kappa - 1 \leq 5$. Thus C_* does not grow with the slot count. The one greedy prefix contributes one mesh, not one mesh per removed slot.

Choose ρ, σ, δ from the bounded real ranges and final ϵ , with $\delta \ll \min(\rho, \sigma)$, so that

$$T_{\text{term}} := \rho + \delta + \rho/6 + 3\sigma + 5\sigma/3 < \epsilon/4.$$

Put $D = 2 + \lceil 2M_{\max}/\sigma \rceil$, and, still before specifying the fixed slot system, choose

$$\begin{aligned} \xi &< \min\{\delta/2, \rho\mathbf{m}(7/10)/2, \sigma/(4C_*), \epsilon/(16C_*D)\}, \\ \eta_{\text{mesh}} &< \min\{\sigma/6, \epsilon/(16C_*D)\}, \\ \epsilon_0 &< \epsilon/(16C_*D). \end{aligned} \quad (\text{B3.81})$$

These choices preserve the comparison margin, padded core and strict width decrease of at least $\sigma/2$. Every strict child leaves its width band; there are at most $D - 2$ such calls on a branch. Same-band comparisons use only the preceding uncentered stage. Zero-slot reflection is done at most

once per plain before centering, and amplifier errors are not amplified again. Thus the induction is acyclic.

Now fix the arithmetic data, profiles and any finite slot count. Form H_N and the finite support boxes through depth D ; choose Z eventually so large that $\theta_N \ll \xi$, the whole-dyad tail inclusions hold and the pool avoids all slot windows. In each stage allocate all divisor, ray, logarithmic and prime-density losses within ϵ_0 . For example, mask deletion can assign each of its at most $N + 2$ masses a share $\epsilon_0/(10(N + 2))$; divisor allocations use global bounds $C_N^{\omega(v)}, d_{N+2}(v)^C \ll_{N,C,a} q_v^a$, with one sufficiently small a , rather than a loss per prime. The prime contour displacement costs $2ez$, so it also fits a share independently of the number of slots. These subsidiary choices affect constants, not the previously fixed mesh.

When at most d strict calls remain, permit exponent error $E_d = T_{\text{term}} + (d + 1)C_*(\eta_{\text{mesh}} + \xi + \epsilon_0)$. The width-floor, diagonal and exceptional calculations fit E_0 ; one strict edge uses its child envelope plus at most the displayed stage allowance, by (B3.69)–(B3.72). Same-band comparisons and reflections have bounded coefficient mass included in that allowance. Cauchy averages child errors, and triangle inequalities take the larger one, so the terminal loss occurs once along a branch. Formula (B3.81) gives $E_{D-2} < \epsilon$. This proves all zero-slot bands and then all positive-slot bands simultaneously for every surviving subset of the original slots.

Finally choose raw kernel-tail orders and all finite profile and height orders backwards by the common-profile lemma; then take one final threshold for their tails, endpoint ratios and logarithmic losses. Each child extra mask is supported on frozen columns or one frozen amplifier prime, so its norm remains polynomial through the finite depth. Exceptional-row masks combine these with the row radical; (B3.77) is uniform over them. Full moving supports remain explicit, including canceled zero powers. This proves (B3.40) with its stated uniformity.

For positive slots the comparison margin is bounded below by $\mathfrak{m}(7/10) > 0$, and $6\kappa - 1 \leq 7/2$ throughout the proved range. The cost-to-capacity ratio is $(\kappa d)/(6\kappa d) = 1/6$. The common mesh therefore works for $7/10 \leq \kappa \leq 3/4$, with the hypotheses and canonical common-profile family in (B3.40). The zero-slot closure is unconditional and restores the fixed nonprincipal rows separately. $\square$

Appendix B4. The probe's local factors and conductor allocation

The physical probe is defined in (B0.21), and its Euler representation and local table are (B0.28)–(B0.30). We derive that table here, including shared primes, then verify the coefficient estimates used to average its numerator. The notation $g_\chi(a, h)$ denotes the unnormalized Gauss sum.

B4.1. Full-modulus Poisson summation

For $A = cn^3$, define the finite convolution

$$\mathcal{F}(a, A, H) = \sum_{\substack{d \bmod a \\ H \equiv b_* A d \pmod{a}}} \chi_a(d) g_{\xi\chi_A} \left(b_* A, \frac{H - b_* A d}{a} \right). \quad (\text{B4.1})$$

Expanding the a -Gauss sum and writing $m = r + b_* A l$ gives

$$\sum_{m \bmod ab_* A} \xi(m) \chi_A(m) q_a^{-1/2} g_{\chi_a}(a, -m) e(Hm/(ab_* A)) = q_a^{1/2} \mathcal{F}(a, A, H). \quad (\text{B4.2})$$

The l -sum imposes the congruence in (B4.1); the r -sum is its remaining Gauss sum. This calculation uses the full modulus and requires no coprimality between a and A .

The planar dilation contributes X/q_A ; with the physical square root it leaves $X^{1/2}/(\sqrt{q_{b_*}} q_A)$. Its Fourier argument is Xq_H/q_A . At each prime of S , the local factor of ξ is primitive and its Gauss

frequency is a unit multiple of H . Thus all frequencies meeting S , including zero, vanish. Write the remaining frequency uniquely as $H = ur^6$, with r primary and u sixth-power-free retaining the unit of H . No factor six is introduced. Mellin inversion, followed by $s = t + 1 - z$, gives (B0.28). On the initial lines $(\Re s, \Re w, \Re z) = (3, 3, 2)$, the trivial finite-sum bound gives the summable majorant

$$C_S \sum_{u,c,n,a,r} q_u^{-2} q_c^{-5/2} q_n^{-7} q_a^{-2} q_r^{-12}. \quad (\text{B4.3})$$

Together with the Gaussian and Mellin decay, this justifies the initial interchanges.

We record the phase cancellation because it determines every local numerator. CRT gives

$$g_{\xi\chi_A}(b_*A, K) = \xi(A)\chi_A(b_*)g_{\xi}(b_*, K)g_{\chi_A}(A, K).$$

At S , $K \equiv H/a$; its factor is $\sqrt{q_{b_*}}\tau\xi(A)\overline{\xi(u)}\xi(a)$. The normalizing factors in the probe leave precisely $\overline{\xi(u)}$. The remaining b_* -symbols cancel with $\chi_a(b_*)\Xi(A)^{-1}$.

For a good prime put $t_p = v_p(A)$, $k_p = v_p(a)$, $H_p = H/p^{v_p(H)}$, $a'_p = a/p^{k_p}$, $A'_p = A/p^{t_p}$. Before cancellation, its CRT-unit factor is

$$\chi_p(H_p)^{k_p-t_p}\chi_p(a'_p)^{t_p}\chi_p(A'_p)^{t_p-k_p}\chi_p(b_*)^{t_p-k_p}.$$

Reciprocity (B1.5) gives the exact identity

$$R(A, a) \prod_p \chi_p(a'_p)^{t_p}\chi_p(A'_p)^{-k_p} = \prod_p R(p, p)^{t_p k_p}. \quad (\text{B4.4})$$

Write $t_p = e_p + 3l_p$, $e_p \in \{0, 1\}$. After dividing by the cross-prime factors in $\gamma_1(c)$, the remaining pair factor for $p < r$ is $R(p, r)^{e_p l_r + e_r l_p + l_p l_r}$. By the quadratic-refinement identity (B1.4), their product is

$$R(c, n)G(n^3) \prod_p \overline{G(p^3)}^{l_p} R(p, p)^{-e_p l_p - l_p(l_p-1)/2}. \quad (\text{B4.5})$$

The first two factors, together with $G(c)$, cancel the inserted $\overline{G(cn^3)}$. Also $\alpha(c)\overline{\alpha(cn^3)} = \overline{\alpha(n)}^3$. These identities allow c and n to share primes.

B4.2. Recovering the six-row Euler table

Fix $p \notin S$, and use $Q, j, \rho, \mathfrak{a}, \mathfrak{b}, V, R$ from (B0.29). Put $\omega_p = \chi_p(-1)$, $j' = j + 6m$, and let $t = e + 3l$, $e \in \{0, 1\}$, be the local completed exponent. The finite sum is exactly $C_p(t, k, j')$ in (B0.30). Prime-power lifting gives

$$g_{\chi_p^h}(p^t, K) = \begin{cases} Q^{t-1} g_{\chi_p^h}(p, K/p^{t-1}), & p^{t-1} \mid K, \\ 0, & p^{t-1} \nmid K. \end{cases} \quad (\text{B4.6})$$

For $t = 0$, $C_p(0, 0, j') = 1$ and $C_p(0, k, j') = \mathbf{1}_{j'=0}$ when $k > 0$. For $t > 0, k = 0$, it is the source Gauss sum. For $t > 0, k = 1, j' \geq 1$, the change of variable in the prime Gauss sum gives

$$C_p(t, 1, j') = \omega_p^{-1} g_{\chi_p}(p, 1) g_{\chi_p^{t-1}}(p^t, p^{j'-1}). \quad (\text{B4.7})$$

All other cases vanish. For $k \geq 2$, the last lift $d = d_0 + p^{k-1}h$ has a nontrivial additive character and sums to zero. At $k = 1, j' = 0$, the defining congruence has no solution.

The complete local summand, after (B4.4)–(B4.5), is

$$(-1)^e \gamma_1(p)^{-e} \eta(p)^e (\overline{\alpha\gamma_3(p)})^l \rho^{k-t} \omega_p^{tk-el-l(l-1)/2} C_p(t, k, j+6m) Q^{-(s+1/2)e-(1+3s)l-wk-6zm}. \quad (\text{B4.8})$$

Here $e = 0, 1$, $l, k, m \geq 0$; there is a single sum over these indices. The bar on $\gamma_3(p)$ follows from (B4.5), and cannot be suppressed unless its square is one.

For clarity, (B4.6)–(B4.7) leave the following cases for $t > 0$:

t	$k = 0$	$k = 1$
$6r + 6$	$j' \geq t - 1$	$j' = t$
$6r + 1$	$j' = t - 1$	$j' \geq t$
$6r + 3$	$j' = t - 1$	$j' = t$
$6r + 4$	$j' = t - 1$	$j' = t$

Increasing r, m together multiplies the summand by R . The strict tails in the first two rows therefore sum respectively to

$$\frac{R(1 - Q^{-1})}{(1 - R)(1 - V)}, \quad -\frac{\eta(p)(Q - 1)Q^{-s-w}V\mathbf{1}_{j \leq 1}}{(1 - R)(1 - V)}. \quad (\text{B4.9})$$

The equalities $j' = t - 1$ or t , using (B1.7)–(B1.10), give the six values $J_j/(1 - R)$ in (B0.29). The $t = 0$ cases give $(1 - V)^{-1} + \mathbf{1}_{j=0}W^\circ/(1 - W^\circ)$. This derives the entire table, including $j = 0, 1$ and the shared-prime terms.

B4.3. Polynomial Euler corrections and their weighted mass

The averaging estimate (B0.41) uses absolute coefficient mass only before the marked moment is applied. We verify the precise weights. For an unramified row prime, put $v = \chi_p(u)$, $d_p = \eta(p)v^{-1}Q^{-s}$, and expand in $t = Q^{-w}$. Equation (B0.29) gives

$$\begin{aligned} H_{p,0} &= \frac{1 - R/Q - d_p(1 - V)}{(1 - R)(1 - d_p)}, \\ H_{p,1} &= \frac{\eta(p)Q^{-s}(1 - QV) - vV + vR/Q}{(1 - R)(1 - d_p)}, \\ H_{p,2} &= \frac{v\eta(p)(Q - 1)Q^{-s}V - v^2R(1 - V)}{(1 - R)(1 - d_p)}. \end{aligned} \quad (\text{B4.10})$$

Thus on $\Re s = a + 16e$, $\Re z = 17/50$,

$$|H_{p,0} - 1| + \sum_{k=1}^2 |H_{p,k}|Q^{k(a-1)} \ll Q^{-1-\epsilon_e}. \quad (\text{B4.11})$$

The leading linear exponent is $-1 - 16e$; the other possible largest exponents are $a - 1 - 6z_0$, $2 - 4a - 6z_0$, $4 - 6a - 6z_0$ and $-a - 6z_0$, all strictly below -1 for $a \geq .51$.

To check the error bound without a local quotient, write $E_p = P_p^* + d_p = E_0 + E_1t$, where

$$E_0 = \frac{R}{1 - R} \left(\frac{1 - Q^{-1}}{1 - V} - d_p \right), \quad E_1 = \frac{-\eta(p)(Q - 1)Q^{-s}V/(1 - V) + vR}{1 - R}. \quad (\text{B4.12})$$

Insert this into (B0.33), using $c_p d_p = v^{-1}$. Its constant coefficient is bounded by $Q^{-a} + Q^{-6z_0} + Q^{4-5a-6z_0} \ll Q^{-a}$. Multiplication by t carries the weight $Q^{a-1} < 1$, and the possible QV term has exponent $1 - 6z_0 < -1$. Hence

$$\sum_{k=0}^3 |\Delta_{p,k}|Q^{k(a-1)} \ll Q^{-a}. \quad (\text{B4.13})$$

At a row prime $p^j \| u$, $1 \leq j \leq 5$, both $\psi_u(p)$ and its conjugate vanish, so $\Delta_p = G_p$. Write $J_j = A_j + B_j t$ using the six-row table, and abbreviate $c = \eta(p)Q^s$. For $j \geq 2$, direct expansion

gives

$$\begin{aligned}
H_0 &= [1 - R/Q + (1 - V)A_j]/(1 - R), \\
H_1 &= [-\eta(p)(Q - 1)Q^{-s} + (1 - V)B_j]/(1 - R), \\
G_0 &= c[R(1 - Q^{-1}) + (1 - V)A_j]/(1 - R), \\
G_1 &= [-Q + R/Q + (1 - V)(cB_j - A_j)]/(1 - R), \quad G_2 = -H_1.
\end{aligned} \tag{B4.14}$$

For $j = 1$,

$$\begin{aligned}
H_0 &= \frac{1 - R/Q}{1 - R}, \\
H_1 &= \frac{\eta(p)Q^{-s}(1 - QV)}{1 - R}, \\
G_0 &= \frac{cR(1 - Q^{-1})}{1 - R}, \\
G_1 &= \frac{R/Q - QV}{1 - R}, \\
G_2 &= -H_1.
\end{aligned} \tag{B4.15}$$

The primitive conductor has exponent one at every such good prime. Allocate $\min(k, j - 1)$ powers of a shift Q^k to $h_r \mid \Lambda_u = q_u/q_{\text{rad}(u)}$; allocate the rest to h_0 . With $\delta = 2a - 1$, (B4.14)–(B4.15) prove

$$\sum_k |H_{p,k}| Q^{-k/2} Q^{\delta(k-j+1)+/2} \leq 1 + CQ^{-ce}, \quad \sum_k |\Delta_{p,k}| Q^{-(k+1)/2} Q^{\delta(k-j+1)+/2} \ll 1. \tag{B4.16}$$

The largest error is $G_1 \asymp Q$ at $j \geq 2$: the factor Q^{-1} cancels its size, and its shift belongs entirely to Λ_u . At $j = 2$, the quadratic error contributes $Q^{1-a}Q^{-3/2}Q^{\delta/2} = Q^{-1}$. The remaining non-tail constant or linear terms have exponents $-\delta$ at $j = 2$, $-3\delta/2$, $-\delta$ at $j = 3$, $-3\delta/2$ at $j = 4$, and $-5\delta/2$ at $j = 5$; at $j = 1$, (B4.15) leaves only negative powers. This checks every row valuation.

Products of (B4.16) cost U^ϵ , as do the divisor choices for finitely many ramified error slots. Off the row, an error slot of norm scale P , after its $P^{z_0-1/2}$ normalizer is removed, has weighted mass at most

$$P^{-1/2} \sum_{q_p \asymp P} q_p^{-a} \ll P^{1/2-a} \leq 1. \tag{B4.17}$$

The ordinary factors are summable by (B4.11). Multiplying these estimates proves (B0.44), uniformly in the row, before any moment is invoked. This is why the subsequent use of MP involves common plain profiles and permitted prime coefficients only.

T2. Primitive reduction and the precise Hecke family

The family in T0 is the finite-order, trivial-infinite-type family represented upstream by **Hecke Family.Character**: a residue character in a nonzero integral modulus, trivial on global units. No restriction to norm lifts, real characters, or bounded conductor is made in the B0–B4 bootstrap. Arbitrary target conductors enter the fixed set S only after the common power saving has been chosen. The unit factors in intermediate sextic rows are retained and need not themselves be principal.

Let η in modulus $\mathfrak{m}$ be induced by the primitive character η^* of conductor $\mathfrak{f} \mid \mathfrak{m}$. For $\Re s > 1$, absolute Euler products give

$$L_{\mathfrak{m}}(s, \eta) = L(s, \eta^*) \prod_{\substack{\mathfrak{p} \mid \mathfrak{m} \\ \mathfrak{p} \nmid \mathfrak{f}}} (1 - \eta^*(\mathfrak{p})(N\mathfrak{p})^{-s}). \tag{T2.1}$$

Indeed the imprimitive character deletes precisely the Euler factors at primes dividing the presented modulus but not the primitive conductor. Analytic continuation gives the same identity

away from the principal pole in the positive half-plane. Each displayed factor is nonzero there: $|\eta^*(\mathfrak{p})(N\mathfrak{p})^{-s}| \leq (N\mathfrak{p})^{-\Re s} < 1$. This proves equivalence of primitive and imprimitive zero sets in $\Re s > 0$, with the pole excluded. It also justifies deleting and restoring any additional fixed Euler factors at S .

For reflected sums, this scalar nonvanishing observation is not enough. One must retain the **combined conductor**: the primitive conductor times the redundant prime ideal product. B0.5 and B4 allocate repeated-prime powers against the row powerful part; B3.4 deletes and restores the redundant factors explicitly. This is why a primitive-conductor-only estimate cannot replace those arguments. The exact upstream finite-polynomial deletion identity and primitive presentation are `CenteredMomentNaturalPrimitive.polynomial_natural_deletion` and `exists_natural_primitive` in `OAI.NumberTheory.DirichletL.Moments.NaturalPrimitive`.

Here is the family argument that closes the proof, including its quantifiers. Under $\beta_* > b_0$, B0.7 fixes a positive exponent Δ from the common gap, geometry, moment mesh and power reserves before fixing η . For each target it then chooses its finite set S , finitely many profile and height orders, a sufficiently small analysis-height exponent, and an eventual threshold. B0.50–B0.51 imply a common power improvement for the Mellin principal term. Its Mellin transform is holomorphic farther left, while on its original half-plane it equals a nonzero Gaussian factor times the nonvanishing Euler correction divided by $L^S(s, \eta)$. The identity theorem rules out zeros in that enlarged half-plane; a zero could not be cancelled by that numerator. Finite factors in T2.1 do not change this conclusion.

Consequently every member of the entire Hecke family has no zero to the right of $\beta_* - \Delta'$, for one $\Delta' > 0$ independent of the member (reduce the reserve if needed). This contradicts the definition of the supremum; no zero attaining the supremum is required. Equivalently use upstream `HeckeZeroSupremum.exists_zero_near_beta` with a positive $\varepsilon < \min(\Delta', \beta_* - 1/2)$. Target-dependent constants and thresholds are harmless; a target-dependent power saving would not suffice. We conclude $\beta_* \leq b_0$, hence the Hecke assertion of T0 for arbitrary height. The classical $\Re s \geq 1$ case and the principal pole are retained throughout.

T3. All-Dirichlet transfer, including imprimitive characters and special points

Fix $q \geq 1$ and a complex Dirichlet character χ modulo q . Its norm lift has modulus $q\mathcal{O}$, and on ideals prime to that modulus is

$$\eta_\chi(\mathfrak{a}) = \chi(N\mathfrak{a}).$$

The field norm maps residue units modulo $q\mathcal{O}$ to units modulo q . Global units of $\mathcal{O}$ have norm one, so this is an allowed finite-order Hecke character of trivial infinite type. It is not assumed primitive. Its continuation is the ordinary Hecke ideal L -function of T0, as proved by the upstream ideal-series bridge.

Let χ_{-3} be the quadratic Dirichlet character of the field K , and regard $\chi\chi_{-3}$ with the common modulus used by the upstream `ShortDraftHeckeBridge.baseChangeChar`. For $\Re s > 1$,

$$L_K(s, \eta_\chi) = L(s, \chi)L(s, \chi\chi_{-3}). \quad (\text{T3.1})$$

To verify every local factor: a rational prime split in K gives two norm- p ideals and the square of $(1 - \chi(p)p^{-s})^{-1}$; an inert prime gives one norm- p^2 ideal and $(1 - \chi(p)^2 p^{-2s})^{-1}$, the product of the factors with signs $+$ and $-$; the ramified prime 3 gives one factor, while the quadratic character vanishes at 3. If $p \mid q$, the original character is zero there and both sides omit the corresponding factors. Thus imprimitive moduli and the ramified prime are included, not hidden in a primitive convention.

The analytic identity away from $s = 0, 1$ is proved in the pinned source by `OAI.SevenEighths.HeckeDirichlet.LFunction_eq_dirichlet_product`. Its norm lift, modulus and coefficient identity

are the declarations `character`, `modulus`, and `idealCoeff_character` in the same file. We use those exact objects.

Now suppose $b_0 < \Re s$ and $\neg(\chi = 1 \wedge s = 1)$.

- If $\Re s \geq 1$, use the classical all-Dirichlet nonvanishing theorem `DirichletCharacter.LFunction_ne_zero_of_one_le_re`, with the supplied pole exclusion converted by `not_and_or`. This includes $s = 1$ for nonprincipal characters. The principal case at $s = 1$ is excluded by hypothesis.
- If $\Re s < 1$, positivity of b_0 gives $s \neq 0$, and the strict upper bound gives $s \neq 1$. T0 and T2 give nonvanishing of $L_K(s, \eta_\chi)$. Equation T3.1 then implies $L(s, \chi) \neq 0$: otherwise its product with the finite complex value of the companion function would vanish.

This proves the all-modulus, all-character, arbitrary-height theorem. In particular, the value at $s = 1$ is never handled by a fallacious zero-times-pole argument; the companion can be principal there. The point $s = 0$ is outside the asserted half-plane. There is no Euler-factor restriction added to the theorem’s statement.

The same proof with $b_0 < \theta$ gives the rational statement

$$\forall \{q : \mathbf{N}\} [\text{NeZero } q] (\chi : \text{DirichletCharacter } \mathbf{C} \ q) \{s : \mathbf{C}\}, \\ \theta < \Re s \rightarrow \neg(\chi = 1 \wedge s = 1) \rightarrow \chi.LFunction(s) \neq 0.$$

The public declarations are `Cycle25.dirichlet_nonzero` and `Cycle25.dirichlet_nonzero_catalogue` in `Cycle25.Assembly.Final.Endpoint`.

T4. Zeta half-plane

For the principal character modulo one, upstream `DirichletCharacter.LFunction_modOne_eq` identifies its L -function with `riemannZeta`. Thus $\zeta(s) \neq 0$ for $\Re s > b_0$, $s \neq 1$. The ordinary analytic zeta function has its pole at one. If a Lean statement includes the assigned value there, Mathlib’s separate `riemannZeta_one_ne_zero` handles that convention; it is not an assertion that the analytic pole is removable.

Background sources used in B0–B4

The formal development uses [OpenAI/math](#) at revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, and Akash Levy’s [weighted-numerator development](#) at revision `2fd60c0926b66ea18d7436f5ed55250fd006ab5d`. Both upstream attributions and Apache-2.0 notices are retained. The public endpoint module is `Cycle25.Assembly.Final.Endpoint`; the coefficient, reflection, moment and source adapters named above identify the analytic dependencies. The following literature supplies the background and normalization conventions:

- [DR] Alexander Dunn and Maksym Radziwiłł, *Bias in cubic Gauss sums: Patterson’s conjecture*, *Annals of Mathematics* 200 (2024), 967–1057, [arXiv:2109.07463v3](#). Only the unconditional theta/cusp material of §5 and Appendix A is used. Its conditional prime asymptotic is not used.
- [GL] Leo Goldmakher and Benoît Louvel, *A quadratic large sieve inequality over number fields*, *Math. Proc. Cambridge Philos. Soc.* 154 (2013), 193–212, [arXiv:1112.1642v2](#), Definition 1 and Theorem 1.1. B2 checks the quadratic family and masks; the corresponding sharp canonical sieve is also located in the pinned Lean source.
- [HB] D. R. Heath-Brown, *Kummer’s conjecture for cubic Gauss sums*, *Israel J. Math.* 120 (2000), 97–124, [Theorem 2](#). The squarefree cubic sieve is also stated in DR, Theorem 9.1, and proved in the pinned `CubicSieve/Sharp.lean`.

- [GZ] Peng Gao and Liangyi Zhao, *Moments of quadratic Hecke L -functions of imaginary quadratic number fields*, J. Number Theory 209 (2020), 359–377, [arXiv:1707.00091v3](#), equation (1.1), for the ordinary primitive trivial-infinite-type functional equation. The formal proof uses the pinned Hecke primitive reflection declarations.
- [TZ] Jesse Thorner and Asif Zaman, *A unified and improved Chebotarev density theorem*, Algebra & Number Theory 13 (2019), 1039–1068, [arXiv:1803.02823v3](#), Theorem 1.1 for the fixed identity extension. Only a fixed-field annular prime-ideal count is needed; no uniformity over growing extensions is invoked.

Authorship and provenance

Hailey Collet initiated and directed the research, chose the targets and sources, and decided what to retain, check and combine. The detailed analytic derivations and much of the drafting were developed with AI assistance, mainly GPT6-Pro and non-Pro Astra, with contributions from GPT 6.1 Sol, Grok 4.7 and Gemini 3.1 Pro. AI assistance was also used for Lean formalization. These model names are the author’s account of the tools used, rather than a per-message telemetry audit. Some research prompts were AI-written and selected or adapted by the author.

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